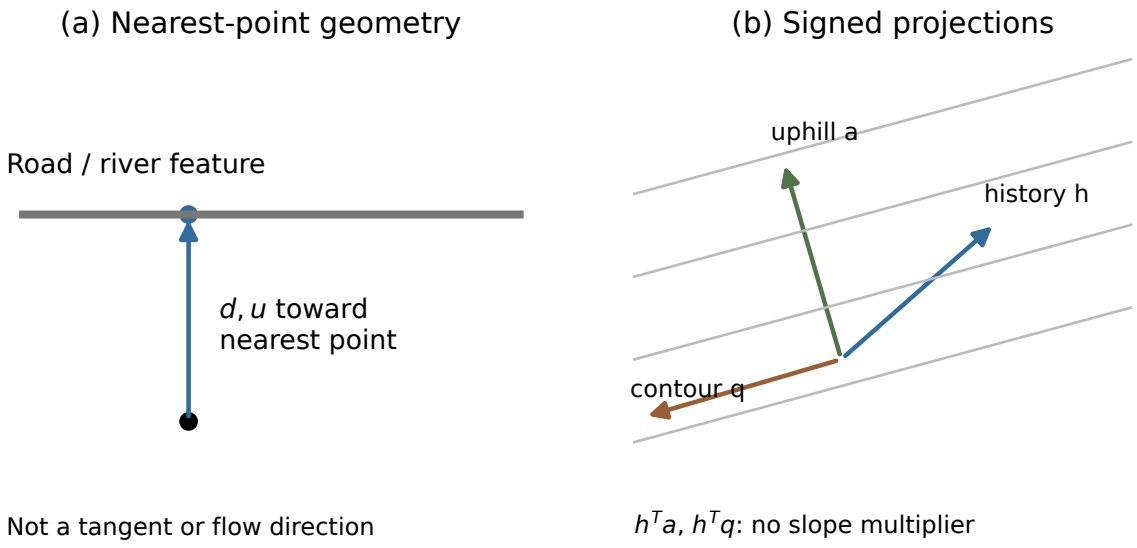
Scientific Supplement: Fitted Predictors, Complete Comparisons and Reproducibility Specifications

Exploratory experiment closeout, 16 October 2026; authorship unconfirmed

This is an independent scientific reading document, not the main paper's appended execution log. The new visual explanations below use unchanged public aggregates. Detailed retained specifications and full tables follow. Historical saved-score captions refer to their capture version; current qualification is the main paper's 14 equivalent/seven inconclusive finite-algorithm states, with resolution-invariant states all inconclusive. Retained failures and scientific gaps are not repaired by reorganising the document. The audit record preserves versions, execution history and review responses.

1 Terrain geometry and the ten configurations



(c) Frozen WorldCover groups and road interaction

built	vegetation	water/wetland	valid other
50	10,20,30,40, 95,100	80,90	other valid codes

$1_{\text{group}} \log(1 + d/(1\,m))$; nodata is missing, not valid other.

Figure 1: Schematic, not a real route. Left: nearest road/river point, distance and direction toward that point, not a tangent or river flow. Centre: signed history h, uphill a and contour q, with distinct dot products; slope magnitude is a separate input and is not depicted. The cover groups are built, vegetation, water/wetland and valid other, with frozen code sets shown. Cover membership and retained-road distance combine; invalid queries are not the valid "other" cover class. The excluded road classes are listed in the retained specifications below.

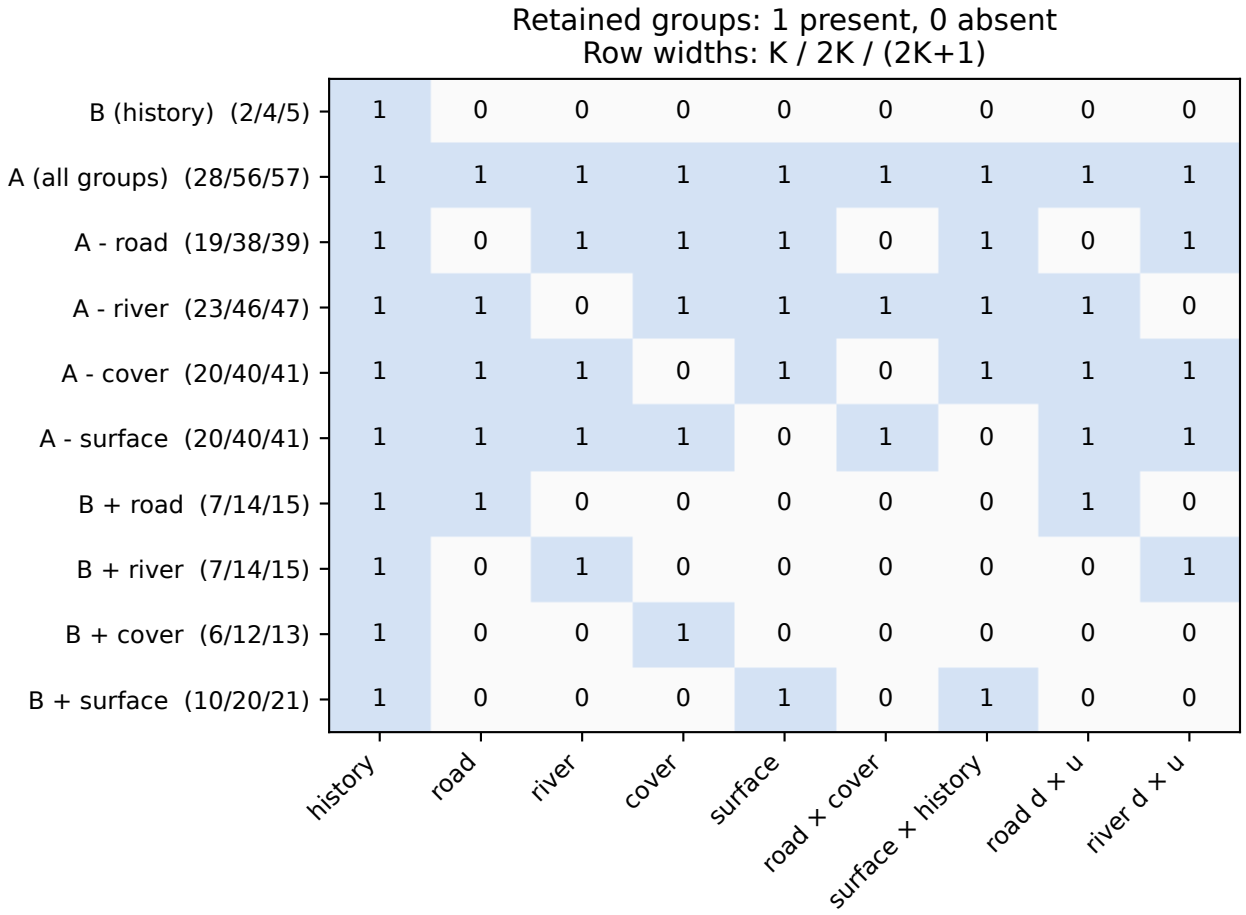


Figure 2: Ten independently fitted configurations. Road–cover interactions require both parents; surface–history remains in LIO-surface. K counts numeric features, not total design columns. For LOO the machine candidate is all-terrain and control is all-minus-group; a factor benefit expressed as removal loss reverses the sign and interval endpoints. LIO compares history-plus-group to base. The families and nonadditive interactions are not redefined by this drawing.

2 Complementary error and uncertainty views

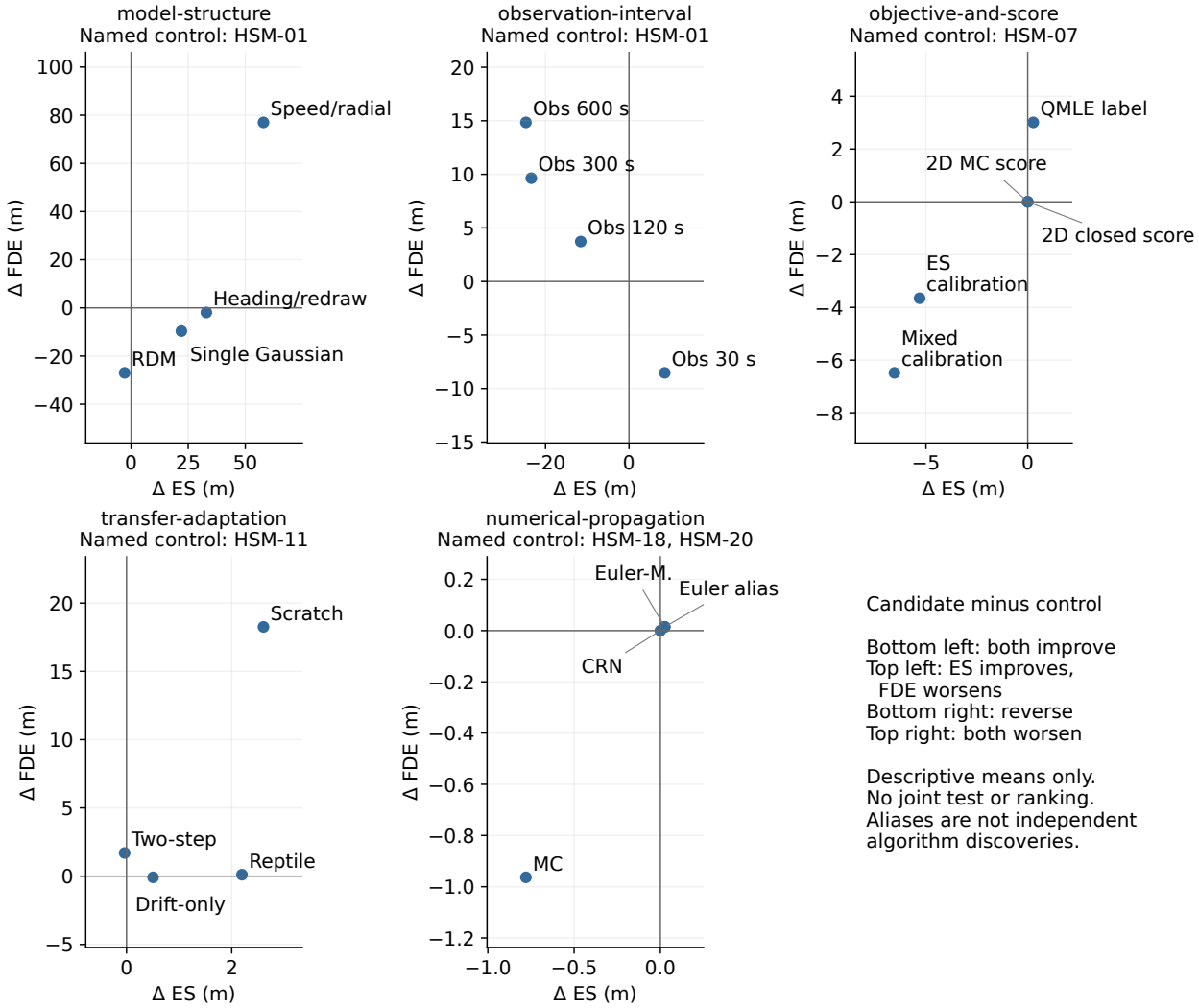


Figure 3: All 21 candidate-minus-named-control contrasts: weighted ES versus mean-position FDE (m). Negative is favourable on each axis. Facets retain original comparison families; named HSM controls are not pooled. These are descriptive paired means, not a new joint test or Pareto ranking.

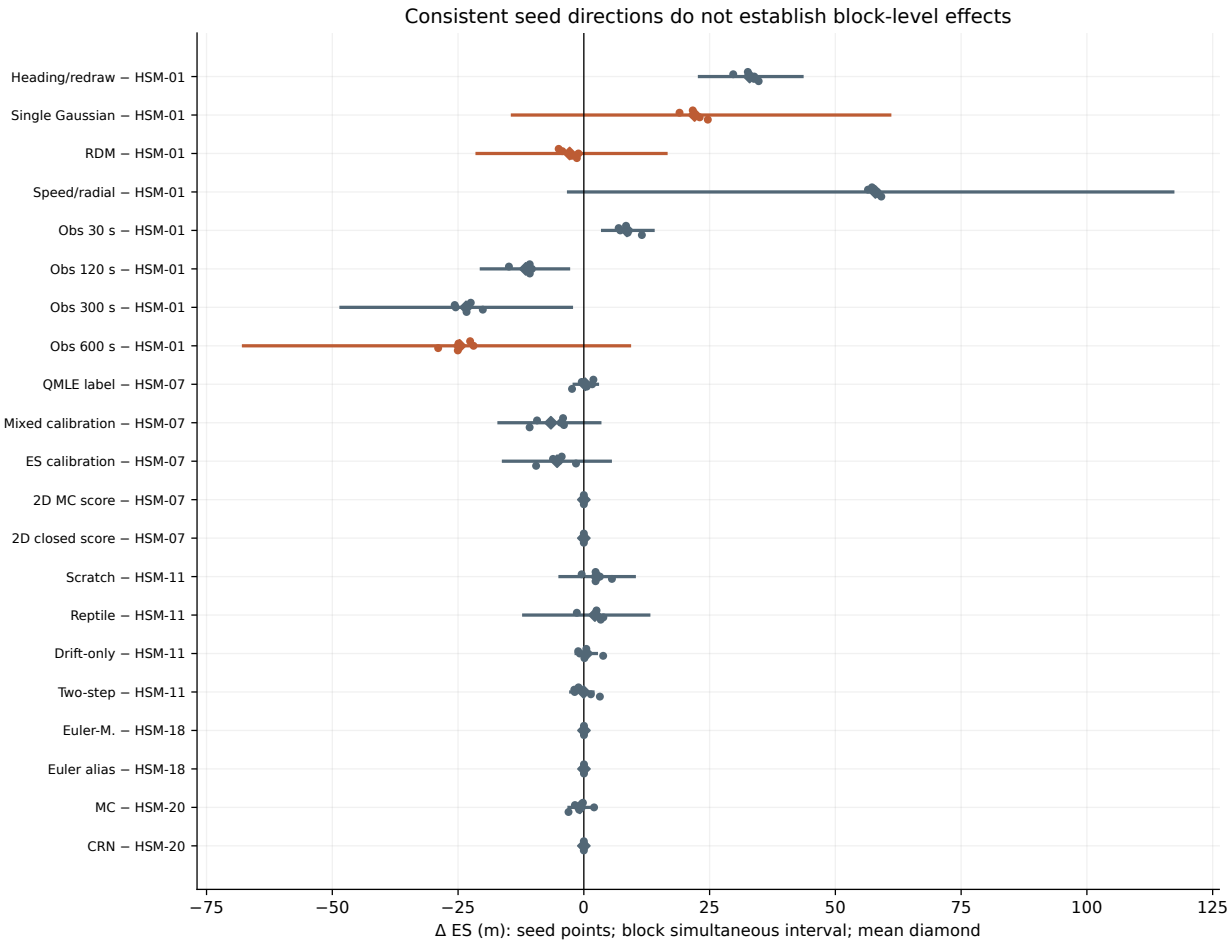


Figure 4: Five saved seed effects versus original record-block simultaneous intervals (m). All comparisons are retained, including the examples with unanimous seed signs and block intervals crossing zero. Seeds vary Monte Carlo simulation, not fitted training samples or independent participants.

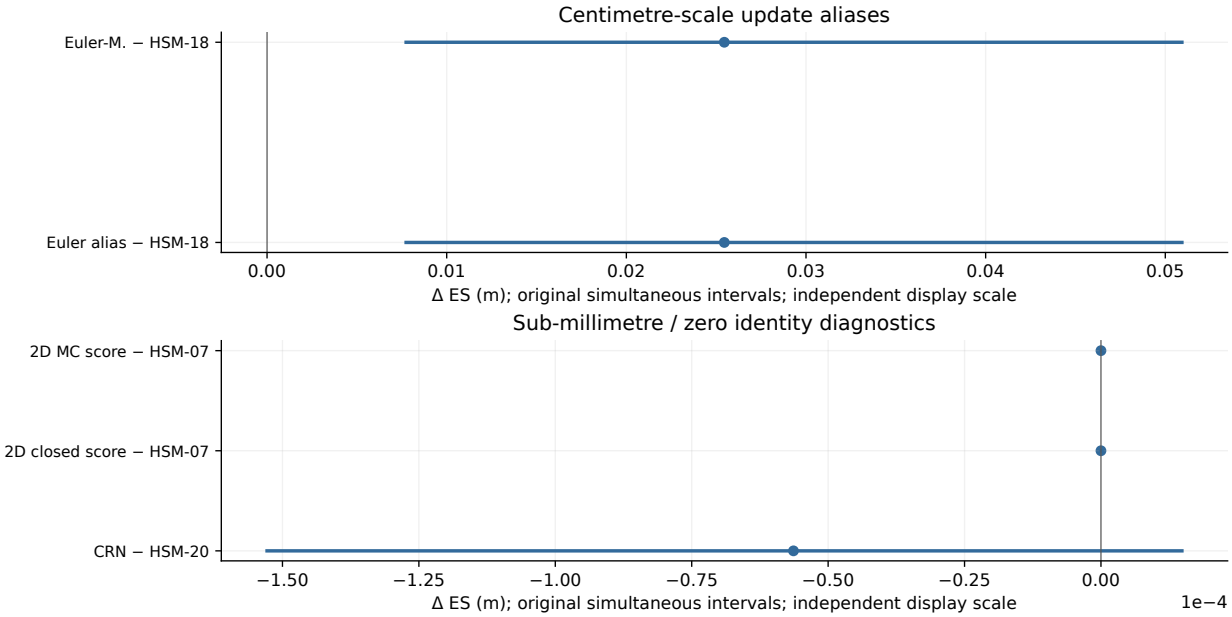


Figure 5: Tiny implementation-consistency effects at their own explicitly labelled scale, complementing the complete practical-effect comparisons below. Original point estimates and intervals are unchanged; exact or near-zero effects do not supply an independent power proof.

3 Fit diagnostics, runtime and the complete method overview

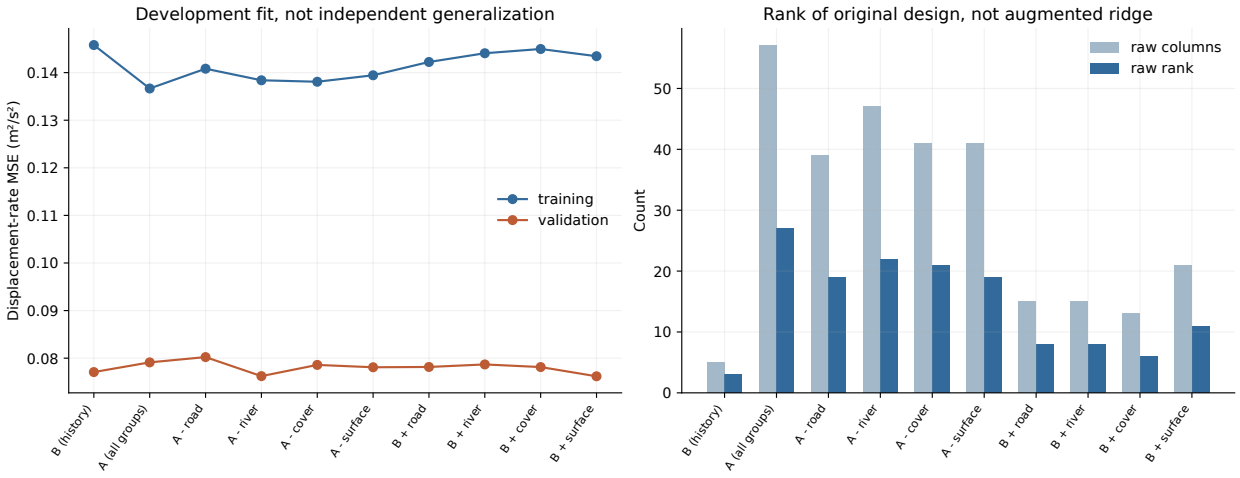


Figure 6: Ten development configurations: training/validation displacement-rate MSE (m^2/s^2), original columns and raw rank. Fit error is not forecast ES. Raw rank deficiency, augmented-ridge full rank and out-of-sample performance are distinct. Rows and regularisation remain unchanged.

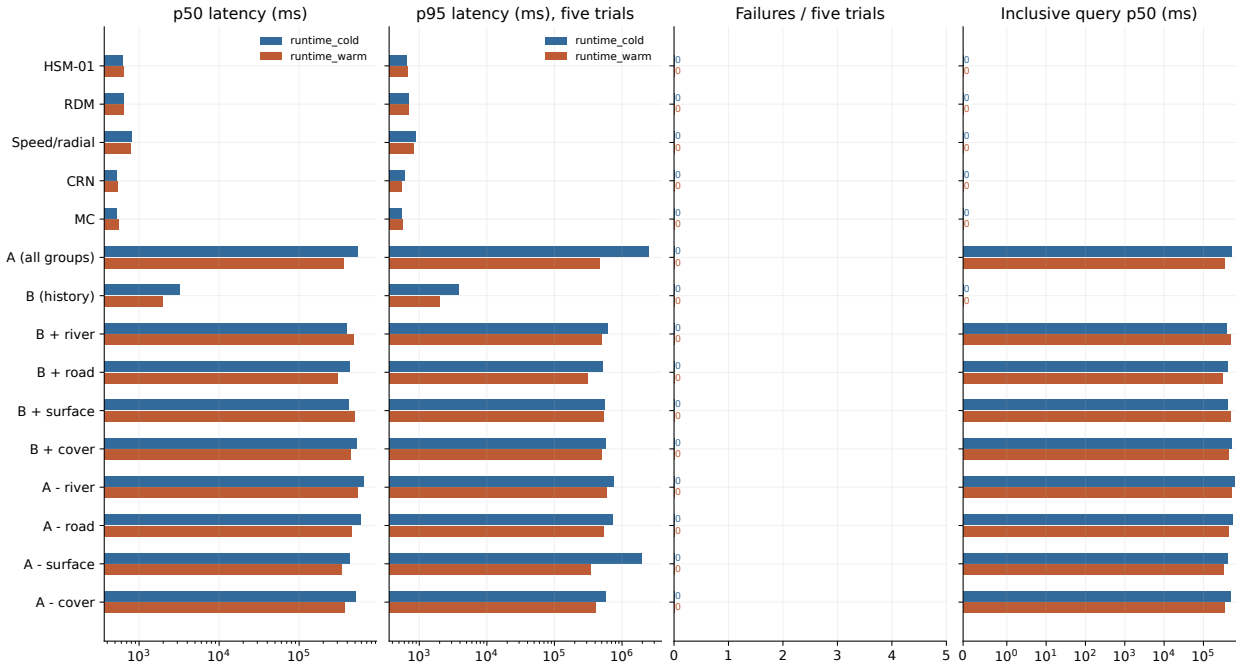


Figure 7: Completed original provider-cold/provider-warm latency p50/p95 (ms) and failure counts for all 15 subjects. Each condition uses five trials; The fourth panel uses original trial records for median inclusive terrain I/O/query milliseconds (zero for no terrain stage), not an additional cost. 15 prewarm runs are excluded from quantiles, not the recorded exposed trials. Provider-cold is not OS-cache-cold. Query costs are inclusive in rollout; five observations do not estimate a stable operational tail.

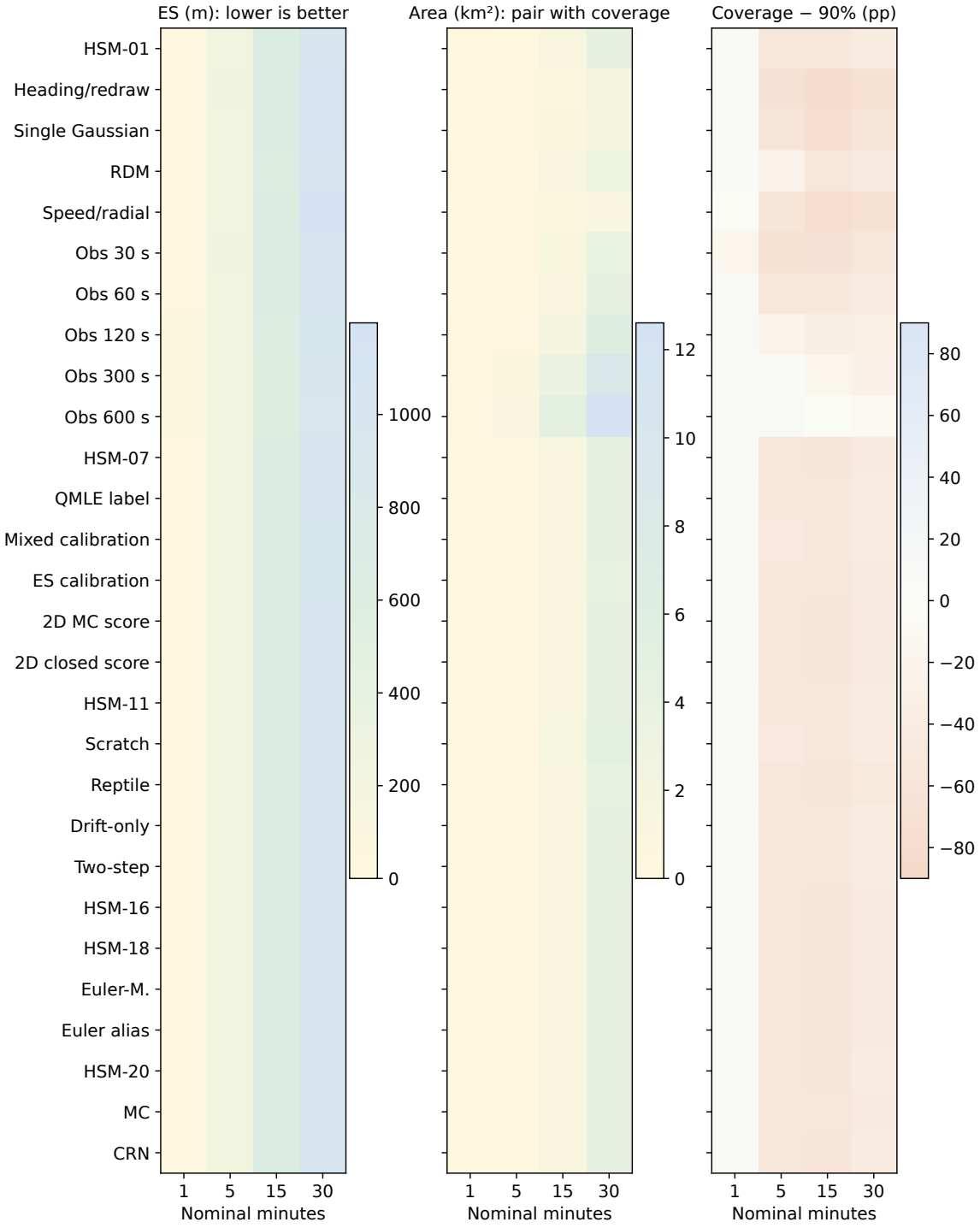


Figure 8: All 28 saved method identities and four horizons: ES (m), mean 90% disk area (km^2), and coverage minus nominal 90% (percentage points). The shared diverging coverage scale is centred at zero, with both overcoverage and undercoverage visible. Complete all-level tables remain below; no configuration is selected as a winner.

4 Family availability, not a terrain benefit ranking

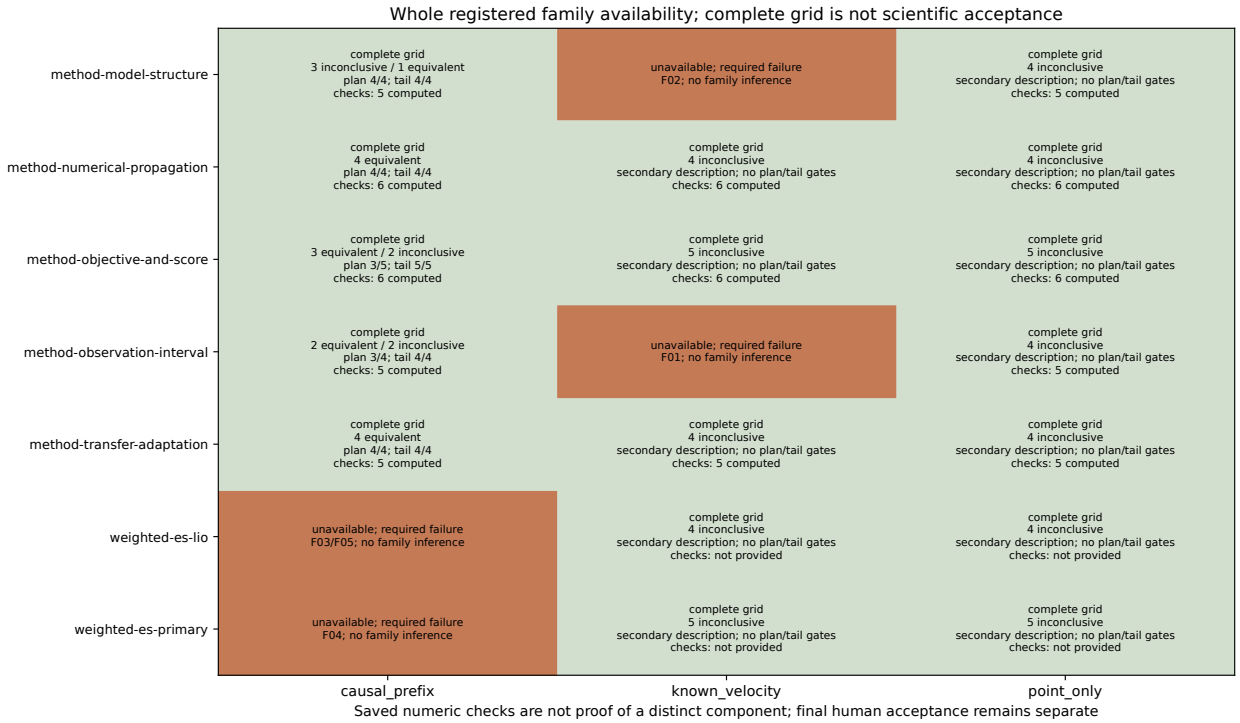


Figure 9: Original seven families by three origin modes. A required failure invalidates its whole registered family; individual successful pairs are not a replacement family. Complete does not mean accepted, practically beneficial, mechanism-qualified or resolution-invariant. The original comparison tables retain each qualification reason.

5 Missing-input and history-feedback limits

Invalid online numeric features are zero-filled with a validity mask. Within a frozen conditioner, the induced drift change is $\Delta b = \theta_{\text{num}}\Delta f + \theta_{\text{mask}}\Delta m$, not necessarily zero. The 404/81 valid development origins do not span these missing-input states. Feature-file counts, cached query counts, absence of an element, invalid geometry and a failed forecast have different denominators. HSM early/late speed and drift traces and per-feature missingness across every forecast were not retained; the four position snapshots cannot reconstruct them. A one-half overlap correlation for constant-noise 10 s secants is a local calculation, not evidence for the full nonlinear feedback process or a measured observation-noise model. No long simulations are rerun here.

6 Independent follow-up designs, not revised evidence

These designs do not replace this cohort, alter its registered families, or repair a retained failure retrospectively. None is executed in this revision.

N01: Task-matched probabilistic reference. Use a constant-velocity state $z = (x_e, x_n, v_e, v_n)$ with process noise, initialised only from the same visible prefix. Choose process and measurement noise on development data, not this final cohort. Match origin eligibility, saved clock, four targets, particle count and scoring conventions. Report ES, mean-position error, marginal coverage and disk area at each target, paired by recording block; distinguish Monte Carlo repeats from people.

N02: Mode correspondence before parameter mixing. Save direction support, component means, weights and uncertainty for each training, adaptation and meta-task role. Specify a matching rule using development data only, compare permuted and matched components before interpolation, and include a no-mixing reference. This tests the specific index-mixing procedure, rather than treating a heading rank as a shared behavioural state or validating meta-learning in general.

N03: Calibration of the evaluated distribution. On development trajectories evaluate the actual multi-mode, multi-step position distribution. Keep component assignments and probability weights consistent between fitting and validation; distinguish marginal disk calibration from simultaneous path coverage. Freeze tuning before new final evaluation. Do not tune a single radius multiplier on the current cohort, whose one-minute 50% and 90% behaviours already differ.

N04: Terrain information versus representation and penalty. Use a separately registered cohort and design. Remove redundant constant mask columns or explicitly match their effective intercept penalty; fit feature scaling on training data only. Compare dimension/penalty-matched controls and keep mode count, shared drift, timing and map eligibility fixed. Specify dependent interaction removal in advance. Report configuration increments unless information, representation and regularisation have actually been isolated; do not select successful pairs from failed families.

N05: Clock and source dependence. First obtain the provenance of relative timestamps and reconstruct a documented source-to-saved-clock transformation; then check absolute-time and UTC interpretation separately. If source identities permit, group near-duplicate routes and repeated visits without using outcomes to choose clusters, and compare source/route-cluster sensitivity with recording-block results. Save early/late drift and speed traces and per-feature validity counts in future runs. These diagnostics cannot be recovered from four retained position snapshots or certify independent participants.

7 Retained detailed scientific specifications and tables

The following mechanically migrated units retain the original formulas, tables and source identities. Their unit-level hashes and old-label destinations are recorded in `revision46-source-migration.json`; the verbatim pre-revision files remain separately named historical sources. Only reading structure, cross-document links and counting-unit clarification are changed. The main paper, not an older snapshot's pending wording, determines the current review state.

8 Related work and scope

9 Continuous-time motion and switching dynamics

Persistent motion has a well-established stochastic formulation. Johnson et al. integrate an Ornstein–Uhlenbeck velocity process to obtain a continuous-time correlated random walk for irregular animal telemetry, including a state-space observation model[1]. The connection is temporal persistence and a distribution over future locations. Here the propagated continuous state is position; velocity and heading summaries are constructed from the visible prefix and, where enabled, simulated history. We do not estimate the same latent-velocity and observation-error model. The animal-telemetry application is a methodological precedent, not evidence that its population or observation process matches these GPX recordings.

Ghahramani and Hinton combine discrete regimes with linear continuous dynamics in switching state-space models and use variational inference for otherwise intractable hidden-state posteriors[2]. This establishes why locally Gaussian dynamics need not produce a single Gaussian trajectory distribution. Our fitted modes and conditional propagators use that broader modelling idea, but ordinary mode weights are saved fit parameters rather than a newly inferred prefix posterior. The specific fitting, switching and propagation rules are described below; neither a mixture nor a switching mechanism is claimed as a new invention.

10 Environment-conditioned and learned dynamics

Environmental information can enter movement models through more than a map image. Integrated step selection analysis jointly estimates movement and resource-selection parameters by comparing observed steps with available alternatives[3]. Its corrected parameter expressions are documented separately[4]; we use the framework only for conceptual positioning, not its likelihood or coefficient formulas. Our road, river, cover and surface features instead modify a position-SDE drift. The terrain comparisons ask about incremental predictive information after refitting within a fixed feature family, not habitat preference or a causal effect of landscape on behaviour.

Neural SDEs enlarge the function class, but their learning tasks differ. Kidger et al. use Brownian input to generate paths with an SDE solver and train against a neural controlled differential-equation discriminator in a Wasserstein-GAN formulation[7]. Gao et al.'s Langevin Graph Network Approach (LaGNA) instead targets equations for coupled network dynamics, separating self, interaction and diffusion contributions with neural modules[8]. Generating a path distribution and recovering a governing network equation are distinct objectives. Our single-record GPX task evaluates finite position forecasts; it does not identify inter-agent interaction laws or validate recovered equations against known ground truth. Social GAN couples recurrent motion histories, inter-agent pooling and adversarial diversity[5]. Trajectron++ combines probabilistic multi-agent forecasting, dynamics and heterogeneous context including semantic maps[6]. These works connect multimodality, dynamics and context, but address different data, interaction structures and training objectives.

We retain the previously selected linear-in-parameters conditioner to make the feature definitions and matched component changes inspectable. This is a fixed representation choice, not evidence that linear models are optimal. Gaussian-mixture modes, history feedback and spatial features also mean that linear coefficients do not imply one globally linear, single-Gaussian path law. The present contribution is the controlled study of method components and environmental inputs in one visible-prefix GPX task, including their limitations; it is not a new neural architecture. The cited external models are not executed baselines. The registered simple CV reference

now has a descriptive same-population comparison in Section 40.1. This is not an independently audited superiority claim, and no superiority over the cited external models follows.

11 Distributional evaluation and finite-computation evidence

A predictor must be assessed as a distribution, not only by the distance between its mean and a realized endpoint. Proper scoring rules provide this foundation[9, 10]. We use a common energy-score estimator alongside mean-position error, empirical coverage and region size: a larger region may cover more outcomes without improving location accuracy. Component effects therefore require paired comparisons and an explicit practical-importance threshold, not a ranking of isolated scores.

Numerical integration and ensemble sampling add distinct approximation errors[11]; increasing particles does not increase the number of recording blocks. Our finite numerical evidence is reported separately from block uncertainty. Familywise confidence intervals use a studentized block bootstrap[14], whereas Holm adjustment concerns zero-null test probabilities[13]. Adaptation is described by the executed first-order Reptile-style updates[12]. Together these distinctions define what the existing comparisons can support without converting a finite check into a convergence theorem, a confidence interval into a test probability, or training diagnostics into held-out predictive performance.

12 Forecast models and terrain ownership

Figure 10 separates the two executed model families before introducing their equations. Both predict future positions; neither predicts a future map. Static geographic features enter the terrain branch, whereas the ordinary method HSM uses local position, solar elevation and sampled modes, not the terrain regression. Their common scoring protocol does not make them one fitted model or one interchangeable control.

Terminology. A terrain *conditioner* is the additive drift correction $c_{\theta_g}(F_g)$, not a separate likelihood or neural network. A *provider* returns map or clock features; an *ensemble* is a collection of simulated forecast paths, not independent observations. A *verdict* is an evidence disposition under the registered rules, not a model ranking. `HSM` denotes the ordinary method reference, whereas `base` and `all-terrain` denote terrain configurations. Code identifiers remain unchanged. In particular, `dt` labels an observation interval, not the integration step, and `Two-step` and `Speed/radial` are separate implemented alternatives.

13 Causal terrain dynamics

The prediction target is a distribution of future positions conditional on the visible origin, not a separate prediction of terrain or a measured future velocity. Let $X_t = (X_{t,e}, X_{t,n})^\top \in \mathbb{R}^2$ be east/north position in metres. Let Z_t collect the causal position-history buffer, its secant velocity v_t (metres per second), unit direction h_t, and direction-validity flag η_t. These auxiliary quantities are computed from visible positions initially and predicted positions thereafter; Z_t has no independently fitted velocity SDE. A static map lookup at the current X_t, combined with this history, is encoded as $F_g(X_t, Z_t) = \phi_g \in \mathbb{R}^{2K_g}$, including values and masks. Here g denotes a selected terrain configuration, not a latent mode. The terrain predictor evolves as

$$dX_t = \{b_0(Z_t) + c_\theta(F_g(X_t, Z_t))\}\, dt + L\, dW_t. \tag{1}$$

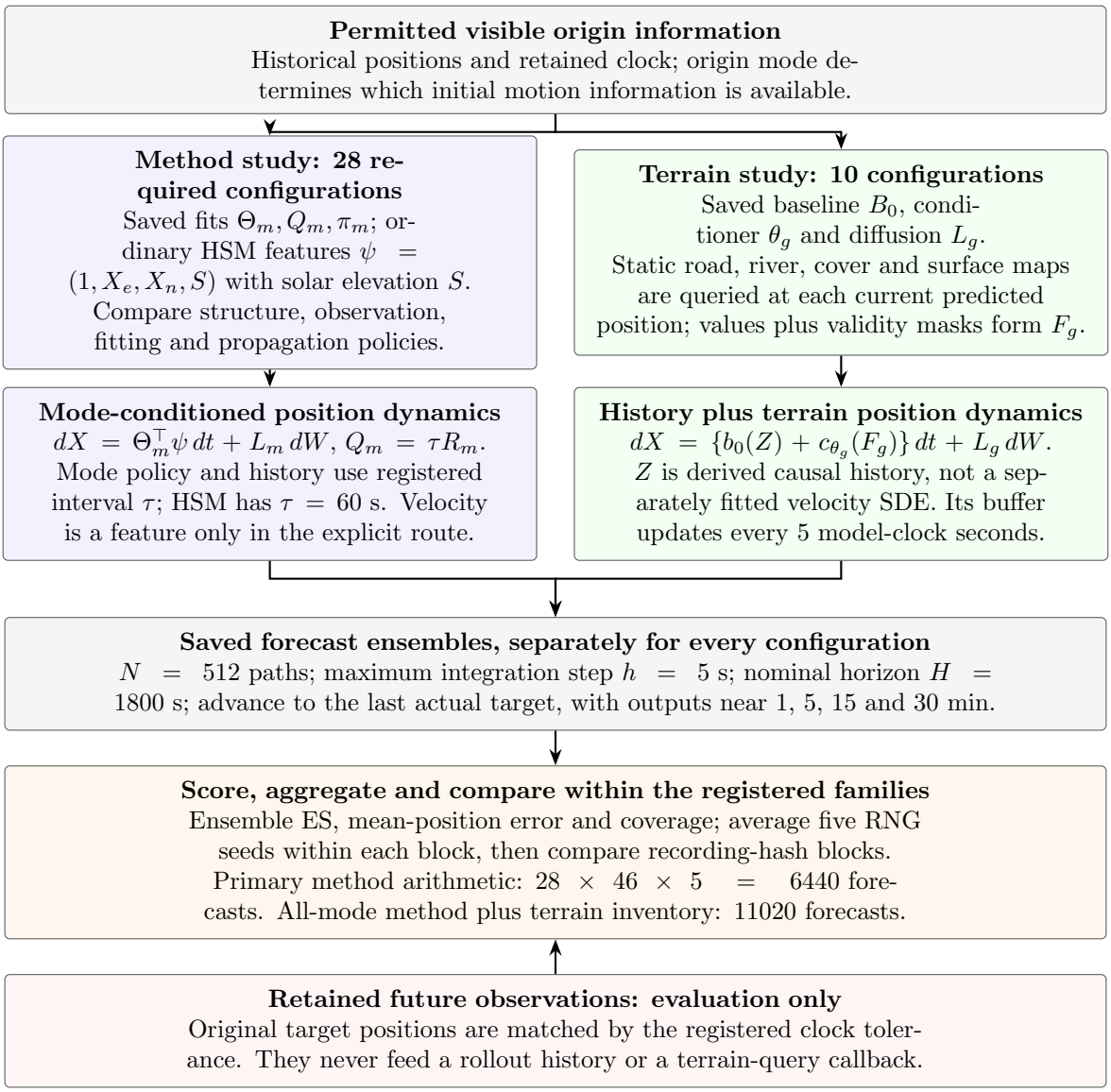


Figure 10: Executed-model and evaluation data flow, not an additional experiment or a neural architecture. The two branches have independent fitted models and controls. The future-observation arrow ends at scoring, not prediction. τ (observation/history interval), h (integration step) and H (forecast duration) are distinct; N and five RNG seeds are simulation resources, not independent observed people. The nominal time axis is conditional on retained clock provenance.

$W_t \in \mathbb{R}^2$ is a standard Wiener process with independent components, $\mathrm{Cov}(W_{t+\Delta} - W_t) = \Delta I_2$. The drift $b_0 + c_\theta$ has units of metres per second; $L \in \mathbb{R}^{2\times 2}$ is a constant noise-amplitude matrix with units metres per square-root second. Its covariance is $Q = LL^\top$, not L itself. The two deterministic parts are explicitly

$$\begin{aligned} r(Z_t) &= (1, v_{t,e}, v_{t,n}, \|v_t\|, h_{t,e}, h_{t,n}, \eta_t)^\top, \\ b_0(Z_t) &= B_0^\top r(Z_t), \qquad c_\theta(\phi_g) = \theta^\top(\phi_g^\top, 1)^\top. \end{aligned} \tag{2}$$

Here $B_0 \in \mathbb{R}^{7\times 2}$ fits the baseline displacement rate by ridge regression with strength 10^{-6}; invalid history direction components are zero with $\eta_t = 0$. Writing R for the stacked rows $r(Z_i)^\top$ and $V_i = \Delta x_i/\Delta t_i$, the actual first-stage fit is

$$B_0 = \arg\min_B \{\|RB - V\|_F^2 + \rho\|B\|_F^2\} = (R^\top R + \rho I_7)^{-1} R^\top V, \qquad \rho = 10^{-6}. \tag{3}$$

This is an unnormalized sum-of-squares objective, including a penalized intercept; ρ is not a mean-loss penalty. The two stages are sequential, not a joint optimization of B_0 and θ. All ten saved terrain fits have exactly identical B_0 coefficients; only the selected correction design and fitted diffusion differ. This check compares existing parameters, not a new fit or a claim that the correction is orthogonal to history. The conditioner coefficients $\theta \in \mathbb{R}^{(2K_g+1)\times 2}$ include an intercept. They fit the residual left by the baseline, rather than regressing observed position directly on a map label. For training transition i, define $y_i = \Delta x_i/\Delta t_i - b_0(Z_i)$ and $x_{\mathrm{aug},i} = (\phi_{g,i}^\top, 1)$. Stacking the rows gives $Y \in \mathbb{R}^{n\times 2}$ and $X_{\mathrm{aug}} \in \mathbb{R}^{n\times(2K_g+1)}$. Each conditioner solves

$$\min_\theta \frac{1}{2n}\|X_{\mathrm{aug}}\theta - Y\|_F^2 + \frac{\lambda}{2}\|\theta\|_F^2, \qquad \lambda = 10^{-4}, \tag{4}$$

The first term is the mean squared residual over $2n$ scalar outputs; $\|\cdot\|_F^2$ sums their squared entries. The second penalizes large coefficients, including the bias, to stabilize a correlated feature design. The value of λ is frozen, not selected on final outcomes. Augmented least squares implements this mean-squared objective in float64; stationarity and conditioning checks accompany the fitted parameters. For two outputs its normal equation is $(X_{\mathrm{aug}}^\top X_{\mathrm{aug}} + n\lambda I)\theta = X_{\mathrm{aug}}^\top Y$. At $n = 404$, the saved diagonal penalty is 0.0404, not 10^{-4}. Thus the baseline and correction regularizers have different loss normalizations; their numeric strengths cannot be compared directly. Each qualified development window supplies one origin-to-first-future row. Both regressions weight displacement-rate rows equally, without duration, inverse-noise, block or participant weights. The observed gap Δt_i varies; it is neither the method fitting interval τ nor the integration step. Table 46 gives its original development distribution, with no regular-time resampling.

This is a linear-in-parameters model on fixed engineered features, with no hidden neural layers. Its drift need not be affine in position: nearest-map queries, log distances, cover indicators and history updates depend nonlinearly or discontinuously on predicted state. Neither the fitted signs nor a physical road-attraction law are imposed. An ablation tests predictive increment under this model, not the universal causal effect of terrain.

The diffusion covariance is estimated from training displacement residuals:

$$LL^\top = \frac{1}{n}\sum_{i=1}^{n} \frac{(\Delta x_i - \widehat{b}_i\Delta t_i)(\Delta x_i - \widehat{b}_i\Delta t_i)^\top}{\Delta t_i}. \tag{5}$$

The residual $e_i = \Delta x_i - \widehat{b}_i\Delta t_i$ estimates the unexplained displacement. Under the frozen-drift Brownian approximation, $e_i \sim \mathcal{N}(0, Q\Delta t_i)$, so $e_i/\sqrt{\Delta t_i}$ has covariance Q, in square metres per second. This explains the division by Δt_i and the residual-moment estimate; it is not a global

likelihood claim for the nonlinear history-dependent path. Equivalently, with rate residual $\zeta_i = \Delta x_i/\Delta t_i - \widehat{b}_i$,

$$Q = \frac{1}{n}\sum_{i=1}^{n} \Delta t_i \zeta_i \zeta_i^\top. \tag{6}$$

Unlike equal-row drift regression, this rate outer product includes duration. Residuals are not centered again, and the denominator is n, not a degrees-of-freedom correction. This is a plug-in second moment after fitting drift, not an unbiased estimator or joint drift–diffusion maximum likelihood. An eigendecomposition $Q = U\,\mathrm{diag}(\lambda_1, \lambda_2)U^\top$ supplies $L = U\,\mathrm{diag}(\sqrt{\max(\lambda_j, 0)})$. The saved terrain loader rejects a smallest eigenvalue below $-10^{-10}\,\mathrm{m}^2/\mathrm{s}$ and asymmetry above the absolute tolerance $10^{-12}\,\mathrm{m}^2/\mathrm{s}$. Only tolerated negative roundoff is set to zero for the square root; no positive floor or jitter is added. All ten saved terrain matrices are symmetric and strictly positive, so none requires this clipping. These absolute thresholds differ from the method branch's relative PSD rule; positivity is not forecast calibration. Each terrain configuration fits its own selected correction columns and diffusion parameters while retaining the shared baseline coefficients. Identical deterministic parameters can be bound to separate forecast seed labels, but this does not represent five independent training realizations. The ten terrain fits and 16 distinct deterministic method fitting identities are restored from saved models during continuation.

Euler–Maruyama integrates Equation (1) with a maximum step of 5 seconds and 512 particles. For substep Δ and independent $\xi_k \sim \mathcal{N}(0, I_2)$, the implemented position update is

$$X_{k+1} = X_k + \{b_0(Z_k) + c_\theta(F_g(X_k, Z_k))\}\Delta + \sqrt{\Delta}\, L\xi_k. \tag{7}$$

This connects the drift to the conditional mean displacement and diffusion to increment covariance $Q\Delta$. History has its own 5-second model-clock cadence. At each history tick, the predictor appends predicted positions and recomputes its secant velocity. Between ticks, history and velocity remain fixed while terrain queries use current predicted positions. Query callbacks have no target trajectory container. Invalid feature values are numerically imputed to zero and accompanied by explicit validity columns; invalid-query counts remain part of the evidence.

The history operator is part of the model, not an integration detail. At terrain tick $t_j = 5j$ seconds after the origin, let $\{(s_{j,r}, x_{j,r}^{(p)})\}_{r=1}^{q_j}$, $2 \le q_j \le 3$, be particle p's retained buffer after appending its new predicted position and keeping at most three entries. Its velocity is

$$v_{t_j}^{(p)} = \frac{x_{j,q_j}^{(p)} - x_{j,1}^{(p)}}{s_{j,q_j} - s_{j,1}}. \tag{8}$$

The initial buffer contains only the visible origin information permitted by the origin mode. Early updates can therefore mix original irregularly timed prefix points with predicted ticks; their denominator is the actual retained time span, not an assumed 5 seconds. Once three equally spaced tick positions are retained, the secant spans 10 seconds. Intermediate integration substeps and requested scoring outputs do not themselves append history. Between history ticks the retained buffer and velocity are held fixed.

14 Observed and simulated history are different inputs

The terrain history is a three-position buffer, initialized with observed positions and subsequently updated with generated positions at 5-second model-clock ticks. If the last observed gap is d, the first update retains the last two observations and the new position, so the secant spans $d + 5$

seconds. The second update retains the origin and the positions at 5 and 10 seconds; subsequent spans are 10 seconds. A fixed number of points therefore does not imply a fixed secant duration or unchanged input distribution. The observed training, validation and primary-prefix timings are reported separately in Section 21; they describe clock support, not measured feedback velocities or an experiment changing this cadence.

There is also a noise-distribution change. For the illustrative local model $dX_t = b\,dt + L\,dW_t$ with constant b, L, define a secant over $T > 0$:

$$\widehat{v}_t = \frac{X_t - X_{t-T}}{T} = b + \frac{L(W_t - W_{t-T})}{T}, \qquad \mathrm{Cov}(\widehat{v}_t) = \frac{Q}{T}, \quad Q = LL^\top. \tag{9}$$

Overlapping buffers reuse the same Brownian increments. For lag $s \geq 0$ in this constant-coefficient illustration,

$$\mathrm{Cov}(\widehat{v}_t, \widehat{v}_{t+s}) = \frac{(T-s)_+}{T^2} Q. \tag{10}$$

At $T = 10$, adjacent 5-second terrain ticks consequently have correlation $1/2$ in directions with nonzero variance, not independent velocity noise. The saved base and all-terrain fits give $\mathrm{tr}(Q) = 2.7387$ and 2.5077 m^2/s. The corresponding two-dimensional local RMS velocity perturbations $\sqrt{\mathrm{tr}(Q)/10}$ are 0.5233 and 0.5008 m/s. These are arithmetic consequences of existing diffusion fits, not measured rollout velocity dispersions or bounds for the history-dependent model.

Observed GPX coordinates may also have measurement error. If one merely illustrates independent endpoint errors of covariance R_{obs}, independent of that local process, the observed-secant covariance becomes $Q/T + 2R_{\mathrm{obs}}/T^2$. This is not a fitted sensor model: neither R_{obs} nor a zero-error observation assumption is established here. Simulation instead feeds generated position increments back into both the baseline velocity and history-dependent features. Normalizing direction, changing map features and drift feedback invalidate the constant-drift variance calculation as a formula for the complete predictor. The kernel does not clip secant speed or apply a velocity-error filter at history ticks. The six illustrated forecast archives retain four position outputs, not full early/late velocity and drift traces. Thus the present evidence does not identify history feedback as the cause of long-horizon undercoverage, nor certify abnormal-speed rates or sensitivity to changing this cadence. The method matrix uses its different adjacent-τ secant (Equation (17)); the 10-second calculation applies to the terrain buffer only.

15 Frozen terrain-feature encoding

The terrain conditioner uses the frozen Heading/redraw transforms, not a newly learned embedding or final-evaluation standardization. For road and river, let d_g be the queried distance in metres and v_g the query-supplied east/north unit vector from the query position towards the nearest point on the selected polyline. The provider selects the closest point in its local metric projection and reports a spherical distance to it. Thus v_g is neither a line tangent, a road travel direction nor a river flow direction; coincident queries have no usable direction. Their features are

$$\ell_g = \log\{1 + d_g/(1\,\mathrm{m})\}, \qquad u_g = v_g/\|v_g\|, \qquad g \in \{\mathrm{road}, \mathrm{river}\}. \tag{11}$$

Distance is usable only when nonnegative. The unit-vector transform requires a norm greater than 10^{-12}. Baseline history direction h uses the same unit-vector transform, with availability determined by the causal history/origin rules. These selected transforms have no fitted centering, scaling or clipping parameters. The logarithm is a frozen engineering choice: it preserves distance ordering, is finite at zero and compresses the large range of map distances, with derivative $1/(1\,\mathrm{m} + d_g)$.

These properties explain the representation, not its empirical optimality. No physical logarithmic response law or comparison proving superiority to raw distance is established by the present ablation; unselected transformations remain outside its result claims.

The four WorldCover indicators w are, in order, built (code 50), vegetated (10, 20, 30, 40, 95, 100), water/wetland (80, 90), and bare/snow/other (60, 70 or another finite unrecognized code). The last category applies only to a usable source value: a missing or nodata query is not recoded as valid "unknown" cover. Surface orientation retains the six supplied components $s = (\beta, \sin\alpha, \cos\alpha, n_e, n_n, n_u)$: slope in radians, downslope aspect east/north, and the three surface-normal components. This is an identity transform, not a fitted terrain representation.

For a valid cover code k, write

$$\begin{aligned} &w_j = \mathbf{1}\{k \in \mathcal{C}_j\}, \quad \mathcal{C}_1 = \{50\}, \\ &\mathcal{C}_2 = \{10, 20, 30, 40, 95, 100\}, \quad \mathcal{C}_3 = \{80, 90\}, \\ &\mathcal{C}_4 = \mathbb{R}_{\text{finite}} \setminus (\mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3). \end{aligned} \tag{12}$$

This grouping cannot identify separate effects of buildings, individual vegetation classes, wetland or snow. Absolute elevation H supplies surface derivatives but is not itself a selected conditioner column. With the east/north DEM gradient $\gamma = (\partial_e H, \partial_n H)$, the geometric definitions are

$$\begin{aligned} \beta = \arctan\|\gamma\|, \quad (\sin\alpha, \cos\alpha) = -\gamma/\|\gamma\|, \\ (n_e, n_n, n_u) = (-\gamma_e, -\gamma_n, 1)/\sqrt{1 + \|\gamma\|^2}. \end{aligned} \tag{13}$$

The provider estimates gradients by central differences over DEM neighbours. Aspect α is clockwise from north and points downhill. Flat terrain has no geometrically defined aspect: the provider records its aspect vector as $(0, 0)$, not an arbitrary northward direction. Thus a flat surface's directional composition can be zero, without evidence of genuine perpendicular travel. Masks still follow source-validity rules.

The four selected compositions are exactly

$$\begin{aligned} c_{\text{road}} &= \ell_{\text{road}} u_{\text{road}}, & c_{\text{river}} &= \ell_{\text{river}} u_{\text{river}}, \\ c_{\text{cover}} &= \ell_{\text{road}} w, & c_{\text{surface}} &= (h^\top a,\ h^\top q), \end{aligned} \tag{14}$$

where $a = -(\sin\alpha, \cos\alpha)$ is uphill and $q = (-a_n, a_e)$ is the chosen contour direction. The surface composition uses a positive 90-degree rotation of uphill in east/north coordinates. There is one uphill direction and its opposite downhill direction, not two distinct uphill choices. A contour has two orientations: the chosen q fixes the sign convention and $-q$ would reverse $h^\top q$. Positive $h^\top a$ denotes uphill alignment; negative denotes downhill. The signed contour projection distinguishes the two orientations but does not give either an intrinsic physical preference. The surface composition therefore adds two directional projections, not all twelve products of a six-component surface vector with two-component history; it does not multiply the projections by slope magnitude. Road, river, cover and surface compositions add 2, 2, 4 and 2 columns, respectively. Together with 18 selected base-variant columns, the full model has 28 numeric columns.

A variant is usable only when all its required canonical inputs are finite and their source status is `valid`, with the additional domain/norm conditions above. Composition masks require their contributing inputs to be usable; the surface projection uses the aspect and history masks. Online canonical numeric inputs are first represented as float32, matching stored snapshot values, before applying these same transform kernels. For numeric features f_j and Boolean masks m_j, the conditioner receives

$$\phi = (\widetilde{f}_1, \ldots, \widetilde{f}_K, m_1, \ldots, m_K)^\top, \qquad \widetilde{f}_j = \begin{cases} f_j, & m_j = 1, \\ 0, & m_j = 0, \end{cases} \tag{15}$$

where $K = 28$ for all-terrain. Invalid values are retained through their masks rather than dropped or interpreted as measured zeros. All LOO/LIO configurations reuse these frozen transforms and change only their selected columns and independently fitted dynamics.

16 Method dynamics and registered alternatives

The method matrix is a separate model family, not the terrain regression with differently named columns. For mode $M_t = m$, its local dynamics are

$$\begin{aligned} dX_t &= b_m(X_t, Z_t, S_t)\,dt + L_m\,dW_t, \\ b_m &= \Theta_m^\top \psi(X_t, Z_t, S_t), \quad L_m L_m^\top = Q_m = \tau R_m. \end{aligned} \tag{16}$$

Here S_t is geometric solar elevation, ψ normally contains $(1, X_{t,e}, X_{t,n}, S_t)$, and Θ_m is the mode-specific ridge fit. R_m is fitted velocity-rate residual covariance and τ the observation/training interval. A frozen-coefficient Euler step has conditional law $\mathcal{N}(X_t + b_m\Delta, Q_m\Delta)$. With three mode probabilities π_m, an initially unobserved mode gives the first-step mixture $\sum_{m=1}^{3} \pi_m \mathcal{N}(X_t + b_m\Delta, Q_m\Delta)$. Segment-constant modes retain that draw; Heading/redraw-mixture and RDM-kernel routes redraw on the τ clock. Single-Gaussian and explicit-feature routes have one mode. The latter appends speed and radial east/north direction to ψ. Nonlinear solar conditions and sampled mode histories mean a multi-step forecast is not asserted to be one exact three-Gaussian density. All executed fits remain linear in their feature coefficients; there is no executed neural-network baseline. The model-mixture labels are distinguished from how their training partitions are actually formed below.

Which visible inputs actually affect the predictor? The ordinary HSM, Heading/redraw-mixture, single-Gaussian and RDM-kernel routes use $\psi = (1, X_e, X_n, S)$, not the supplied velocity or a signed history direction. In a base fit, $\pi_m = n_m / \sum_k n_k$, where n_m is the number of training transitions assigned to mode m. HSM stores the parameter blend $\pi_m = 0.35\pi_m^{\text{pretrain}} + 0.65\pi_m^{\text{adapt}}$. Prediction reads the saved probabilities without estimating a $\Pr(M_0 = m \mid \text{new visible prefix})$ posterior. Its coefficients and probabilities are not refitted from a final-evaluation prefix. Consequently, fixing numerical position, clock, condition field, saved parameters and random-stream identity makes these ordinary routes invariant to a change in initial velocity alone. Updating the velocity buffer does not establish that ordinary HSM uses a velocity feature.

The dependency chain is nevertheless not equivalent to discarding every visible position. The primary method coordinate frame is anchored at the first visible point; it determines X_0 and the geographic conversion used by the solar field. The two secondary modes instead use the forecast-origin frame, avoiding hidden-prefix information in point-only inputs. Changing a prefix can therefore change numerical position or the frame even when its final geographic point is fixed. Different origin identities also select different random streams. The secondary-mode comparisons do not isolate velocity sensitivity under the fixed-input experiment above. The explicit-feature route adds speed $\|v\|$ and position-radial direction $X/\max(\|X\|, 10^{-9})$, with zero direction at $X = 0$; reversing v alone preserves speed. Terrain models directly use velocity and history direction through b_0 and the retained features. Software-only reversed-prefix tests with a fixed frame/stream check this distinction over four short output times; they are not new empirical forecasts or an independent audit of the final matrix. Required configurations compare mode structure, explicit decomposition, observation intervals, fitting objectives, training or adaptation policies, and numerical inference. Solar conditioning is retained in the required HSM configuration; its alternative-condition arm is

exempt and cannot establish a solar effect. Method noise uses a velocity-rate residual covariance R with fixed increment covariance $Q = R\tau$ in the retained time coordinate, where τ is the registered observation/training interval. It is not rescaled by substituting the numerical integration step for τ. Method history also follows its registered interval. Repeated HSM anchors remain separate configuration bindings but are not counted as independent replications.

More specifically, for method history tick $t_j = j\tau$, the per-path velocity update is

$$v_{t_j}^{(p)} = \frac{U_{j\tau}^{(p)} - U_{(j-1)\tau}^{(p)}}{\tau}, \qquad U_t^{(p)} = \begin{cases} X_t^{(p)}, & \text{MC or CRN,} \\ \mu_t^{(p)}, & \text{conditional Gaussian-moment propagation.} \end{cases} \tag{17}$$

Here positions and conditional centres $\mu_t^{(p)}$ use the method's own coordinate frame; a centre is conditional on that sampled mode path, not the ensemble mean. The first update uses the origin as $U_0^{(p)}$; until then the origin-supplied velocity is held. This is an adjacent-tick secant, unlike the terrain buffer in Equation (8). The HSM cadence is 60 seconds; observation-interval variants use their own τ. The velocity channel supplies the explicit-feature route's speed feature; it is not an additional fitted velocity SDE or an extra column in the ordinary HSM drift.

The HSM method uses three segment-persistent modes, solar elevation, global pretraining followed by target adaptation, energy-score covariance calibration, split integration and conditional Gaussian-moment propagation. Its reference observation interval is 60 seconds. The five registered comparison families contain 21 candidate-to-HSM contrasts (Table 1); six HSM anchors and the identical 60-second replay make up the other seven required configurations. Different anchors are not additional independent controls.

Table 1: Registered primary method comparisons, without empirical verdicts. Each count denotes candidate configurations, not recording-hash blocks.

Family	Variants and matched HSM reference	Count
Model structure	Heading/redraw, one Gaussian, residual mixture and explicit features; HSM-01	4
Observation interval	30, 120, 300 and 600 seconds; HSM-01	4
Objective and score	QMLE, mixed, pure energy, two-dimensional MC and closed score routes; HSM-07	5
Training and adaptation	Scratch, Reptile, drift-only and two-step; HSM-11	4
Numerical propagation	Two Euler labels (HSM-18), and MC/CRN propagation (HSM-20)	4

Appendix 44 maps these scientific names to the complete configuration inventory. The `dt` labels denote observation/training and history cadence in seconds, not integration step or forecast duration. The solar-condition, animal-transfer and bridge alternatives are excluded from this study's effect claims; the separate ten-configuration terrain study is retained.

Mode fitting and features. Training modes are deterministic tertiles of segment displacement heading, with segment identity breaking ties; they are not inferred behavioural states from a likelihood-mixture EM fit. Separate ridge regressions use intercept, local position and solar elevation. The residual-mixture route instead partitions pooled residuals by the largest of east, north and minus their sum. The one-Gaussian route pools all transitions. The explicit-feature route is also single-mode and adds speed and the two components of radial direction; it is not a separately fitted radial/tangential dynamical decomposition. At prediction time, segment-persistent modes remain fixed, while switching routes redraw modes on the reference-interval clock, not at each numerical

substep. Solar elevation is recomputed from predicted geography and the retained clock using its model UTC interpretation and the NOAA fractional-year approximation[15]. Acquisition UTC is not independently certified (Section 25); this is geometric elevation, not a weather measurement or a refraction model.

Base regression and covariance estimation. For one prepared batch and component m, let D_m contain n_m feature rows and Y_m their two-dimensional displacement-rate targets. The base regression minimizes $\frac{1}{2}\|D_m\Theta - Y_m\|_F^2 + \frac{\rho}{2}\|\Theta\|_F^2$, with $\rho = 10^{-6}$. All coefficients, including the intercept, are penalized; the data term is not divided by n_m. With fitted residuals $e_{mi} = y_i - \widehat{\Theta}_m^\top \psi_i$ and their mean $\bar{e}_m$, the actual base parameters are

$$\widehat{\Theta}_m = (D_m^\top D_m + \rho I_d)^{-1} D_m^\top Y_m, \tag{18}$$

$$\widehat{R}_m = \frac{1}{n_m - 1}\sum_{i=1}^{n_m}(e_{mi} - \bar{e}_m)(e_{mi} - \bar{e}_m)^\top + \epsilon I_2, \qquad \epsilon = 10^{-8}. \tag{19}$$

The implementation solves the linear system rather than forming the inverse; d is the feature count. It symmetrizes the sample covariance before adding the jitter in $\mathrm{m}^2/\mathrm{s}^2$. Thus this method covariance is centred and uses $n_m - 1$, unlike the terrain rate-residual second moment; it is not a regression degrees-of-freedom correction or an unbiased SDE diffusion estimator. Base proportions are $\widehat{\pi}_m = n_m / \sum_k n_k$. Each component requires at least two transitions: otherwise fitting fails, without pooling, deleting or replacing that component. These are base fits, not the later adapted/calibrated parameters. Written as a mean-loss objective, the base penalty is ρ/n_m; the normalized Reptile inner objective uses ρ instead (Equation (22)). The same symbol therefore does not mean the same sample-size-scaled penalty in those two procedures.

What the residual-mixture label actually means. The RDM-kernel route first fits the same ridge model to all batch transitions, then computes $r_i = y_i - \widehat{\Theta}_{\text{pool}}^\top \psi_i$. Its component label is

$$g_i = 1 + \operatorname*{argmax}_{j \in \{0,1,2\}} (r_{i,e}, r_{i,n}, -r_{i,e} - r_{i,n})_j. \tag{20}$$

Ties select the first maximum in that displayed order; an exactly zero residual therefore belongs to component 1. Separate regressions and centred covariances are then fitted to each group using Equations (18)–(19). The pooled coefficients are only used to form labels, not retained as a common drift with added residual centres. Grouping occurs once per base-fit batch, independently for training and adaptation, without iterative likelihood maximization, responsibilities, or expectation–maximization. Here RDM denotes the resulting Gaussian-mixture transition route, not a standard fitted latent Gaussian-mixture likelihood. At prediction time it uses the saved blended probabilities and redraws on the τ clock; these are not state-dependent posterior responsibilities. Original group memberships and transition counts were not saved, so this algorithmic specification does not supply their empirical frequencies or matched behavioural regimes across batches.

Heading-rank labels. More precisely, sort the K segments in a particular batch by $\operatorname{atan2}(X_{\text{end,n}} - X_{\text{start,n}}, X_{\text{end,e}} - X_{\text{start,e}})$, then by segment identity. For zero-based rank r_s, the label is

$$m_s = \min\{2, \lfloor 3r_s/K \rfloor\}, \qquad r_s = 0, \ldots, K-1. \tag{21}$$

The implementation labels 0, 1 and 2 correspond to mixture component indices 1, 2 and 3 above; equivalently, $m = m_s + 1$. The sorting angle is in radians on the $[-\pi, \pi]$ branch, with the cut on the

westward axis; the implementation uses `math.atan2`, including its signed-zero convention. There is no circular unwrapping or zero-displacement heading filter. The label is an ordering partition, not an observed movement state. Every transition from a segment inherits this label. The retained role accounting of all sixteen fits gives the same 328 training, 76 adaptation and 81 validation segments. Table 2 reports the cardinalities implied by this rule for each complete role batch; no headings, residuals or fits are reconstructed. Segment lengths differ, so these cardinalities do not give transition counts n_m or probabilities π_m. They do not describe the residual-mixture partition. Ranking is repeated independently in each batch, including eligible Reptile region tasks: the table is not a per-region task census, and matching mode indices does not establish matching physical headings or behavioural states.

Table 2: Deterministic heading-rank segment counts for complete prepared role batches. These are not per-mode transition frequencies, final blended probabilities, RDM residual-group counts or Reptile region-task counts.

Prepared role	Segments	Mode 0	Mode 1	Mode 2
Training	328	110	109	109
Adaptation	76	26	25	25
Validation	81	27	27	27

Fitting and adaptation scope. Training observations are resampled at the registered interval; at most one incomplete terminal interval is removed, without changing scoring targets. Global and adaptation ridge fits are blended with adaptation weight 0.65. Drift-only blending retains the pretrained covariance at that stage, while both the `all` and `Two-step` labels use the same covariance blend. Subsequent validation calibration can still rescale that covariance. Thus two-step is a retained execution identity, not evidence of a distinct sequential optimizer. Scratch uses the adaptation fit alone. Reptile groups development segments by the adapter's recorded location label, requires at least two tasks with at least three segments each, and uses four outer epochs and five inner gradient steps before the same target-adaptation blend.

Tasks use exact source-location strings, without a country key or case normalization; these labels are not certified cities, regions or independent participants. Of 84 label groups, 30 meet the three-segment threshold: 258 segments and 20,029 transitions per complete pass through those tasks. The other 54 groups (70 segments, 5,980 transitions) remain in the global initialization but do not receive task visits. Appendix 49 gives the exact source-field rule, complete task list and saved exposure accounting.

Executed finite Reptile updates. The initial model is the global training ridge fit. Eligible label groups are visited in lexicographic order once per epoch, without random task sampling or shuffling. Heading tertiles are recomputed within each label-group task; mode indices are not matched behavioural labels across tasks. For task r and mode m, let D_{rm}, Y_{rm} and n_{rm} denote its feature rows, velocity-rate targets and transition count. With $\rho = 10^{-6}$, each of five inner steps is

$$\begin{aligned} L_{rm} &= \|D_{rm}\|_2^2/n_{rm} + \rho, \\ \Theta_m^{k+1} &= \Theta_m^k - \frac{0.5}{L_{rm}} \left\{ \frac{D_{rm}^\top (D_{rm}\Theta_m^k - Y_{rm})}{n_{rm}} + \rho\Theta_m^k \right\}. \end{aligned} \tag{22}$$

This is full-batch gradient descent on the normalized task ridge objective, not a closed-form re-solve or Adam. The initialization's base ridge uses an unnormalized squared loss, so equal ρ does not

imply equal effective penalties. A mode with fewer than two transitions skips its weight update and retains its prior covariance. Otherwise the adapted covariance is the centred post-update residual sample covariance with divisor $n_{rm} - 1$, plus $10^{-8}I$; mode probabilities are task transition proportions. For each task visit in epoch $e = 0, 1, 2, 3$, the outer update is

$$\xi \leftarrow \xi + \frac{0.35}{\sqrt{e+1}}(\xi_r^{\text{inner}} - \xi), \qquad \xi = (\{\Theta_m\}, \{R_m\}, \pi). \tag{23}$$

Probabilities are renormalized. Tasks have equal visit counts and a common outer step, not transition-count-weighted pooled outer updates. Covariance and probability interpolation is part of this finite implementation, not a claim that these are gradient parameters of the original Reptile objective. The target adaptation fit is independently solved and blended:

$$\Theta_m^{\text{final}} = 0.35\Theta_m^{\text{meta}} + 0.65\Theta_m^{\text{target}}. \tag{24}$$

The `all` route applies the same blend to R_m and π before validation covariance calibration. Thus initialization influence is not erased by fully re-solving the target problem from that initialization.

The original saved Reptile fit reports 30 eligible tasks and mean task-visit residual MSE 1.595 before and 1.260 after the inner updates (m^2/s^2). These are repeated training-task diagnostics, not held-out performance or 120 independent observations. Its final saved weights, covariances and probabilities each differ from ordinary HSM's. Intermediate pre-meta and pre-target parameter snapshots were not saved; we do not reconstruct them through new fitting. These facts establish the executed parameter flow, not a general meta-learning benefit. Saved-source correspondence is given with the task accounting in Appendix 49.

What the finite objective comparison actually changes. Let $\Theta_m^0, R_m^0, \pi_m^0$ denote the model after training and target adaptation but before validation calibration. All four routes here use 26,009 train, 5,762 adaptation and 6,149 validation transitions from 328, 76 and 81 windows, respectively, at $\tau = 60$ seconds. For mixed and pure-ES only, a second ridge fit on the validation transitions provides Θ_m^V, using that role's deterministic heading-tertile labels. Blending corresponding mode indices does not establish matched behavioural regimes across the two fits. Their candidate parameters are

$$\Theta_m(\eta) = (1-\eta)\Theta_m^0 + \eta\Theta_m^V, \qquad R_m(s) = s^2 R_m^0, \qquad \pi_m = \pi_m^0. \tag{25}$$

The scale grid is $\mathcal{S} = \{0.55, 0.75, 1, 1.3, 1.7\}$ and the drift grid is $\mathcal{A} = \{0, 0.1, 0.25, 0.5\}$. The validation fit's covariance and mode probabilities are not substituted into these candidates.

For validation rate target y_i, features ψ_i and its deterministic label ℓ_i, set $r_i(\eta) = y_i - \psi_i^\top \Theta_{\ell_i}(\eta)$. For three-component models, $\ell_i = m_{s(i)} + 1$ is the validation segment's heading-rank label; a single-component model always selects component 1. The residual-mixture route also uses these heading labels in this validation surrogate, not its residual-group labels g_i: the validation residual routine does not reassign RDM groups or infer mixture responsibilities. This retained label mismatch is an implementation boundary, not a matched-component mixture-likelihood calibration. It is disclosed here without refitting, relabelling or changing any saved prediction. Calibration pools covariance by the unweighted mean $\overline{R}^0 = M^{-1}\sum_m R_m^0$, not by π_m^0, and uses $S_s = s^2\overline{R}^0 + \epsilon I_2$, $\epsilon = 10^{-8}$. This jitter is additional to that already included in each fitted R_m^0. With independent $U, U' \sim \mathcal{N}(0, S_s)$, the two validation surrogates are

$$\begin{aligned} E(\eta, s) &= \frac{1}{n_V}\sum_i \left\{ \mathbb{E}\|U - r_i(\eta)\| - \tfrac{1}{2}\mathbb{E}\|U - U'\| \right\}, \\ N(\eta, s) &= \tfrac{1}{2}\left\{ 2\log(2\pi) + \log\det S_s + \frac{1}{n_V}\sum_i r_i(\eta)^\top S_s^{-1} r_i(\eta) \right\}. \end{aligned} \tag{26}$$

The code uses a pseudoinverse for the quadratic term; the jittered covariance is positive definite. E is evaluated approximately by order-10 tensor-product Gauss–Hermite quadrature. If g_k are the 100 two-dimensional nodes, w_k their normalized product weights and $C_s C_s^\top = S_s$, its terms use $\sum_k w_k \|\sqrt{2} C_s g_k - r_i\|$ and $\sum_k w_k \|2 C_s g_k\|$, respectively. The underlying population expression is the energy score (Equation (37)) of this Gaussian rate surrogate; finite quadrature is not a proof of exact propriety. It is not the multimodal predictive-position or joint-path energy score used for final evaluation. Likewise N is the negative log density of a pooled Gaussian in the fixed velocity-rate units, not a mixture log-likelihood or the continuous-time SDE likelihood.

Writing $E_0 = E(0,1)$, $N_0 = N(0,1)$, the actual selections are

$$\begin{aligned} \text{HSM:} &\ \min_{s \in \mathcal{S}} E(0,s), \\ \text{pure-ES:} &\ \min_{\eta \in \mathcal{A}, s \in \mathcal{S}} E(\eta,s), \\ \text{mixed:} &\ \min_{\eta \in \mathcal{A}, s \in \mathcal{S}} \left\{ \frac{E(\eta,s)}{2\max(E_0, 10^{-12})} + \frac{N(\eta,s)}{2\max(N_0, 10^{-12})} \right\}. \end{aligned} \tag{27}$$

QMLE leaves the adapted ridge parameters unchanged and records N_0; it does not run another likelihood optimization. Mixed uses the registered weight 0.5. The pure-ES configuration's `infinity` weight marker is ignored by its energy-only branch, not used in infinite arithmetic. Ties are broken by objective, then drift fraction, then scale; HSM breaks ties by objective and scale. These are finite development procedures, not unconstrained end-to-end score fitting.

Table 3: Original saved calibration choices and objective values. Before/after values use different objectives across rows and are not directly comparable. ES here has units m/s; mixed is normalized.

Route	Candidates	η	s	Saved objective, before → after
HSM	5	0	1	ES: 0.7317 → 0.7317
QMLE label	0	0	1	Gaussian NLL: 2.4534 → 2.4534
Mixed calibration	20	0.5	1	normalized: 1.0000 → 0.9885
pure-ES	20	0.5	1	ES: 0.7317 → 0.7214

The saved weights, covariances and mode probabilities are exactly equal between HSM and QMLE, and between mixed and pure-ES. Mixed/pure-ES differ from HSM in weights only among these parameter arrays. Both select the largest allowed validation blend, not evidence that 0.5 is globally optimal. Validation improvements are selected in-sample values, not unbiased external performance. The comparison changes both data use and candidate selection, so a final difference cannot identify a loss-function-only effect. The aggregate receipt is `paper/pirc17/estimator-calibration-description-v1.json`. It does not reconstruct the full candidate objective surface or verify saved forecast arrays.

Numerical and diagnostic identities. Gaussian-moment propagation samples mode histories and evaluates nonlinear conditions at conditional means; MC and CRN evaluate them on individual paths. Affine kernels are only locally exact with coefficients held fixed. The EM and Euler labels implement the same update. The closed/MC energy routes and MC/CRN pairing have separate estimator diagnostics; both sides of every predictive comparison still use the common ensemble score. Shared random streams alone do not prove variance reduction, and different step grids are not a coupled-Brownian refinement oracle. These scope limits remain explicit even if a forecast or mechanism check succeeds.

The actual local kernels and joint output paths. At a step beginning at t_n, write $b_m(x) = \mathsf{A}_m x + a_{n,p}$, where A_m is the transpose of the two position rows of Θ_m. The offset contains the intercept and all non-position features, evaluated at the step-start conditional centre for FP or noisy path position for MC/CRN. Solar elevation and any speed/radial terms are held fixed through both split half-steps, not recomputed between them. Let $Q_m = \tau R_m$, $H = \exp(\Delta \mathsf{A}_m/2)$ and $J_{1/2} = \int_0^{\Delta/2} \exp(u\mathsf{A}_m)\, du$. The implemented kernels are

$$\begin{aligned}(F, J, \Omega)_{\mathrm{EM}} &= (I + \Delta \mathsf{A}_m, \Delta I, \Delta Q_m),\\ (F, J, \Omega)_{\mathrm{split}} &= (H^2, (I+H)J_{1/2}, H(\Delta Q_m)H^\top).\end{aligned} \tag{28}$$

Split means exact half-drift, additive noise, exact half-drift. The affine reference instead uses

$$\begin{aligned}F &= \exp(\Delta \mathsf{A}_m), \qquad J = \int_0^{\Delta} \exp(u\mathsf{A}_m)\, du,\\ \Omega &= \int_0^{\Delta} \exp(u\mathsf{A}_m) Q_m \exp(u\mathsf{A}_m)^\top\, du.\end{aligned} \tag{29}$$

Augmented matrix exponentials compute J without assuming an invertible A_m; a block matrix exponential computes Ω. The eigenfactor $G = V \operatorname{diag}(\sqrt{\max(\nu, 0)})$ satisfies $GG^\top = \Omega$; only negative roundoff is removed, while materially indefinite or nonfinite results fail rather than trigger a fallback. For each particle, the continually updated path and FP moments obey

$$\begin{aligned}X_{n+1}^{(p)} &= F X_n^{(p)} + J a_{n,p} + G z_{n,p},\\ \mu_{n+1}^{(p)} &= F \mu_n^{(p)} + J a_{n,p},\\ C_{n+1}^{(p)} &= F C_n^{(p)} F^\top + \Omega, \qquad z_{n,p} \sim \mathcal{N}(0, I_2).\end{aligned} \tag{30}$$

Initially $X_0^{(p)} = \mu_0^{(p)} = X_0$ and $C_0^{(p)} = 0$. The moment equations are used only in FP. Conditional on sampled mode histories and initial velocity, its four saved positions are snapshots of the same noisy path, not independent draws from four Gaussian marginals. If $P_{s,t}$ is the chronological product of intervening F's, their cross-covariance is $C_s P_{s,t}^\top$. The eight-dimensional path score therefore uses the saved common particle index across times. This is an assumed-density approximation when features depend nonlinearly on position, not an exact global nonlinear transition law.

The event grid is the union of absolute numerical ticks (maximum step 5 seconds), history ticks, mode ticks and the actual four scoring times. At a boundary the order is: propagate with the old mode; update a due history secant; save a due output; then redraw a switching mode for the next step. No redraw is needed after the last output. HSM keeps its initial mode. Random streams separately index initial velocity, mode-tick uniforms and Gaussian increments; the latter use the hexadecimal representations of both interval endpoints. FP/CRN paired streams share origin, pair identity and seed, whereas ordinary MC uses its independent propagation/run identity. These are identical-grid couplings, not evidence of variance reduction or Brownian consistency after changing the grid. The source-bound description above identifies the frozen algorithms without producing another rollout.

Saved modes and noise. HSM stores probabilities 0.3350, 0.3302 and 0.3348 for mode indices 0–2. These are adapted model parameters, not new-prefix posterior weights or observed mode frequencies. Ordinary adaptation blends the separately normalized training and adaptation proportions with weights 0.35 and 0.65; Reptile also interpolates task-local proportions. Consequently,

multiplying the final probabilities by the training-exposure label does not recover per-role mode counts. Those counts are not retained in the saved artifacts. Table 44 reports all 16 fits without inventing them.

Let $\widetilde{R}_m$ be a mode's rate-residual covariance after training and adaptation, and s the selected validation multiplier. The saved covariance has already undergone the scaling

$$R_m = s^2 \widetilde{R}_m \quad (\mathrm{m}^2/\mathrm{s}^2), \qquad Q_m = \tau R_m \quad (\mathrm{m}^2/\mathrm{s}), \qquad \lambda_j(Q_m) = \tau \lambda_j(R_m). \tag{31}$$

The embedding does not apply s^2 again, and τ is not the numerical step h. For HSM, the three Q_m eigenvalue pairs are $(44.133, 53.105)$, $(44.163, 53.470)$ and $(35.195, 47.111)$ in m^2/s. The RDM multiplier $s = 1.7$ means a factor 2.89, already included in its saved R_m, not an extra simulation-time factor. Different τ fits change both the rate residual covariance and this embedding; their differences are not an integration-step effect.

The rate regression adds $10^{-8}I$ before validation scaling. This is distinct from the prediction binder's numerical PSD rule: reject a negative eigenvalue below $-128\epsilon_{\mathrm{mach}} \max(1, \|R_m\|_2)$ and replace only smaller-magnitude negative roundoff by zero, not by a positive floor. All 44 saved method-mode covariances are symmetric and strictly positive; none requires that negative-roundoff repair. This does not prove covariance calibration or stable long-horizon forecasts. A mixture's predictive covariance also contains between-mode mean dispersion and evolving dynamics; the tabulated spectra are not forecast-region sizes. The complete per-mode values and source bindings are in `paper/pirc17/method-noise-description-v1.json`.

17 Ten terrain configurations

Write B for the baseline that retains `history.direction`, and A for B plus road, river, WorldCover and surface groups. The ten configurations are B, A, four $A - g$ and four $B + g$. Removing a group removes all selected features and compositions whose ownership includes that group. A composition involving two terrain groups has coupled ownership; its effect cannot be uniquely assigned to one owner without an additional registered interaction experiment.

Table 4 connects factor names, symbols and expressions to the ownership and widths in Table 5; it is not an additional experiment.

Table 4: Considered factors and frozen expressions. Eighteen base numeric columns plus ten compositions give 28, each with a validity column. Velocity belongs to the separate baseline fit; elevation is a derivation source. Neither adds to these 28 columns. Group-level cover ablations do not identify separate subclass effects.

Factor	Symbol	Expression or directional meaning	Input role
Motion	$v_t, \|v_t\|$	Causal secant velocity; its norm is speed.	Baseline
History	h_t, η_t	Unit causal-history direction and validity; both east/north components retained.	2 numeric
Road distance	ℓ_{road}	$\log(1 + d_{\mathrm{road}}/(1\,\mathrm{m}))$; distance to nearest road point.	1 numeric
Road direction	u_{road}	Unit vector towards nearest road point; not a tangent.	2 numeric
River distance	ℓ_{river}	$\log(1 + d_{\mathrm{river}}/(1\,\mathrm{m}))$; distance to nearest river point.	1 numeric
River direction	u_{river}	Unit vector towards nearest river point; not flow direction.	2 numeric
Built cover	w_1	$\mathbf{1}\{k = 50\}$.	1 numeric
Vegetated cover	w_2	$\mathbf{1}\{k \in \mathcal{C}_2\}$; six codes grouped.	1 numeric
Water/wetland	w_3	$\mathbf{1}\{k \in \{80, 90\}\}$.	1 numeric

Table 4 (continued)

Factor	Symbol	Expression or directional meaning	Input role
Bare/snow/other	w_4	$\mathbf{1}\{k \in \mathcal{C}_4\}$ for valid finite codes.	1 numeric
Elevation	H	DEM value used to derive surface geometry; no direct selected column.	Derived only
Slope	β	$\arctan \|\gamma\|$, radians.	1 numeric
Downslope aspect	$\sin\alpha, \cos\alpha$	East/north downhill unit vector; flat convention is zero.	2 numeric
Surface normal	n_e, n_n, n_u	Upward three-dimensional unit normal, Equation (13).	3 numeric
Road/river interaction	$\ell_g u_g$	Log-distance times towards-feature direction, two columns per group.	2 + 2
Cover–road interaction	$\ell_{\text{road}} w$	Four products, jointly owned by road and cover.	4 numeric
Uphill alignment	$h^\top a$	Positive uphill, negative downhill; no slope-magnitude multiplier.	1 numeric
Contour alignment	$h^\top q$	Signed orientation for the fixed rotated contour vector.	1 numeric

Table 5 records the declared conditioner inputs. Every numeric column has a separate validity column, so the input width is $2K$. These widths exclude the ridge-solver intercept and the separate base-drift regressors; they are not parameter counts or evidence of a successful fit. All ten configurations retain history and have their own registered fit, rather than masking inputs of one fitted full model.

Table 5: Frozen terrain design. K is the number of numeric columns; the same number of validity columns is retained. The group column lists only terrain groups, with history present in every row. No predictive effect is implied by the declared width.

Configuration	Retained terrain groups	Numeric	Validity	Input
`base`	None	2	2	4
`all-terrain`	Road, river, WorldCover, surface	28	28	56
`loo-road`	River, WorldCover, surface	19	19	38
`loo-river`	Road, WorldCover, surface	23	23	46
`loo-worldcover`	Road, river, surface	20	20	40
`loo-surface`	Road, river, WorldCover	20	20	40
`lio-road`	Road	7	7	14
`lio-river`	River	7	7	14
`lio-worldcover`	WorldCover	6	6	12
`lio-surface`	Surface	10	10	20

The selected compositions are road log-distance by direction, river log-distance by direction, grouped WorldCover by road log-distance, and surface orientation by history direction. The WorldCover–road composition is removed in both `loo-road` and `loo-worldcover`, and is absent from both corresponding single-group additions. The surface–history composition remains in `lio-surface` because baseline history is retained. Thus LIO describes adding a group to the history-conditioned baseline, not a context-free group effect, and overlapping LOO deletions cannot be summed as independently attributable feature contributions.

The full-model leave-one-out comparison is the primary incremental question. The baseline leave-one-in comparison supplies supporting information. For a loss ℓ, define the positive-benefit

presentation

$$D_{\text{all}} = \ell(B) - \ell(A), \tag{32}$$

$$D_g = \ell(A - g) - \ell(A), \tag{33}$$

$$D_{\text{in,g}} = \ell(B) - \ell(B + g). \tag{34}$$

The machine inference ledger stores candidate-minus-control loss differences, so its negative differences indicate benefit. A positive-benefit plot must negate both the estimate and the interval, reversing the endpoints. This explicit convention prevents sign changes during paper preparation.

Table 6: Actual registered terrain roles. A is all-terrain and B is base; $\Delta = \ell(\text{candidate}) - \ell(\text{control})$. These roles, not the LOO name alone, determine the sign conversion.

Comparison	Candidate	Control	Positive factor benefit
Overall	A	B	$D_{\text{all}} = -\Delta$
LOO	A	$A - g$	$D_g = -\Delta$
LIO	$B + g$	B	$D_{\text{in,g}} = -\Delta$

For example, $\ell(A) = 100$ and $\ell(A - g) = 160$ give $\Delta = -60$, hence a positive factor benefit $D_g = 60$. The seed differences must use these same roles, and a machine interval $[l, u]$ becomes $[-u, -l]$. Choosing the deleted model as candidate would instead give $D_g = +\Delta$, but that is not this registry. Synthetic regression cases cover beneficial, harmful, practical-equivalent and boundary-crossing effects; they check direction and endpoint order, not empirical qualification. The public comparison tables and current forest plots retain the machine convention: their negative values favour the candidate. The positive-benefit D notation above is a separately labelled presentation, not permission to reinterpret an unchanged negative-axis plot. Descriptive configuration profiles do not replace complete paired comparisons when a family has failed or missing rows.

18 Experimental setup

19 Cohort and development population

The experiment uses a frozen trajectory cohort, feature specification and selected representation/capacity. Their implementation-record names are mapped in Appendix 44. Training and adaptation use only the registered development population. The task-qualified population contains 404 outer-training windows and 81 validation windows; the method assignment divides the outer-training population into 328 training and 76 adaptation windows. Terrain fitting uses one first-future transition per outer-training origin. Validation transitions supply diagnostics rather than final-evaluation fitting rows.

What a recording block represents. The release field named `independent_block_id` is a recording hash, not a participant identifier or geographic cluster. The builder computes SHA256 over the literal UTF-8 coordinate string, a NUL separator, the literal speed string and a second NUL separator. Identical strings under different file names receive the same block. Equivalent coordinates encoded differently, near-duplicate routes, repeat visits and recordings by the same person are not thereby identified. The retained harvest has no user identifier; an R1 segment is a derived cleaning unit, not a preserved original GPX track-segment boundary. We therefore call these *recording-hash blocks*, without claiming independent participants or unseen routes.

For block hash g, the release assignment is determined by

$$u_g = \mathrm{uint64}_{\mathrm{BE}}(\mathrm{SHA256}(\texttt{20260912:}\|g)[0:8])/2^{64}.$$

Training if $u_g < 0.70$, validation if $0.70 \le u_g < 0.85$, and final evaluation otherwise. All files sharing g inherit the same assignment. Exact file, recording, segment and window identities have zero cross-split intersections; the identity-only reconciliation is documented in Appendix audit record. It does not establish participant, approximate-route or physical-time independence.

Table 7: Released identities and sampled windows. Assigned files/blocks include records without an eligible midpoint window; counts are not participants.

Identity unit	Training	Validation	Final evaluation
Assigned recording files	5,373	1,120	1,125
Assigned recording-hash blocks	5,254	1,101	1,094
Released refined segments	52,176	10,236	12,849
Files with midpoint windows	5,370	1,118	1,125
Blocks with midpoint windows	5,251	1,099	1,094
Midpoint windows	49,845	9,831	12,370

Each refined segment contributes one midpoint window with disjoint history and target indexes. Training/adaptation roles are assigned by recording hash before task qualification. The selected 328/76/81 training/adaptation/validation windows occupy 259/62/62 recording-hash blocks. Exact index and assignment rules are in Appendix audit record.

Exact source-sequence check. Table 8 describes the complete source recordings serving the selected windows, not just their fitting transitions. Across the six pairs of roles (training–adaptation, training–validation, training–evaluation, adaptation–validation, adaptation–evaluation and validation–evaluation), both coordinate-only and coordinate-plus-clock comparisons found zero identical complete sequences among respectively 17,290, 17,556, 12,236, 4,290, 2,990 and 3,036 possible file pairs. The check hashes the row count and full ordered float64 longitude/latitude sequence; the second comparison appends the retained integer-nanosecond clock. Signed zero and NaN representations are normalized, with a separate missing-clock sentinel. No coordinate rounding, tolerance, sorting, reversal, translation, point deletion or resampling is used. All read coordinates are finite and no retained clocks are missing. Within-role matches already belong to the same original recording-hash block, so no new cross-block matches were found; we neither remove files nor change fitted weights or parameters. These checks do not detect approximate routes, partial overlaps or repeated visits, identify participants, or certify physical UTC or coordinate provenance. Thus zero cross-role matches do not establish independent people or routes. The saved aggregate is `source-sequence-description-v1.json`.

Previously used evaluation data. This evaluation cohort was used in earlier work; it is not a first-use, design-blind holdout. Current role separation does not erase that exposure. Historical cohort counts and the distinction between legacy directory splits and current assignments are in Appendix audit record.

Table 9 distinguishes the released development partitions from the much narrower task-qualified subset. The frozen development audit requires complete original windows, at least two history points, positive timestamp gaps no greater than 60 seconds, and at least 1800 seconds of follow-up. Joint validity additionally requires available surface, road, river, cover and historical-motion features

Table 8: Selected windows and their complete source recordings. Equal pairs (groups) count identical files within each role; coordinate-only and coordinate-plus-clock checks give the same counts. Whole-source rows include duplicate files and unused extents: they are not fitted transitions, unique observations or independent participants.

Role	Windows	Files	Source rows	Equal pairs (groups)
Training	328	266	1,407,171	9 (5)
Adaptation	76	65	388,784	4 (2)
Validation	81	66	360,616	4 (4)
Final evaluation	46	46	342,836	0 (0)

at every original point from the origin through the registered target end. This is a development-population filter, not a guarantee of validity at future predicted positions or a substitute for the separate final-input rules.

Table 9: Frozen development selection denominators: windows (recording-hash blocks). Rows are nested filters, not independent cohorts or forecast success counts.

Population stage	Outer training	Validation
Released development partitions	49,845 (5,251)	9,831 (1,099)
30-minute temporal support	509 (395)	100 (76)
Also joint feature validity	404 (321)	81 (62)

Final-evaluation selection. Table 10 describes the original saved dispositions of every released final-evaluation window; it is not the development funnel. Metadata completeness precedes temporal support, which requires a visible prefix, positive gaps no greater than 60 seconds, at least 1800 seconds of follow-up, and four distinct original score targets within tolerance. Joint validity then requires all five feature groups from origin through the original target end. This selection uses feature validity along the observed future route, retrospectively restricting the evaluation population; it is not a criterion available from the visible prefix alone at deployment. Future-route validity is used for admission, not passed to the predictor. This report groups already recorded reasons; it does not rerun raw qualification or alter any eligibility decision.

Table 10: Original final-evaluation funnel. Windows and recording-hash blocks are different denominators; nested rows do not count repeated forecasts.

Stage	Windows	Recording-hash blocks
Released final evaluation	12,370	1,094
Temporal support	127	89
Also joint feature validity	106	73
Primary selected origins	46	46

All 12,370 dispositions have complete identity/index metadata. Temporal conditions exclude 12,243 windows; feature validity excludes another 21 among the 127 remaining windows. Their mutually exclusive *first recorded* reasons are nonpositive gaps (106), insufficient follow-up (12,137), and road validity (21). First reason is a counting convention, not a unique physical failure attribution. Multiple flags overlap: for example, 1,497 windows have a road-validity flag overall, but only 21 are

additional exclusions after temporal checks. These counts must not be added. The 73 eligible blocks contain 53 with one qualified window, 15 with two, four with three, and one with eleven. Selection retains one origin per chosen block: 27 eligible blocks and 60 eligible windows are not used as primary origins. This is the frozen design, not 60 failed predictions. Saved dispositions alone contain no geographic, speed or duration distributions. The source joins below supply recorded duration, saved interval-speed and geographic descriptions for included and excluded windows. Aggregate counts and original source bindings are in `paper/pirc17/final-cohort-description-v1.json`.

Duration and map coverage enter eligibility before performance is computed. Eligible recording-hash blocks are ordered by the hash of the fixed domain `pirc17-final-block-order-v1` and their block identity (identity breaks hash ties); the lexicographically smallest eligible sample supplies each block's origin, without looking at forecast error. The frozen final selection records 73 eligible recording-hash blocks, of which the primary design selects 46. Both secondary modes reuse the same first six blocks of that ordered primary selection. Thus 58 origin–mode cases do not constitute 58 distinct recording-hash blocks. These block counts are identity-grouping units, not verified participant or independent-route counts. Coverage-selected windows define a task-specific population. The results therefore cannot be read as estimates for all raw trajectories or all geographic regions.

Population support and selection limits. The original selection admits a restricted long-track, map-supported task, not a representative deployment population. Median retained-clock follow-up is 69.8 s in the 12,370 released windows and 2,323 s in the 46 selected origins; only 130 released windows have at least 1,800 s of follow-up. The frozen temporal and map-validity rules, rather than duration alone, determine eligibility (Table 10). The released and selected window-median saved interval-speed summaries are 1.621 and 1.486 m/s under the retained construction script. These post-hoc descriptions include complete windows, including future observations, but do not become predictor inputs or revised eligibility. They are not distributions of online secant or simulated velocities; neither retained-clock differences nor saved scalars certify physical acquisition time or speed.

Source-country label coverage decreases from 63 released countries to 12 selected countries; GB, BE and LU supply 33/46 selected origins (71.7%). Labels are not independently verified point locations or participant identities, and these counts do not establish participant, near-route or physical-time independence. All original stage denominators are retained; no window is removed or replaced by a successful-subset estimate. HSM stage distributions and exclusions remain in Tables 35, 36 and 37 in Appendix 43. Source-scalar and clock correspondence limits remain in Appendix audit record.

The planner chose 46 primary blocks from development variance estimates for the registered terrain contrasts, with an assumed improvement of twice the practical margin. Those estimates used only three development blocks and remain a planning assumption. A sensitivity analysis with doubled standard deviation required substantially more blocks. The selected sample size is not a guarantee of adequate power for every terrain or method contrast.

20 Development data and fitting checks

All method intervals use the same 328 training, 76 adaptation and 81 validation windows. Sixteen saved parameter fits supply the 28 prediction configurations; changing the observation interval changes transition counts, not the number of windows. The ten terrain models each use 404 first-future training transitions and 81 validation transitions. Their complete counts, fitting errors and numerical checks are in Appendix 45.

Two findings matter for interpretation. All selected terrain inputs are valid at the development origins, leaving constant-one masks alongside the penalized intercept. The full raw design has width 57 but training/validation ranks 27/25. Adding features therefore changes the regularized representation as well as the available information; Equation (47) shows the constant-subspace penalty. Development validity does not guarantee support at predicted positions. Also, full-terrain training rate MSE is 0.136666 versus the baseline's 0.145796, whereas validation MSE is 0.079117 versus 0.077077 (m^2/s^2). Better training fit is therefore not itself evidence of better forecasts. The position-distribution results are evaluated separately in Sections 34 and audit record.

21 Observed history support and model-clock mismatch

The mathematical buffer and local noise model are defined in Section 14. Here we report the observed clock support of the original development and primary evaluation inputs.

The 46 saved primary causal-prefix records each retain three observed points. Their initial secant spans are 2–61 seconds (median 7); none is 10 seconds. If the last observed gap is d, the first 5-second update keeps the last two observed points and the new simulated position, giving span $d + 5$. The second update keeps the origin and the 5- and 10-second positions; subsequent spans are 10 seconds. Table 11 summarizes these saved final-input timings, *not* a distribution of training prefixes or a sensitivity experiment.

Table 11: Secant spans from all 46 saved primary prefixes, in seconds. The first-tick spans follow the unchanged buffer rule; the final row is the model-clock schedule, not measured trajectory duration.

Terrain history stage	Minimum	Median	Maximum
At the observed origin	2	7	61
After the first 5-second tick	6	8	36
At the second tick and thereafter	10	10	10

For the original terrain-development population, a separate clock-only description of the same 397 selected feature files recovers all 404 training and 81 validation prefixes. Every retained prefix contains three observed points, selected as the last three visible points within its registered window. Table 12 reports their actual secant spans, not the first-future fitting interval, an ordinary-method resampling interval τ, or an integration step. Training spans have median 6 seconds (interquartile range 2–12) and validation spans median 4 seconds (2–11). Of 404 training spans, 254 are shorter than 10 seconds, 13 equal 10 seconds and 137 longer; validation counts are 54, 3 and 24. These are window-origin counts, not counts of independent participants.

Table 12: Original terrain-development secant spans in seconds, using only saved clock metadata. First-tick rows add 5 seconds to the last observed gap according to the unchanged buffer rule; they are not simulated outputs or a sensitivity experiment. Quartiles use linear interpolation.

Role and history stage	n	Min	Q25	Median	Q75	Max
Train: observed origin	404	2	2	6	12	62
Validation: observed origin	81	2	2	4	11	61
Train: first tick (derived)	404	5.992	6	8	11	36
Validation: first tick (derived)	81	6	6	7	11	35

The observed-training, observed-final and stable simulated histories therefore have different time support: keeping three points does not keep the secant duration fixed. The clock-only description

neither reconstructs coordinate measurement errors nor measures the velocity, direction or drift distribution after feedback. It does not establish training–rollout distribution equivalence or identify an effect of the history schedule.

22 Missing-input extrapolation and retained query counts

A validity column records an input condition; it does not by itself train a response to that condition. Every selected mask is one in the 404 training and 81 validation origin rows described above. If the correction is written $c_\theta = a_0 + \sum_j (a_j \widetilde{f}_j + \gamma_j m_j)$, with all coefficients two-dimensional, those rows contain $a_0 + \sum_j \gamma_j$ only as a combined constant, not separately estimated missing-input effects. Regularization selects a coefficient decomposition without supplying missing-input observations.

Holding the history, other feature pairs and coefficients fixed, changing a set J of feature pairs from $(f_j, 1)$ to the implemented zero-imputed $(0, 0)$ changes drift by

$$b_{\text{invalid}} - b_{\text{valid}} = -\sum_{j \in J} (a_j f_j + \gamma_j). \tag{35}$$

Here the valid counterpart is hypothetical when the source is unavailable; its f_j is not known from an invalid query. The equation explains the operation, not an observed counterfactual or a measured failure effect. All affected composition masks belong in J. If history changes too, the baseline and other history features must also change; this fixed-history formula no longer describes the whole update. Invalid input does not simply switch the predictor to a freshly fitted baseline or change its constant diffusion Q.

For the unchanged three illustrated cases, original receipts record zero invalid encoded rows for both base and all-terrain, out of 185856, 186368 and 186368 particle-step encoding rows respectively. These are repeated evaluations, not independent points or unique map cells. The base still encodes history but makes zero raw map queries; all-terrain's raw map query-row count equals its encoding-row count. The saved invalid counter means at least one selected feature is unusable in that row, not a per-column map-nodata or provider-cause count. These six successful examples therefore contain no observed missing-input events and cannot certify robustness to missing features elsewhere. Complete forecast per-column rates and missing-event drift traces are not retained in this diagnostic. The descriptive projection `paper/pirc17/history-missingness-description-v1.json` binds the original metadata and diffusion arithmetic; it neither repairs failed families nor replaces the independent saved-output audit. Missing-input behaviour remains unsupported extrapolation, with no post-outcome encoding or penalty changes.

23 Geographic reporting and real case-study availability

The source recording-to-cluster mapping has now been joined to all 46 sealed primary blocks, without selecting on prediction outcome. The original raw harvest file matches the release's recorded SHA-256. The cluster metadata supplies country, region and harvest-area labels; it is separately hashed, not a geographical annotation already certified by the original release. The resulting route-free projection is saved as `paper/pirc17/preliminary-geography-v1.json`. Table 13 reports recording-hash blocks, not participants, seeds or forecast jobs.

For example, the source labels include Bayswater (England, seven blocks), Boechout (Flanders, five), Wilhelmshaven (Lower Saxony, three), and Freemansburg (Pennsylvania, one). The full 27 harvest-area labels and counts are in the projection. These are archive cluster labels, not independently verified point-level city boundaries, neighbourhoods or street addresses. The source

Table 13: Source-country labels of the 46 primary recording-hash blocks.

Country	Blocks	Country	Blocks	Country	Blocks
United Kingdom	17	Belgium	10	Luxembourg	6
Germany	3	Spain	2	Poland	2
Austria	1	France	1	Ireland	1
Netherlands	1	Russia	1	United States	1

road assets used here do not provide a verified street-name assignment for each trajectory. The broader archive or map inventory is not the selected evaluation population: twelve source countries, dominated by three countries, do not establish global applicability or country-specific predictive performance. No new environment-stratified hypothesis tests are introduced from these small, outcome-inspected groups.

The three stage figures show aggregate measurements, not case maps. Three additional local-review cases have been rendered from saved HSM forecasts and source observations: the first block from each of the first three encountered source countries in sealed order (ranks 0, 3 and 4), at the first registered RNG seed. Selection does not inspect completion status or error; unavailable or failed selected cases are not replaced. Each case aligns ESA WorldCover, SRTM elevation, Overture roads and HydroRIVERS in the same origin-anchored frame; masks and missing layers remain explicit. The visible three-point prefix is separate from the retained future observation route. Only four forecast times were saved: connecting their means is a visual guide, not a continuous saved predictive trajectory. Illustrations generate no new forecasts, fits or primary inferential results and do not substitute for the now-completed independent saved-output audit or establish general forecast reliability. At the owner's explicit request, selected rendered case panels are now included in this local working manuscript (Section 42); raw GPX and coordinate rows remain excluded. Public redistribution, privacy and map-licence review are still pending. A source area label or a geographic frame is not a licence to publish a private route.

24 Environment coverage at the forecast origin

All 46 admitted origins have been joined to their exact frozen terrain-feature rows using the release recording, segment, recording-hash block and origin clock. The selected feature-file hashes and condition-source hashes match the sealed manifest and prefix. This bounded description reads 46 origin rows, not all snapshot rows or future-route environments. It does not independently establish equality with online queries during prediction. The route-free projection is `paper/pirc17/preliminary-origin-environment-v1.json`.

At the origin, WorldCover labels are trees (27 blocks), grass (9), cropland (3) and built-up (7). The other seven registered classes—shrub, bare, snow/ice, water, wetland, mangrove and moss/lichen—have zero origin blocks, not zero predictive error. Thus a large map-layer inventory does not imply balanced experimental coverage. An origin pixel labelled trees is not a classification of the whole route as forest; zero water pixels at origins does not mean that routes never approach rivers or water. All 46 origin rows have valid cover, surface, road and river descriptors. Table 14 gives their actual continuous ranges, without introducing outcome-selected distance or slope thresholds.

Table 14: Frozen offline descriptors at 46 forecast origins; not full-route or online predicted-position summaries.

Descriptor	Minimum	Median	Maximum
Slope angle (degrees)	0.19	3.01	35.28
Road distance (m)	0.04	113.32	1273.48
River distance (m)	9.45	1229.95	3936.09

25 Clock provenance and case-map sources

Two data clocks must be distinguished. The upstream R1 pipeline reconstructs relative intervals from inter-point distances and stored speeds. The cohort builder, however, reads absolute nanosecond timestamps from the retained condition column `t`, aligns recording and source-point indexes, and refines segments on that condition clock. R1's reconstructed intervals are not substituted for it. The forecast origin, target matching, secant velocities and nominal 30-minute horizon use the retained condition clock. Source alignment proves which stored points and times are used, not that those times have been independently verified against sensor logs or original GPX timestamps. Consequently, real-time horizon and solar-angle interpretations retain this provenance limitation. The 5-second predicted-history cadence and maximum integration step are model settings, not claims about sensor sampling or clock accuracy. No clock, target or prediction is changed by this reporting correction.

Table 15 distinguishes saved and algorithmic clocks. Exact tick matching or timestamp format does not certify UTC provenance. The limited three-file footer check is documented in Appendix audit record.

Table 15: Clock quantities and their executed uses; units refer to the registered clock, whose physical provenance remains uncertified.

Quantity	Construction	Use
R1 relative time	Inter-point distance/stored speed, clipped intervals	Cleaning and source-segment reproduction; not forecast time
Condition nanosecond ticks	Saved `t`, matched by file/point index	Origin, target matching, observed gaps and secants
Origin-relative seconds	Integer tick difference divided by 10^9	Fitting and prediction time coordinate
Solar calendar input	Origin ticks plus rounded elapsed nanoseconds, interpreted as UTC	Calendar phase at predicted positions
τ, history cadence, h	Registered preprocessing/simulation settings	Different model/numerical controls, not sensor frequency

The formal visible-prefix loader requires strictly increasing retained times; it does not silently deduplicate new inputs. The earlier release anomaly counts and legacy handling, with their different denominators, are collected in Appendix audit record.

Executed solar condition. For an integer origin tick a_0 and elapsed seconds s, the provider uses $a(s) = a_0 + \mathrm{round}(10^9 s)$. Calendar decomposition treats this as UTC, preserving subsecond remainders. All sixteen saved method fits record the same solar policy and `solar_elev` condition. During fitting, solar is recomputed at the resampled development positions/ticks, not interpolated from stored historical solar. During rollout it is recomputed at the step-start FP centre or sampled

path position; the next step does not receive a future observed position. The NOAA fractional-year approximation is implemented as follows[15]. Let u be fractional UTC hours, D day of year, $Y \in \{365, 366\}$ year length, east-positive longitude λ in degrees and latitude ϕ in radians:

$$\begin{aligned}
\gamma &= \frac{2\pi}{Y}\left(D - 1 + \frac{u - 12}{24}\right), \\
E &= 229.18\big(0.000075 + 0.001868\cos\gamma - 0.032077\sin\gamma \\
&\qquad - 0.014615\cos(2\gamma) - 0.040849\sin(2\gamma)\big), \\
\delta &= 0.006918 - 0.399912\cos\gamma + 0.070257\sin\gamma \\
&\quad - 0.006758\cos(2\gamma) + 0.000907\sin(2\gamma) \\
&\quad - 0.002697\cos(3\gamma) + 0.001480\sin(3\gamma), \\
H_\odot &= \frac{\pi}{180}\left(\frac{60u + E + 4\lambda}{4} - 180\right), \\
S &= \frac{180}{\pi}\arcsin\big[\sin\phi\sin\delta + \cos\phi\cos\delta\cos H_\odot\big].
\end{aligned} \tag{36}$$

Here E is the equation-of-time correction in minutes, δ solar declination in radians, $H_\odot$ the solar hour angle in radians and S geometric elevation in degrees (the code clips the sine argument to $[-1, 1]$). This is not atmospheric refraction, measured illumination, terrain shadow or a fitted weather model. The formula's reproducibility does not verify source UTC, historical-generator parity or ephemeris accuracy. If source ticks are reconstructed or time-shifted, the model uses their induced calendar phase; these results do not establish a physically validated daylight or time-of-day effect. Time-based velocity and nominal horizon interpretations likewise remain conditional on the retained clock.

Case backgrounds use fixed map vintages, not verified contemporaneous maps for each trajectory. Layer-specific acknowledgements are collected in Appendix audit record. Acquisition, privacy and case publication permissions remain unverified; map access is not permission to redistribute private routes.

26 Model map support and query rules

Table 16 lists model assets, distinct from the three case backgrounds. The catalog contains 1405 admitted files: 1354 receipt assets plus 51 additional parents (1283 snapshot parents, 1232 already in the receipts). These counts do not measure visited assets, successful queries or support at predicted positions; full reconciliation is in Appendix audit record.

Raster selection is by containing extent, with SRTM before Copernicus, priority-folder assets before other assets, then lexical path order. Once a containing tile is selected, an invalid sample does not trigger another-product fallback. DEM heights use bilinear sampling, whereas WorldCover uses the containing pixel and rejects masks or class zero. Road and river queries use a separate cell-centred metric frame to select the nearest loaded line, then return spherical distance and a projected towards-line unit vector; they are not globally nearest across absent cells. Appendix 51 specifies the frames, raster stencils, road-class filter and missingness gates.

Receipt declarations give SRTM's 2000 acquisition interval, Copernicus's 2011–2015 interval and 2021 release, WorldCover's 2021 class vintage, and Overture's 2026 release. These are not verified per-route temporal matches. The saved description binds the catalog and ten source files without reading raster pixels or rerunning queries. It does not recover all-forecast per-factor invalid-query rates or certify each visited raster's geometry, parent-only product version, acquisition accuracy or

Table 16: Retained model map catalog, not case backgrounds or measured use. Files are receipt entries plus additional snapshot parents. Versions and nominal resolutions describe receipt metadata; parent-only files do not gain missing product metadata by inclusion.

Layer	Files	Receipt specification	Query operation
SRTM / Skadi	833+41	GL1 snapshot; 1 arcsecond	Bilinear height; metric finite differences
Copernicus DEM	44+0	GLO-30, 2021 release; 30 m nominal	GeoTIFF CRS and affine; bilinear height
ESA WorldCover	396+10	2021 v200; 10 m nominal	Containing stored pixel; no class interpolation
Overture roads	76+0	2026-08-19.0; vector	Nearest admitted road geometry
HydroRIVERS	5+0	v1.0; vector	Nearest admitted river geometry

publication licence. Thus terrain effects remain conditional on this restricted spatial and temporal support, rather than evidence of map-independent performance.

Data references and their roles. Model inputs use Skadi elevation[20], Copernicus GLO-30[22], WorldCover v200[16], Overture roads[23] and HydroRIVERS[18]. Skadi is not certified pure NASA SRTM for every tile. Product/version, documentation and attribution distinctions, and missing upstream identities, are detailed in Appendix audit record; neither citations nor access dates certify route-matched acquisition or publication permission. Attribution documents[21, 17, 19] describe separate obligations, not publication clearance; documentation access dates are not acquisition dates or immutable data versions.

27 Three origin modes

The primary `causal_prefix` mode estimates initial velocity from the last three visible observations using their retained condition timestamps. Available history stops at the origin; no observed future position updates the predictor. Initial velocity measurement error is unknown and is not set to zero by interpretation.

The descriptive `known_velocity` mode supplies position and that same causal secant velocity to the predictor. Its name denotes the supplied input, not an independently measured exact sensor velocity. It adds no future information. The `point_only` mode supplies the initial position and draws particle velocities from an empirical prior fitted only on the 404 outer-training prefixes. Its direction channel is prior-initialized and then predicted, rather than an unobserved prefix silently recovered from the data. The two secondary modes do not support population factor verdicts or hypothesis tests with only six blocks.

28 Horizon, observations and experiment size

The nominal horizon is 1800 seconds, with scoring slots at 60, 300, 900 and 1800 seconds. Each target is a distinct original observation nearest the slot, within 30 seconds, with the earlier observation breaking a tie. Targets are not interpolated. Adjacent observation gaps must satisfy the registered 60-second limit, and forecasts are evaluated at the actual selected elapsed times; the final target can be between 1770 and 1830 seconds.

Actual target clocks. Table 17 summarizes all 46 primary recording blocks. The 690 already-bound HSM-01/RDM/Obs 300 s score records agree exactly on target identities and four elapsed times across configurations and five seeds. Repeated forecasts therefore count each recording block once: 184 target slots, not 690 independent targets. Thirteen last targets exceed 1800 seconds, and the maximum is 1814 seconds. The formal method and terrain calls receive the actual target-time vector; the method event grid stops at its last requested target, rather than clipping at the nominal horizon. This saved-score description does not independently audit prediction arrays or certify the source sensor clock (Section 25).

Table 17: Actual saved target times on the original primary 46-block cohort, in seconds. Each row counts 46 recording blocks, not repeated seeds. Nominal times label the task; no target interpolation is performed.

Nominal	Minimum	Median	Maximum	Max. absolute offset
60	49	60	67	11
300	294	300	305	6
900	889	900	910	11
1800	1785	1800	1814	15

The allowed 30-second tolerance is 50% of a nominal one-minute horizon. The observed maximum first-slot offset is 11 seconds, or 18.3%; it is not a negligible timing difference. Complete arithmetic and source bindings are in `paper/pirc17/target-time-description-v1.json`.

Training resampling is a different operation. For interval dt, including 30 seconds, preprocessing creates a grid from the segment's first time in dt increments and appends the original endpoint when necessary. Development positions are linearly interpolated; solar values are recomputed at those positions and calendar times. Before fitting, at most one short terminal interval is removed. All other intervals must equal dt within 10^{-9} seconds, and at least four points must remain; violations fail rather than silently exclude a window. Neither this training grid nor its interpolated positions supplies evaluation truth.

Registered forecast workload. Every required configuration uses five forecast seeds, 20260814 through 20260818. The 58 origin–mode cases reuse 46 recording blocks. Table 18 maps the 11020 work items to data units. A work item is one configuration–origin-mode–seed forecast, containing 512 simulated paths at four scoring times. It is not 512 observed trajectories. The two six-block modes reuse primary blocks; their 12 cases do not add 12 recording-hash blocks.

Table 18: Planned scientific forecasts, not independent sample counts.

Matrix / origin scope	Configurations	Cases × seeds	Forecasts
Methods / primary	28	46×5	6440
Methods / two secondary modes	28	12×5	1680
Terrain / primary	10	46×5	2300
Terrain / two secondary modes	10	12×5	600
Total			11020

This yields 8120 method and 2900 terrain forecasts. The score inventory also contains 290 registered same-grid references and 58 deterministic CV paths, totalling 11368 rows. Runtime trials and forecast replays are diagnostic work, not additional real observations.

29 Distributional quality and diagnostics

Energy score is an evaluation rule for a predictive distribution, not a model-specific Gaussian loss or an SDE drift score. It applies to every executed single-mode, mixture and terrain ensemble using the same definition. For a distribution P_t on position with a finite first moment, independent $U, V \sim P_t$, and observed target y_t, define the lower-is-better convention

$$\mathrm{ES}(P_t, y_t) = \mathbb{E}\|U - y_t\|_2 - \tfrac{1}{2}\mathbb{E}\|U - V\|_2. \tag{37}$$

This is the energy score of Gneiting and Raftery (their 2007 Equation 22 uses the opposite, reward orientation) and the multivariate forecast formulation of Gneiting et al.[9, 10]. For the true outcome law G, its expected excess loss is

$$\mathbb{E}_{Y\sim G}\{\mathrm{ES}(P, Y) - \mathrm{ES}(G, Y)\} = \tfrac{1}{2}D_E(P, G) \geq 0, \tag{38}$$

where $D_E = 2\mathbb{E}\|U - Y\| - \mathbb{E}\|U - V\| - \mathbb{E}\|Y - Y'\|$, with independent draws from their respective laws. The energy-distance property gives equality only for identical distributions under the stated moment conditions. It explains why accuracy and calibrated spread are jointly assessed, not why any particular fitted model is correct.

For $N \geq 2$ Monte Carlo paths X_{it}, the registered estimator is

$$\widehat{\mathrm{ES}}_t = \frac{1}{N}\sum_{i=1}^{N}\|X_{it} - y_t\|_2 - \frac{1}{2N(N-1)}\sum_{i\neq j}\|X_{it} - X_{jt}\|_2. \tag{39}$$

There are $N(N-1)$ ordered distinct pairs, each estimating the independent pair expectation; self-pairs are omitted. Thus this U-statistic is unbiased for the model-law score under independent identically distributed sampling. In contrast, evaluating the finite empirical distribution itself uses N^2 in the denominator and is a different, finite-ensemble convention. The forecast-generating law, its numerical approximation and finite Monte Carlo noise are not identified by the score formula alone. A deterministic point forecast reduces to its Euclidean error. The same registered operator, not a slot's closed-score training/calibration formula, scores every arm. The primary score is $\sum_t w_t\widehat{\mathrm{ES}}_t$, with $w_t = 1/4$ for the four registered targets. Its unit is metres and smaller values are better. Equal weights do not turn the score into a continuous-time integral.

Secondary diagnostics include displacement error of the ensemble mean, terminal error, coordinatewise CRPS, particle endpoint error quantiles and path energy score. Coverage and area use ensemble-mean-centred disks whose radii are the registered empirical order statistics at levels 50, 80, 90 and 95 percent. These disks are not highest-density regions or joint path credible regions. Particle endpoint quantiles are within-ensemble quantities; they differ from population quantiles of per-origin forecast error.

Position entropy is a plug-in categorical entropy in nats on a fixed local grid spanning $[-10000, 10000]$ metres in both coordinates, with 250-metre cells and one overflow bin. It measures spread on that grid. Lower entropy alone is not evidence of higher accuracy. Negative log likelihood remains unavailable when no predictive density estimator has been qualified.

For reproducible calculation, let $T = 4$, $\bar{X}_t = N^{-1}\sum_i X_{it}$, and $e_t = \|\bar{X}_t - y_t\|_2$. Grid ADE is $T^{-1}\sum_t e_t$, FDE is e_T, and coordinate k's CRPS replaces each Euclidean norm in Equation (39) by the absolute difference in coordinate k, reported separately for east/north. Path ES applies that same operator to the eight-dimensional vectors $(\sqrt{w_t}X_{it})_{t=1}^{T}$ and $(\sqrt{w_t}y_t)_{t=1}^{T}$; it is not the sum of marginal ES values. At level p, sort $r_i = \|X_{it} - \bar{X}_t\|_2$ and take $r_{p,t} = r_{(\lceil pN\rceil)}$. Disk area is $\pi r_{p,t}^2$, and coverage is $\mathbf{1}\{e_t \leq r_{p,t}\}$, subsequently averaged over registered seeds and equally weighted blocks.

If n_j particles occupy grid bin j, including overflow, then $H_t = -\sum_{j:n_j>0}(n_j/N)\log(n_j/N)$. These separate calculations prevent endpoint error, coverage, spread and joint-path quality from being treated as interchangeable metrics.

30 Block aggregation and multiplicity

The statistical unit is the registered recording-hash block. Scores are averaged over the five fixed seeds within each block, then equally across blocks. Neither particle count nor seed count increases the number of independent participants or recording-hash blocks. The unchanged registry calls this count `independent_blocks`; that field name is not proof of stochastic independence. The block bootstrap assumes independent, exchangeable recording blocks for its population interpretation. Zero identity intersections do not establish this assumption: shared people, near routes or shared context can remain dependent. Intervals below are conditional on that assumption, not participant-level or unseen-route generalization guarantees. The original at-least-30-block count gate is retained but does not certify 30 independent people. Required paired rows form the full configuration–origin–seed grid; failed forecasts are not removed by intersecting successful subsets.

For candidate c, control a, block b, seed s, let S_{bsc} be its four-time ES (averaging matched origins within a block, if present). The effect actually estimated is

$$\Delta_b = \tfrac{1}{5}\sum_{s=1}^{5}(S_{bsc} - S_{bsa}), \qquad \widehat{\Delta} = \tfrac{1}{B}\sum_{b=1}^{B}\Delta_b, \qquad \widehat{\mathrm{se}} = \frac{\mathrm{sd}(\Delta_1,\ldots,\Delta_B)}{\sqrt{B}}. \tag{40}$$

Negative $\widehat{\Delta}$ favours the candidate. Resample the B block indices with replacement, sharing each draw across all comparisons in its family; do not resample particles as independent people. For draw r, compute $T_r^* = (\widehat{\Delta}_r^* - \widehat{\Delta})/\widehat{\mathrm{se}}_r^*$. If its lower/upper tail quantiles are q_l, q_u, the interval is $[\widehat{\Delta} - q_u\widehat{\mathrm{se}}, \widehat{\Delta} - q_l\widehat{\mathrm{se}}]$. The two-sided zero-null probability is estimated as $(1+\#\{|T_r^*| \geq |\widehat{\Delta}/\widehat{\mathrm{se}}|\})/(2000+1)$ before Holm adjustment. Zero bootstrap variance is handled explicitly, including unbounded pivots, rather than silently made precise.

Inference uses 2000 paired block bootstrap draws, significance level 0.05, and a practical energy-score margin $\delta \approx 41.10$ metres. The margin was defined as five percent of an existing validation CV score before the final effects were read. Bootstrap resampling seeds are fixed separately from forecast seeds. Simultaneous bootstrap-t bands use $\alpha/(2m)$ tails within a family of m contrasts; Holm adjustment is applied separately to centred two-sided bootstrap-t null tests. The five terrain primary contrasts, four terrain supporting contrasts, and five method comparison families retain their separate scope. There is no single global familywise guarantee across the entire study.

The second fixed bootstrap stream checks sensitivity of interval endpoints. Endpoint changes exceeding $\delta/4$ yield an inconclusive result, rather than automatically increasing resamples until a preferred verdict appears. Degenerate bootstrap uncertainty is retained explicitly. Conclusive primary effects require at least 30 recording-hash blocks and the registered seed agreement, mechanism and power conditions. Secondary origin modes supply descriptive summaries only.

31 Practical margin and comparison-specific planning

The registered arithmetic retains $\delta = 41.100259$ m, five percent of the previously recorded equal-block validation CV ES rounded to six decimals. Ordinary prose and figures display $\delta \approx 41.10$ m; display rounding does not change any comparison or qualification rule. It is a task-scale statistical/engineering convention, not a validated search-success or rescue-utility threshold. Final effects

and variances do not redefine it or expand the 46-block cohort. For development paired-block SD $\widehat{\sigma}_d > 0$, assumed improvement $d_\star = 2\delta$, family size m, target $p_\star = 0.8$, and SD multiplier $a \in \{1, 2\}$, the registered normal planning approximation is

$$P_a(B) = \Phi\left\{\frac{(d_\star - \delta)\sqrt{B}}{a\widehat{\sigma}_d} - z_{1-\alpha/(2m)}\right\},$$
$$n_a = \left\lceil \left\{\frac{(z_{1-\alpha/(2m)} + z_{p_\star})a\widehat{\sigma}_d}{d_\star - \delta}\right\}^2 \right\rceil. \tag{41}$$

Here Φ is the normal CDF and $z_p = \Phi^{-1}(p)$. SD uses denominator $B_d - 1$, after averaging origins and five forecast seeds within each of only $B_d = 3$ calibration-exposed development blocks. Terrain SD comes from the finite $N = 512,\ h = 0.3125$ s reference, not final coarse-grid variance; methods use their own comparison-specific differences. Observed improvement is not substituted for $d_\star$. The 1x/2x SD scenarios are not confidence bounds, bootstrap power validation or representativeness certificates.

Table 19 gives the already frozen $B = 46$ values; n_1, n_2 are planning counts, not authorization to add blocks. Planning additionally requires primary origins, at least 30 blocks, present strictly positive SD and $P_1(46) \geq 0.8$. A displayed 1.000 is rounded. Obs 600 s is below target. Although the arithmetic has a zero-SD limiting branch, the policy rejects zero SD as power evidence for the two d2 score-only aliases. The four LIO comparisons lack comparison-specific SD. These restrictions limit protocol-permitted categorical claims, not the truth of an observed effect. No larger experiment is launched to satisfy an unfavourable row.

Table 19: Frozen comparison-specific planning, not final effects or scientific qualification. SD uses three development blocks; target 0.8 at assumed improvement twice the practical margin. P=planning only passes, L=below target, Z=zero SD, M=missing SD.

Comparison	SD (m)	n_1	n_2	P_1/P_2	Gate
`all-vs-base`	65	30	117	0.957/0.333	P
`road`	68.4	33	130	0.933/0.295	P
`river`	81.1	46	182	0.806/0.196	P
`worldcover`	38.6	11	42	1.000/0.850	P
`surface`	35	9	34	1.000/0.920	P
`Heading/redraw`	18.9	3	10	1.000/1.000	P
`Single Gaussian`	73.9	37	145	0.898/0.270	P
`RDM`	54.2	20	78	0.996/0.529	P
`Speed/radial`	66.2	29	116	0.956/0.347	P
`Obs 30 s`	9	1	3	1.000/1.000	P
`Obs 120 s`	14	2	6	1.000/1.000	P
`Obs 300 s`	48	16	61	1.000/0.658	P
`Obs 600 s`	84.4	47	188	0.790/0.199	L
`QMLE label`	9.7	1	3	1.000/1.000	P
`Mixed calibration`	1.22	1	1	1.000/1.000	P
`ES calibration`	6.3	1	2	1.000/1.000	P

Table 19 (continued)

Comparison	SD (m)	n_1	n_2	P_1/P_2	Gate
`2D MC score`	0	1	1	1.000/1.000	Z
`2D closed score`	0	1	1	1.000/1.000	Z
`Scratch`	9.05	1	3	1.000/1.000	P
`Reptile`	32.9	8	29	1.000/0.959	P
`Drift-only`	6.14	1	1	1.000/1.000	P
`Two-step`	10	1	3	1.000/1.000	P
`Euler-M.`	0.0167	1	1	1.000/1.000	P
`Euler alias`	0.0167	1	1	1.000/1.000	P
`MC`	11.8	1	4	1.000/1.000	P
`CRN`	0.000371	1	1	1.000/1.000	P
Four LIO vs B (history)	–	–	–	–	M

32 Finite bootstrap resolution and saved diagnostics

The primary/second-stream seeds are 20260926/20260927; each uses $R = 2000$ NumPy `default_rng` draws of B block indices, shared across its family. Both observed and resampled SDs use denominator $B - 1$. For $m = 5$, each simultaneous tail has probability 0.005 and only 10 expected draws; for $m = 4$, 0.00625 and 12.5. Nominal family confidence is not an exact finite-sample coverage theorem. Quantiles use `inverted_cdf` order statistics, not interpolation through infinities. The add-one raw p-value has spacing 1/2001, approximately 0.00050, before Holm adjustment. A smallest-grid p-value is not practical importance.

The executed degenerate branch sets $\eta = 32\epsilon_{64}\max(1, |\widehat{\Delta}|)$. For $\widehat{\mathrm{se}}_r^* > \eta$ it uses the usual pivot; otherwise residual absolute value at most η gives zero, and larger positive/negative residual gives $+\infty/-\infty$. The observed zero-null statistic uses the same routine centred at zero, with tolerance $32\epsilon_{64}$. Nonfinite bounds are missing numeric endpoints, not clipped intervals. Exact zero differences can produce a point interval and $p_0 = 1$; that alone is not predictive-equivalence evidence. Zero-SD planning still excludes the score-only aliases.

Table 43 supplements the stage intervals in Table 25 with saved p-values and endpoint sensitivity. All 21 shifts are within $\delta/4 = 10.27506475$ m, but this does not certify forecasts, mechanisms or planning. The frozen engine retains the maximum shift and pass flag, *not* second-stream endpoints, index draws or pivots. Those missing values are not fabricated or reconstructed by rerunning bootstrap here. The surviving primary/marginal intervals, five seed effects and full-precision probabilities/shifts are in `paper/pirc17/inference-description-v1.json`. Their values reconcile with the complete 6440-row method stage; this existing-record projection is not a new final analysis or new hypothesis tests.

The complete 21-row tail-diagnostic table is in Appendix 47. Its entries supplement the primary effect intervals; the missing second-stream endpoints remain a reproducibility limitation.

33 Executable comparison states and terrain decisions

Let $[l, u]$ be the candidate-minus-control simultaneous interval, p_H its Holm zero-null probability, and e_s five seed differences. The decision-only roundoff guard is

$$g = 128\epsilon_{64}\max\{1, \delta, \max_i |S_i|\}. \tag{42}$$

It changes no effects, intervals, draws or scientific margin, and only downgrades otherwise conclusive boundary decisions. For a complete primary comparison, Q denotes: at least 30 blocks; available and passed recomputed mechanisms; comparison-specific planning passes; both intervals finite; and endpoint shift at most $\delta/4$.

Table 20: Comparison conditions after scope/missing-evidence checks. Conclusive rows also require Q; negative difference favours the candidate.

State	Boolean condition
beneficial	$u < -\delta - g,\ p_H \le 0.05,\ \#\{s : e_s < -g\} \ge 4$
harmful	$l > \delta + g,\ p_H \le 0.05,\ \#\{s : e_s > g\} \ge 4$
equivalent	$l > -\delta + g,\ u < \delta - g,\ \#\{s : \lvert e_s\rvert < \delta - g\} \ge 4$
inconclusive	No conclusive row above; or a computed gate fails.

Equivalence has no zero-null significance requirement. Seeds are forecast RNG streams, not five studies or retrainings. Methods use these predictive states in their own five families, never terrain retention labels.

Missing/failed required scores first make the whole prespecified family `unavailable`; no successful intersection is used. With complete scores, unavailable required mechanisms make that comparison `unavailable`, whereas a computed mechanism failure means `inconclusive`. Secondary origins remain descriptive. Low/missing/zero-SD planning, too few blocks, unbounded/unstable tails and practical/seed boundary overlap prevent conclusive predictive states. Computable raw statistics/states remain separate from qualified states/reasons. Missing evidence is not a zero effect.

Numerical applicability is separate: the bounded terrain-primary pilot does not cover LIO, all methods, secondary origins or every horizon; method mechanisms are not a total numerical-error bound. An allowed predictive state concerns the registered finite-budget algorithm only. No current artifact certifies a globally converged or resolution-invariant effect. The earlier phrase *accepted numerical evidence* must not imply a hidden blanket convergence gate.

For terrain, benefit $D = -\Delta$ has interval $[-u, -l]$. Factor decisions also require matching final primary populations, available LOO/LIO evidence, and independent fits/owner closure verified for base, full, LOO and LIO. Missing required evidence takes priority: `unavailable`; unresolved ownership/direction conflict: `inconclusive`. After these checks and primary LOO planning, the implemented mapping is Table 21.

Table 21: Conditional terrain mapping. B/H/E/I mean beneficial/harmful/equivalent/ inconclusive. Opposite raw LOO–LIO practical directions override any cell to inconclusive, even when LIO was downgraded for missing planning SD.

LOO \ LIO	B	H	E	I
B	retain	inconclusive	retain	retain
H	inconclusive	harmful	harmful	harmful
E	redundant	redundant	redundant	redundant
I	inconclusive	inconclusive	inconclusive	inconclusive

LIO must be available and checked for opposing raw direction; the frozen mapping does not require an independently powered favourable LIO state to retain a factor on qualified LOO evidence. Missing LIO SD remains disclosed, not passed. Redundancy is incremental equivalence in this full model, compatible with standalone LIO information. Joint interactions do not acquire unique/additive/causal attribution; history is a baseline diagnostic.

The five retained failures already make both primary terrain families unavailable (Table audit record). Method tail flags and planning rows alone are not final qualification. The independent saved-output audit and fixed-budget aggregate replay have completed; their evidence bindings and the 112-card inventory are given in Section audit record. Availability, mechanisms, planning reasons, numerical applicability, raw/qualified states and tail diagnostics remain separate fields. Their successful arithmetic replay does not establish global numerical convergence or human acceptance.

34 Method ablation results

We compare model structure, observation interval, learning objective, training/adaptation, and numerical inference on the same 46 recording-hash blocks and five forecast seeds. The complete primary method grid contains 6440 forecasts; it is not an average over the method and terrain matrices. Table 22 identifies each question and its actual control. The identity ledger has 22 arms and 36 execution slots, not 36 independent algorithms. Paired distributional changes, numerical diagnostics and computation time answer different questions. The results below retain the original saved scores, now checked by the completed independent output audit and fixed-budget aggregate replay. Seed/mechanism diagnostics and all 165 runtime records are available; their scope and interruption limits are not waived by successful replay. Table 18 in Section 28 gives the planned workload; Appendix audit record retains the separate captured completeness census.

35 How to read the method experiments

The control is another registered predictor evaluated on the same real targets, not the observed trajectory itself. Table 22 groups candidates by the question they change; the comparison group is also the scope of its multiple-comparison adjustment. Names denote the actual bounded implementations described in Section 16, not unrestricted versions of every algorithm bearing that name.

Table 22: Method comparison questions, technology names and registered controls.

Question / control	What the candidates actually change
Model structure / HSM-01	Pointwise redraws a mode on the reference clock; Single Gaussian removes the three-mode mixture. RDM is a Gaussian-mixture transition kernel grouped by residual direction, not mixture-likelihood EM fitting. Explicit decomposition adds speed and radial features in a single-mode regression.
Observation interval / HSM-01	Obs 30 s/120/300/600 use observations spaced by that many seconds; Obs 60 s is the HSM-01 alias. They change fitting data, history cadence and noise binding, not just the maximum prediction integration step.
Objective and score / HSM-07	QMLE labels the linear-Gaussian ridge fit. Mixed combines energy and negative-log-likelihood development calibration; Pure ES uses energy alone. d2 MC/closed are score-estimator routes with the same forecast, not two new predictors.
Training and adaptation / HSM-11	Scratch uses the adaptation fit only. Reptile uses regional development tasks and finite inner/outer updates. Drift only retains pretrained covariance at mixing; Two step shares the covariance mixing used by HSM, not a separate sequential optimizer.
Numerical propagation / HSM-18, HSM-20	EM means Euler–Maruyama here, not expectation–maximization; EM/Euler are one update under two labels versus HSM-18 split integration. MC (Monte Carlo) samples individual paths; CRN (common random numbers) shares indexed streams with the HSM-20 conditional-Gaussian-moment control. Shared draws alone do not establish variance or speed improvement.

One forecast work item is a configuration–origin-mode–seed prediction with 512 simulated paths at four scoring times, not 512 observed trajectories. The secondary origin modes reuse primary recording blocks; their detailed workload counts are in Section 28.

For a primary configuration c, let E_{cbs} be the equally weighted average of its four saved energy scores at block b and forecast seed s. The reported absolute score and paired difference use

$$\bar{E}_c = \frac{1}{46}\sum_{b=1}^{46}\frac{1}{5}\sum_{s=1}^{5}E_{cbs}, \qquad \bar{D} = \frac{1}{46}\sum_{b=1}^{46}\frac{1}{5}\sum_{s=1}^{5}(E_{\text{candidate},bs} - E_{\text{control},bs}). \tag{43}$$

The uncertainty procedure resamples the 46 paired blocks, preserving their matched seeds; it does not treat 230 forecasts or particles as independent real outcomes. Missing or failed required rows make a family unavailable, rather than reducing its denominator.

Table 23: Metric reading guide; different columns answer different questions.

Metric	Meaning and favourable direction
ES (m)	Distributional error, including location and dispersion; lower is better, but it is not mean position error.
ADE / FDE (m)	Ensemble-mean position error averaged over four targets / at the last target; lower is better. Neither describes every simulated path.
Δ ES (m)	Candidate minus its named control; negative favours the candidate. A small difference is not automatically practically useful.
Simultaneous interval	Within-family uncertainty for the paired difference. Crossing zero leaves the sign unresolved; crossing the practical boundary prevents the corresponding qualified claim.
Holm p	Adjusted evidence against zero mean difference, not the probability that a model is correct, a measure of improvement size or proof of equivalence.
90% disk coverage	Fraction of real targets inside the ensemble's nominal 90% disk; calibration aims near 90%, not simply the largest value. Size also matters; an arbitrarily large disk is not a useful forecast.
Prediction seconds	Recorded computation, not predicted physical duration. Lower cost alone does not establish better forecast quality.

36 Structure, observation interval and fitting strategy

Relative to HSM-01, Heading/redraw propagation has an ES increase of 32.95 m with an interval excluding zero. This interval is not entirely above the 41.10 m practical boundary, so a zero-difference finding is not by itself a registered harmful-component verdict. Single Gaussian and RDM differences have intervals spanning zero. The explicit-decomposition mean is 57.96 m worse, but its wide interval also spans zero: a large mean alone does not establish a reliable practical effect.

The Obs 120 s and Obs 300 s configurations reduce ES by 11.56 and 23.40 m, respectively, with within-family intervals below zero. Neither improvement interval lies wholly beyond the registered practical margin. The Obs 600 s mean is lower still, but its interval spans zero; it is not an established optimum. These dt labels change the fitted method's observation interval, not the frozen prediction integration policy. More favourable ES also need not mean a more accurate forecast mean: HSM-01's mean endpoint error is 1326.09 m, compared with 1335.73 m for Obs 300 s and 1340.94 m for Obs 600 s.

QMLE, mixed and pure-ES objectives differ from HSM-07 by 0.28, -6.55 and -5.31 m, with all intervals spanning zero. The scratch, Reptile, drift-only and two-step variants likewise have small differences relative to HSM-11. These data do not establish a large predictive gain for each additional component. Conversely, non-rejection of zero is not proof of exact equality.

Table 3 establishes identical dynamics for HSM/QMLE and mixed/pure-ES. HSM-07 shares the score-only stream; candidates use distinct slot-indexed streams. Equal seed numbers do not make those streams identical. Mixed/pure-ES additionally fit validation data. These comparisons cannot establish loss-only superiority or replace the independent output audit.

The two d2 score-only routes have exactly the same common predictive scores as their HSM-07 anchor. Their zero paired spread does not establish powered independent replication or validate the native score estimators. EM and Euler are implementation aliases with identical saved scores; both differ from HSM-18 by about 0.02545 m. Their small adjusted zero-test probability does not make this centimetre-scale difference practically important. The MC and CRN mean differences do not, by themselves, prove variance reduction or runtime speedup; those questions require their registered diagnostics.

37 Sensitivity to the five simulation seeds

For each registered seed, we average the candidate-minus-control weighted ES over the same 46 blocks, before averaging those five values for the reported paired estimate. Table 42 gives all 21 comparisons, the five original seed values and their sign counts. Six rows are positive for every seed, seven negative for every seed, six change sign, and two score-only routes are exactly zero. These are counts of comparison rows, not 21 independent algorithms: EM/Euler remain aliases. Seeds repeat simulation of deterministic fitted configurations; they are not five refits, new participants or independent held-out studies. Equal numeric seeds also do not imply shared slot-indexed random streams.

The residual mixture's five differences range from -5.00 to -1.08 m, yet its block-level simultaneous interval still spans zero. Conversely, the explicit-feature configuration is worse by 56.49–59.13 m for every seed, but its block interval also spans zero. Thus simulation-seed agreement does not resolve uncertainty across recordings or establish a practical component verdict. QMLE changes sign (two negative, three positive values), as do all four training/adaptation candidates and MC propagation. This small-seed descriptive check does not estimate Monte Carlo convergence, between-fit variability or precision-invariant effects. It adds no scores, resampling, adjusted tests or qualification to the existing intervals.

38 Complete primary method results under the fixed simulation budget

The primary results use all 28 configurations on the same 46 recording-hash blocks and five forecast seeds: 6440 saved-score rows, or 230 per configuration. Repeated seeds do not increase the number of observed recording blocks. The estimates come from the fixed 6 October scoring snapshot; its exact capture time and reconciliation with the later execution census are documented in Appendix audit record.

The independent saved-output replay and subsequent anonymous aggregate replay reproduce the estimates and registered decision states for the original fixed simulation budget. This qualification does not establish global numerical convergence or effects invariant to particle count and integration step. All 21 prespecified method contrasts are retained in Table 25. The unchanged

2000-draw paired-block procedure and independent fixed-seed tail check provide nominal 95% simultaneous intervals within each of the five families. There is no combined across-family coverage claim. The table uses $\Delta = \mathrm{ES}_{\mathrm{candidate}} - \mathrm{ES}_{\mathrm{control}}$: negative differences favour the candidate. HSM-01, HSM-07, HSM-11, HSM-18 and HSM-20 are the registered arm-specific anchors, not interchangeable pooled replications. The original practical margin remains $\delta \approx 41.10$ m, using the unrounded registered value for all calculations. The integration and CRN rows retain extra digits solely to show numerical diagnostic scale, not centimetre-level predictive accuracy or practical improvement.

Reproduced finite-budget decision states. The original registered analysis has now completed once on all 58 closed score blocks. Its anonymous scalar projection is `paper/pirc17/analysis-stage-summary-v1.json`; no fits, forecasts or new bootstrap draws were added by this projection. The 21 primary contrasts retain exactly the estimates, intervals and Holm values already shown below. The saved decision procedure records 14 practical-equivalence states and seven inconclusive states, with no practically beneficial or harmful state (Table 24). The saved-output audit and the pinned aggregate replay now verify this arithmetic; they are not independent empirical replications or final human acceptance. The external qualification is recorded in `paper/pirc17/qualification-v1/test02.json`: 21 evidence/provenance checks and 104 frozen-rule software tests, without new forecasts or bootstrap draws. All 112 evidence-card/disposition references are checked in `paper/pirc17/qualification-v1/card-inventory.json`. All three origin modes and four whole unavailable comparison families remain in the analysis; the two six-block secondary modes do not acquire hypothesis tests.

Equivalence here requires strict simultaneous-interval containment in the prespecified $[-41.100259, 41.100259]$ m band and the applicable planning, tail and mechanism conditions; it is not inferred from a nonsignificant zero-effect test. For example, RDM's saved interval $[-21.22, 16.33]$ m is inside this band, although it crosses zero. Conversely, Obs 300 s's interval $[-48.27, -2.47]$ m excludes zero but crosses the practical boundary. The two d2 aliases remain inconclusive despite identical predictions and zero intervals: zero development SD is not planning-power evidence. Obs 600 s retains insufficient planning power. These states do not establish exact algorithm identity, calibrated uncertainty or a general best model.

The original mechanism consumer records all 28 required gates in each of the three modes as computed and passed. This includes weak diagnostic gates: the Euler/EM condition is merely error ≥ 0, not an accuracy ceiling. The primary same-grid split diagnostic is 2.19×10^{-5} m against its 2 m ceiling; the maximum closed-versus-Monte-Carlo relative discrepancy is 0.1937 against 0.2, using the registered finite scoring-only draws. Passing these particular checks is not a global N/step convergence bound. Every primary method comparison still records resolution-invariant inference as inconclusive; final manuscript and human acceptance remain pending. The stage captions retained in the following two original tables are historical: their pending-replay wording does not describe the completed replay. The numeric cells and the practical-equivalence versus inconclusive distinction are unchanged.

Table 25: Preliminary complete-primary-method arithmetic; not final qualified verdicts.

Candidate	Control	Δ (m)	Simultaneous CI (m)	Holm p
Pointwise	HSM-01	32.95	[22.99, 43.37]	0.0020
Single Gaussian	HSM-01	22.01	[-14.19, 60.81]	0.3278
RDM	HSM-01	-2.83	[-21.22, 16.33]	0.7111

Table 25 (continued)

Candidate	Control	Δ (m)	Simultaneous CI (m)	Holm p
Explicit decomposition	HSM-01	57.96	[-3.03, 117.08]	0.0540
Obs 30 s	HSM-01	8.56	[3.73, 13.76]	0.0020
Obs 120 s	HSM-01	-11.56	[-20.37, -3.04]	0.0030
Obs 300 s	HSM-01	-23.40	[-48.27, -2.47]	0.0230
Obs 600 s	HSM-01	-24.69	[-67.65, 9.08]	0.1159
QMLE label	HSM-07	0.28	[-1.92, 2.70]	1.0000
Mixed objective	HSM-07	-6.55	[-16.86, 3.18]	0.5497
Pure ES	HSM-07	-5.31	[-15.98, 5.26]	0.8396
d2 MC	HSM-07	0.00	[0.00, 0.00]	1.0000
d2 closed	HSM-07	0.00	[0.00, 0.00]	1.0000
Scratch	HSM-11	2.60	[-4.73, 10.02]	1.0000
Reptile	HSM-11	2.19	[-11.95, 12.91]	1.0000
Drift only	HSM-11	0.50	[-1.52, 2.49]	1.0000
Two step	HSM-11	-0.04	[-2.60, 1.81]	1.0000
Euler-M.	HSM-18	0.02545	[0.00774, 0.05092]	0.0280
Euler	HSM-18	0.02545	[0.00774, 0.05092]	0.0280
MC	HSM-20	-0.78	[-2.95, 1.49]	0.3653
CRN	HSM-20	-0.000056	[-0.000153, 0.000015]	0.1669

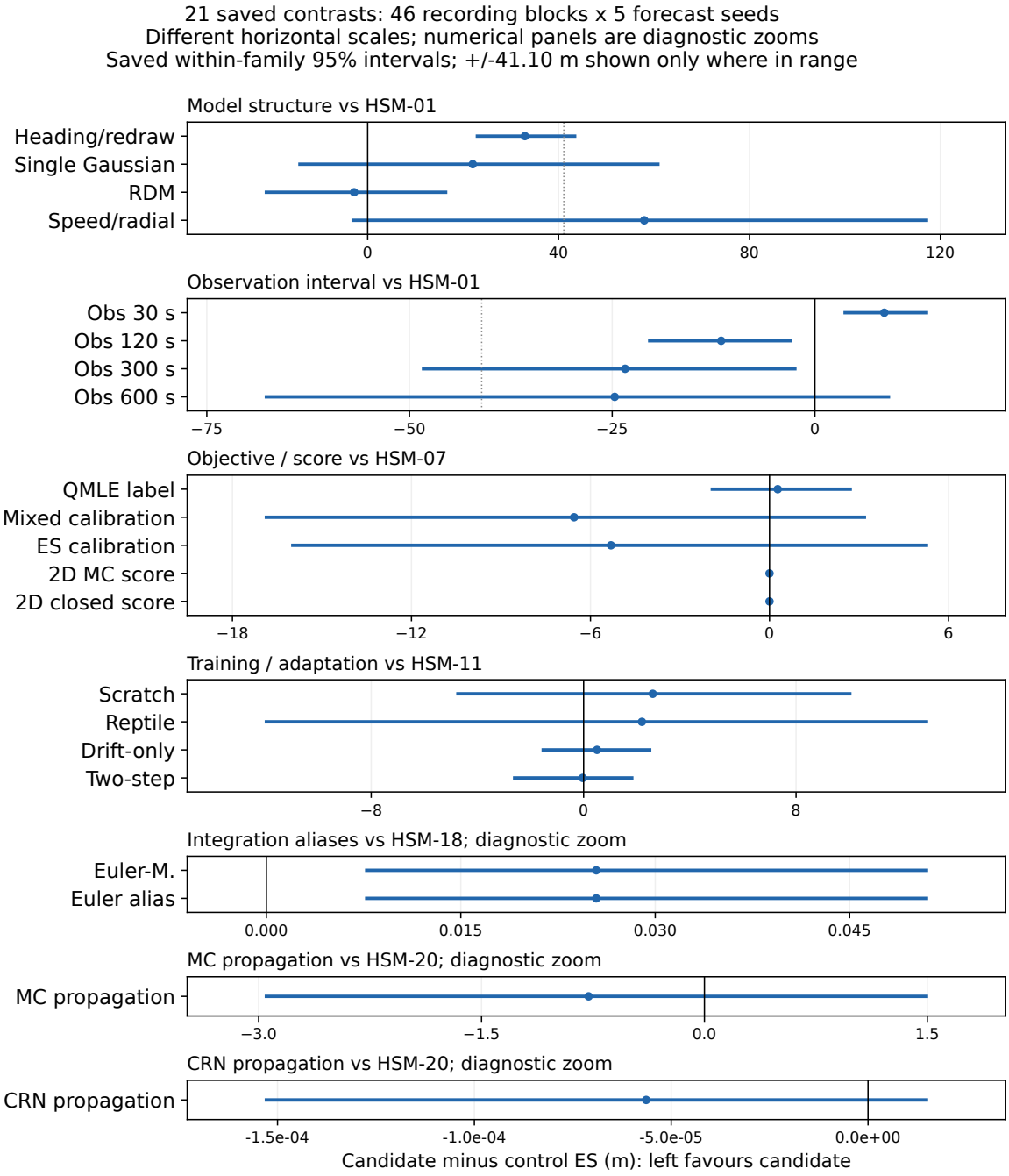


Figure 11: All 21 saved paired contrasts, with explicitly different horizontal scales. Points are recording-block means of candidate-minus-control ES; bars are saved within-family simultaneous 95% intervals. Each panel names its HSM control; left of zero favours the candidate. Numerical panels are diagnostic zooms, not enlarged practical effects. Dotted ±41.10-m boundaries appear only where within the axis range; they are outside the smaller panels. Neither zoom nor crossing a line qualifies a conclusion. The seven panels represent five method families: integration aliases, MC propagation and CRN propagation are three axes of the numerical family. Exact small differences and aliases remain in Table 25.

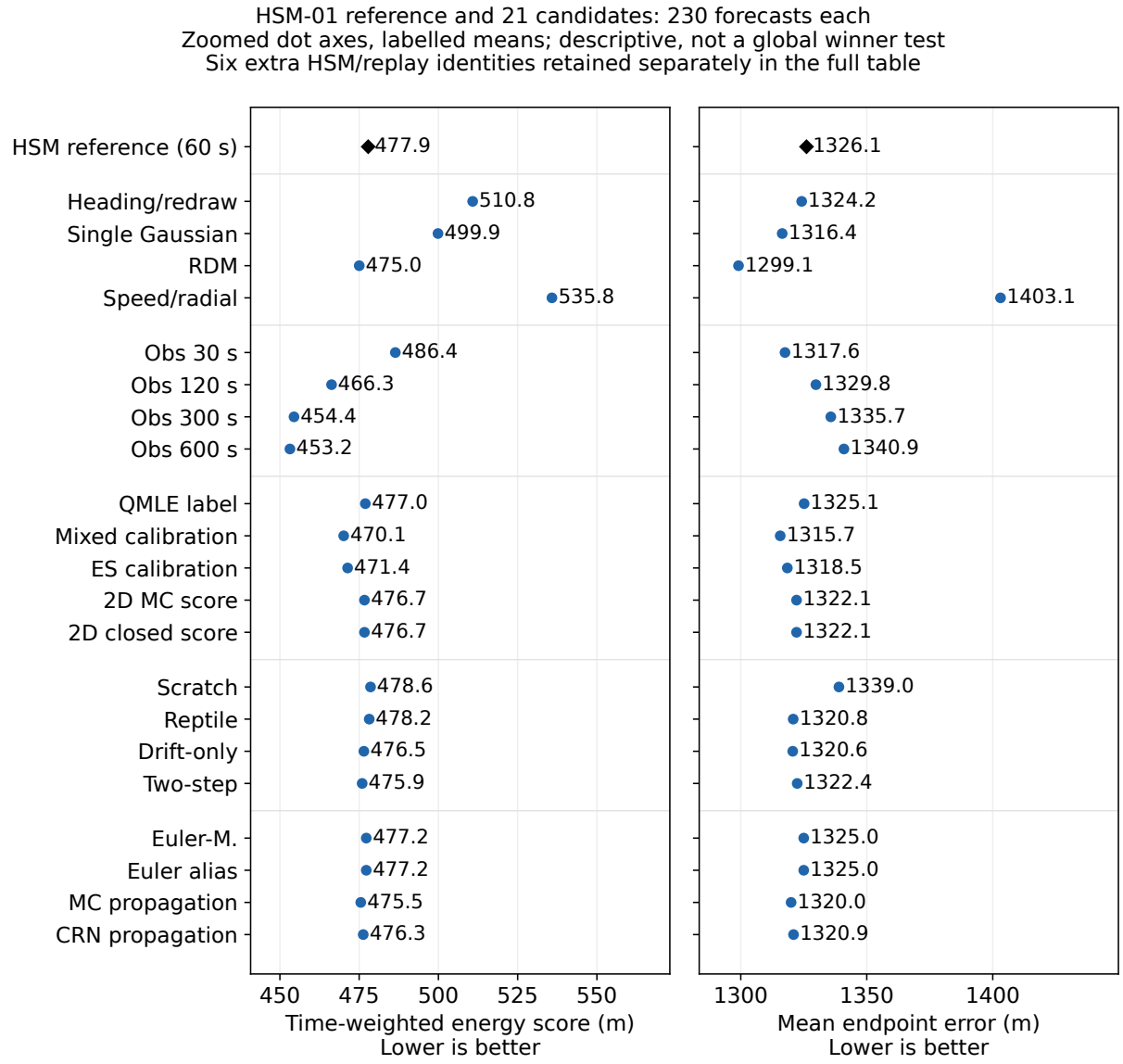


Figure 12: HSM-01 reference and all 21 candidates, with numeric labels for saved absolute ES and endpoint-error means in metres. Both dot axes are zoomed and lower is better; differences in dot position are not paired effect estimates. Observation variants follow increasing interval order. Six extra HSM/replay roles are omitted only from this display and are retained separately, not averaged, in Table 47. Each row uses all 230 predictions; scoring routes and aliases are not independent algorithms. This is descriptive, not a global winner test.

Table 24: Historical saved primary method decision states: replay was pending at capture, and completed on 8–9 October. Human acceptance remains open. All 21 contrasts are retained; equivalence is within the fixed practical band, not exact equality. No family has a beneficial or harmful state.

Registered family	Equivalent	Inconclusive
Model structure	1	3
Observation interval	2	2
Fitting and score route	3	2
Training and adaptation	4	0
Numerical propagation	4	0
Total	14	7

39 Horizon-dependent error and empirical coverage

HSM-01's weighted ES is 477.85 m, whereas its mean 30-minute endpoint error is 1326.09 m; these are different quantities. The retained all-method aggregate reports marginal ES at each target, but only the four-target average ADE and last-target FDE for point accuracy. Those two summaries do not identify the earlier mean-position errors, and ES cannot be converted into them. We do not reconstruct a complete point-error profile from these summaries. The existing map cases separately retain four-time mean-position errors, described in Section 42; they are terrain illustrations, not a substitute for all-method accuracy. Table 26 shows increasing marginal ES and substantial undercoverage of the nominal 90% radial prediction disk at the longer times. Coverage is averaged over five simulated ensembles for each of the same 46 targets, not over 230 independent observed targets. These proportions are descriptive; no new coverage-test family or confidence interval is introduced. The pattern does not identify a unique drift, diffusion or mixture failure.

Table 26: Descriptive saved-score horizon profile for the complete primary HSM-01 grid; actual original target times follow the registered tolerance.

Nominal horizon (min)	HSM-01 ES (m)	90% disk coverage (%)
1	50.57	95.65
5	243.23	37.83
15	609.27	35.22
30	1008.35	46.52

The full configuration-by-horizon pattern. Figure 13 extends the view to all 28 original primary configurations, with full values in Table 48. At the thirty-minute slot, their saved mean marginal ES spans 941.89–1197.62 m; all nominal 90% disk coverage means are below 90%, ranging from 21.74% to 79.13%. These are ranges across configurations, not confidence intervals, population bounds or a global best-model test. The weighted primary summary can conceal a horizon-dependent tradeoff: the residual mixture has lower ES than HSM-01 at five and fifteen minutes (228.87 versus 243.23 m; 598.70 versus 609.27 m), but higher ES at thirty minutes (1022.19 versus 1008.35 m). No per-horizon significance test is introduced. The marginal ES is a distributional score, not mean position error; higher coverage alone is not better calibration without considering the nominal probability and region size. These are original nominal target slots, not new interpolated exact-time predictions. The saved target-clock description binds the three illustrated models,

not every prediction array of all 28 configurations. The separate completed whole saved-output audit covers the complete score grid; neither this narrow description nor that audit certifies a physically sourced target clock.

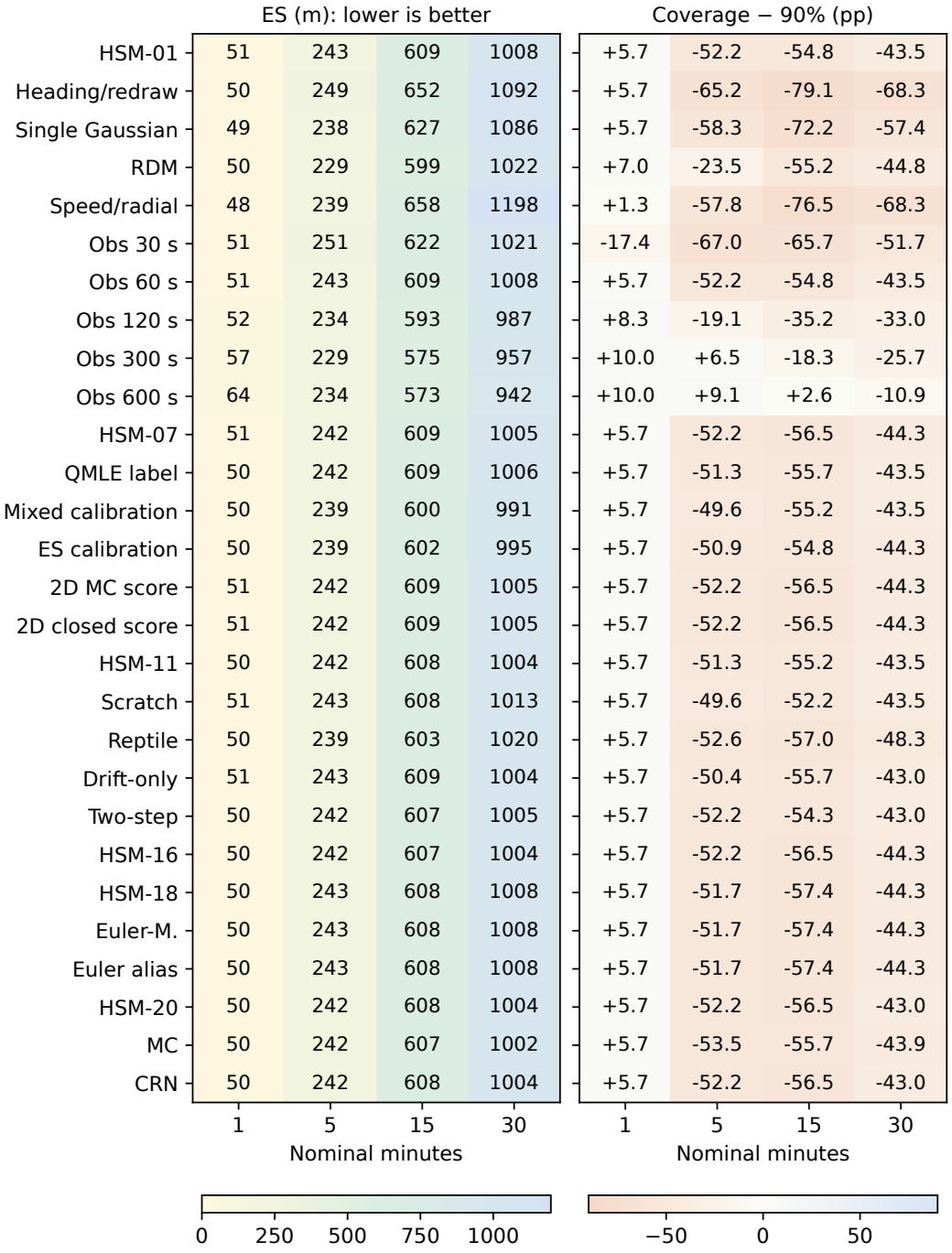


Figure 13: 28 configurations by four horizons by two metrics: 224 scalar cells. Left: ES (m), lower is better. Right: observed coverage minus nominal 90% (percentage points), where zero is the reference and equal positive/negative departures use equal colour distance. Each cell averages five seeds within each of 46 blocks, then blocks equally. Original configuration ordering and separate HSM roles are retained; no performance ranking or new test. Black numbers and pale linear scales match the main overview. HSM coverage percentages and precision remain in Table 48.

Region size and coverage across all configurations. Figure 14 and Table 49 extend the joint region description to every original primary configuration, using the same 6440 saved scores, 46 blocks and five seeds. Each nominal 90% disk cell reports mean area and target containment; the table also gives mean radius. We average saved disk areas themselves, not the square of an averaged radius. The full saved aggregate retains all four probability levels at all four target slots. This extension reads saved score numbers only; it does not rescore particles, interpolate targets or add predictions.

At thirty minutes, the 28 mean 90% disk areas span 1.247–12.609 km^2, while empirical coverage spans 21.74–79.13%, always below nominal 90%. Obs 600 s has the largest area and highest observed coverage in this finite display: 12.609 km^2 and 79.13%, versus HSM-01's 4.534 km^2 and 46.52%. Its area is about 2.78 times HSM-01's, yet its mean endpoint error is slightly higher (1340.94 versus 1326.09 m). Thus its higher coverage does not simultaneously establish sharper regions, better point accuracy and calibration. Obs 600 s changes fitted observation spacing, historical inputs and noise estimation, not only the prediction integration step; these descriptive comparisons do not identify which change produces the pattern.

Nor is the area–coverage relation uniform across horizons. At five minutes, the residual mixture has a larger mean area than HSM-01 (0.400 versus 0.294 km^2) and higher coverage (66.52% versus 37.83%); at thirty minutes its area is smaller (3.079 versus 4.534 km^2) and coverage is slightly lower (45.22% versus 46.52%). The display therefore reveals horizon-dependent descriptive tradeoffs, not a universal size–quality rule, global winner test or causal mechanism. No new per-horizon interval or calibration adjustment is introduced; block-resolved diagnostics remain the three-model description below. The completed independent output replay does not turn those three illustrative models into all-configuration block-resolved calibration evidence.

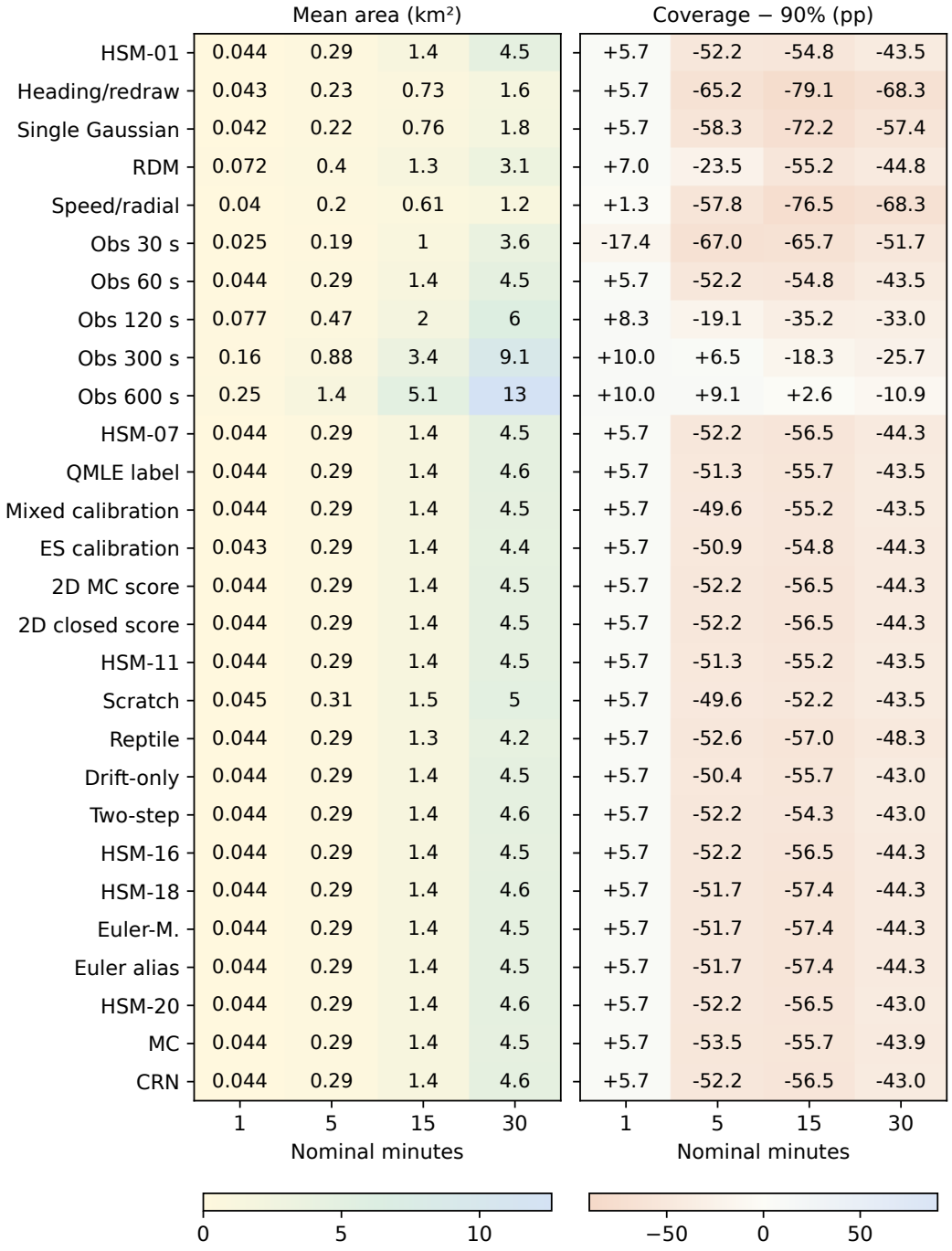


Figure 14: 28 configurations, four horizons, two metrics: 224 scalar cells. Left: mean 90% disk area (km^2), with a linear scale shared with the main overview. Right: coverage minus 90% (percentage points), with a symmetric zero-centred scale. Neither area nor coverage is a standalone quality rank. Each cell averages seeds within each of 46 blocks, then weights blocks equally. Area averages individual disk areas, not the square of mean radius. The same cohort and forecasts underlie the other overviews; repeated views are not independent evidence. HSM values are in Table 49.

All probability levels and the cost of wider regions. The complete 28-configuration profiles also show that the coverage mismatch depends on nominal probability, not only horizon (Table 50). At one minute, 24 configurations have 50% disk coverage below 50%, whereas 27 have 90% disk coverage above 90%. Thus high short-horizon coverage at one probability level does not establish calibration of the predictive distribution. At thirty minutes, all 28 are below nominal at the 80%, 90% and 95% levels; observed 95% coverage spans 26.52–87.39%. At the 50% level, 27 are below nominal and one equals it, without certifying that configuration's other levels. These are counts and ranges of saved configuration means, not independent algorithm replications or confidence intervals: aliases and HSM controls remain in the frozen inventory. The table reports area ranges alongside coverage; widening the same predictor's disk can raise containment without repairing its mean-position error. No probability level is changed, and no new calibration test or fitted correction is made.

We additionally describe the same 690 previously displayed primary scores for HSM, RDM and Obs 300 s, with no new forecasts, particle rescoring or inference tests. Each original score stores all four marginal disk levels. Table 27 reports HSM's coverage and mean disk area together. Mean area averages $\pi r_{p,t}^2$ over seeds within each block and then equally over 46 blocks; it is not $\pi\{\text{mean}(r_{p,t})\}^2$.

Table 27: HSM's complete primary saved-region profile. Each cell is observed coverage (%) / mean disk area (km^2); column headings are nominal probabilities, not observed coverage.

Horizon (min)	50%	80%	90%	95%
1	33.04 / 0.013	87.39 / 0.031	95.65 / 0.044	96.52 / 0.057
5	6.52 / 0.094	25.65 / 0.210	37.83 / 0.294	50.43 / 0.378
15	9.57 / 0.539	24.35 / 1.062	35.22 / 1.415	43.91 / 1.753
30	33.48 / 2.002	37.39 / 3.552	46.52 / 4.534	52.17 / 5.496

The long-horizon mismatch persists beyond the 90% level: at thirty minutes, HSM's nominal 95% disk covers only 52.17% of targets on average over ensembles. At one minute, nominal 50% and 90% disks cover 33.04% and 95.65%, respectively, showing that the mismatch depends on probability level as well as horizon. These are descriptive measurements, not new significance statements or proof of a particular failure mechanism.

Table 28: Thirty-minute nominal 90% disk profiles on the same original 46 blocks and five ensembles per model. A smaller area is not preferable without adequate coverage; larger coverage alone need not imply calibration.

Model	Coverage (%)	Mean area (km^2)	Mean radius (m)
HSM	46.52	4.534	1183.6
RDM	45.22	3.079	989.5
Obs 300 s	64.35	9.109	1692.8

The Obs 300 s disk has higher thirty-minute coverage than HSM, but its mean area is about twice as large and its coverage remains well below nominal 90%. Its mean endpoint error is also slightly higher (1335.73 versus 1326.09 m). Thus improved ES or increased coverage cannot be interpreted as a sharper, more accurate and calibrated forecast simultaneously. RDM's smaller thirty-minute region similarly does not establish better precision when coverage remains low. This three-model display is post-outcome descriptive reporting of already selected examples, not a global ranking, an additional hypothesis family or a model-selection rule. The projection `paper/pirc17/calibration-description-v1.json` retains the complete original 690-score bindings without

private trajectory IDs or coordinates. Its ES, FDE and 90% coverage reconcile to the unchanged original stage table. Missing, failed, duplicate or wrong-source rows reject the whole projection. The separate independent saved-output audit is complete; this descriptive projection does not itself establish calibration or participant independence.

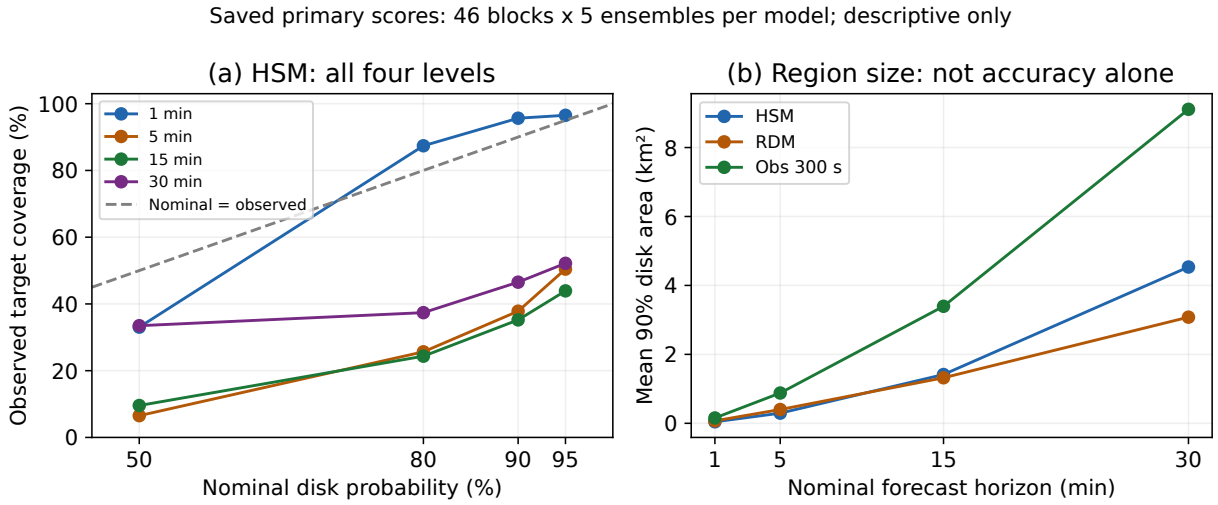


Figure 15: Saved calibration and region-size diagnostics, not new experiments. Left: HSM's observed coverage against nominal probability, separately for each horizon; the diagonal denotes nominal agreement. Right: mean 90% disk area for the three previously displayed models. Connecting lines only guide the eye between four measured levels or horizons; there is no intermediate target evaluation, simultaneous path coverage, or new statistical test.

Variation across recording blocks. For a fixed model, target slot and disk probability level, we first average the five forecast seeds within each of the same 46 recording blocks:

$$C_b = \frac{1}{5}\sum_s \mathbf{1}\{y_b \in D_{bs}\}, \qquad a_b = \frac{1}{5}\sum_s \frac{|D_{bs}|}{10^6}. \tag{44}$$

Here D_{bs} is the saved marginal disk, C_b is a descriptive containment fraction and a_b is mean area in square kilometres. Five seeds repeat prediction of the same real target; they are not five independent observations. Individual C_b values lie on a 0.2 grid. Table 29 reports medians and interquartile ranges over blocks; linear-interpolated quantiles at $(46-1)p$ need not lie on that grid and are not confidence intervals. All four saved levels and horizons are retained in `paper/pirc17/calibration-block-description-v1.json`; the table and Figure 16 show nominal 90% disks only.

At 30 minutes HSM has 23 of 46 blocks with no covered seed and 20 with all five covered, despite overall coverage of 46.52%. Its area median is 4.022 km^2 [3.405, 4.966], with range 2.963–11.551 km^2. The Obs 300 s median containment is 100%, but 16 blocks still have zero covered seeds; its area median is 8.518 km^2 [7.707, 9.564]. Neither an overall mean nor a high median establishes reliable conditional calibration, deployment readiness or which component causes the observed misses.

For a coordinate-free example, anonymous block B13 is the first saved block in the original ordering with HSM 0/5 and Obs 300 s 5/5 containment at 30 minutes. HSM, RDM and Obs 300 s

Table 29: Original 46-block distributions of nominal 90% disks. Coverage and area are median [25th, 75th percentile]; the last columns count blocks with zero/all five seeds covering their target.

Model	Min	Coverage (%)	Area (km^2)	0/5	5/5
HSM	1	100 [100, 100]	0.043 [0.042, 0.046]	2	44
HSM	5	0 [0, 100]	0.280 [0.261, 0.308]	26	16
HSM	15	0 [0, 100]	1.285 [1.150, 1.557]	29	15
HSM	30	10 [0, 100]	4.022 [3.405, 4.966]	23	20
RDM	1	100 [100, 100]	0.072 [0.070, 0.074]	1	44
RDM	5	100 [5, 100]	0.401 [0.389, 0.411]	12	28
RDM	15	0 [0, 100]	1.322 [1.300, 1.357]	28	14
RDM	30	0 [0, 100]	3.088 [3.000, 3.179]	24	20
Obs 300 s	1	100 [100, 100]	0.155 [0.150, 0.159]	0	46
Obs 300 s	5	100 [100, 100]	0.871 [0.832, 0.894]	1	44
Obs 300 s	15	100 [20, 100]	3.264 [3.060, 3.497]	11	31
Obs 300 s	30	100 [0, 100]	8.518 [7.707, 9.564]	16	29

have 0/5, 0/5 and 5/5 containment and mean areas 3.489, 3.204 and 8.260 km^2, respectively. This example is selected after seeing outcomes to illustrate size/containment, not to infer superiority. No route or geographic identity is released; the separate permissions for real map cases remain unverified.

The known-velocity structure and observation-interval families each contain one failed required row and remain unavailable as complete families. The other complete six-block secondary summaries are descriptive only. Missing rows and failed comparisons are not filled with zero, deleted to form a favourable successful intersection, or replaced by primary-mode data.

40 Descriptive method results by origin cover

To make environmental coverage visible, Table 30 describes HSM, RDM and Obs 300 s using all 690 existing scores from their original $3 \times 46 \times 5$ primary grid. Within each block, five simulation seeds are averaged first; block means are then averaged within the original WorldCover origin category. ES still averages the four registered target times, while FDE measures the last target. The category-weighted totals reconcile to the unchanged primary method table. Missing, failed, duplicate or wrong-source rows reject the whole descriptive projection; no successful-subset summary is substituted.

Table 30: Descriptive origin-category means, ES/FDE in metres. No subgroup confidence intervals, hypothesis tests or environmental-effect verdicts.

Origin pixel	Blocks	HSM	RDM	Obs 300 s
Trees	27	427.66/1184.64	420.85/1153.52	414.59/1203.95
Grass	9	556.86/1597.16	557.16/1587.72	514.09/1579.13
Cropland	3	623.14/1678.49	589.40/1591.57	600.89/1775.32
Built-up	7	507.60/1372.16	529.38/1364.23	468.75/1342.70

These are post-outcome descriptions, not preregistered subgroup inference. For example, Obs 300 s has a lower mean ES but a higher FDE in the tree and cropland origin categories; RDM has a higher ES but a slightly lower FDE in the built-up category. These patterns illustrate

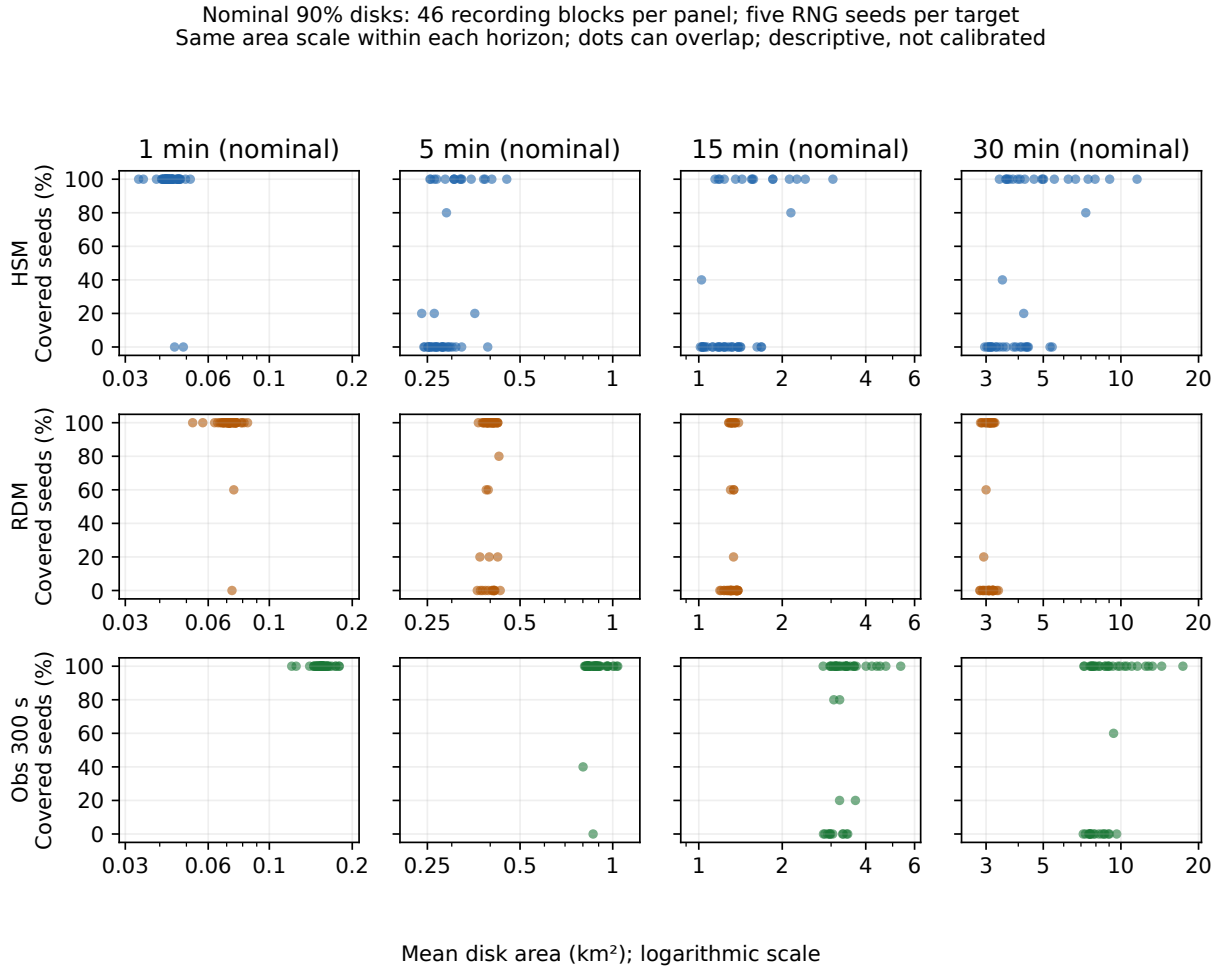


Figure 16: Paired block-level size and containment for the three previously displayed models at four nominal horizons. Each point is one recording block averaged over five forecast seeds; points can overlap. Area uses a logarithmic scale, shared across models within each horizon, while containment is the fraction of five saved disks containing the same target. These are not 46 estimates of population calibration; no simultaneous path coverage, new hypothesis test or post-hoc radius rescaling is performed.

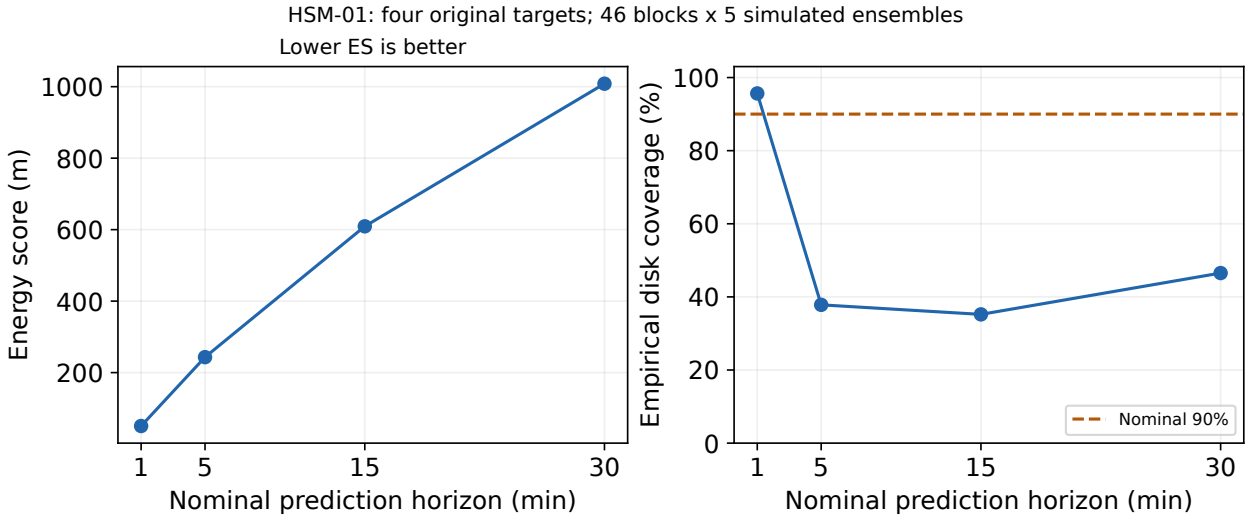


Figure 17: HSM-01's actual aggregate saved scores and coverage at the four original targets. The dashed line is nominal 90% coverage; observed 30-minute coverage is only 46.52%. Lines guide the eye, not intermediate observations or a survival curve. These are distribution-level summaries, not trajectory/map overlays or independent 230-target measurements.

different metric rankings, not causal terrain effects or robust category-specific model superiority. The cropland group contains only three recording-hash blocks, not 15 independent trajectories. The seven absent origin classes retain null metrics in the projection. Formal method inference remains on all 46 blocks; actual terrain effects require the separate registered terrain comparisons and their evidence qualification.

40.1 A completed constant-velocity reference

The registered CV reference extrapolates the saved causal initial velocity: $\widehat{x}^{\mathrm{I}}(t) = x_0 + t\widehat{v}_0$. It is a deterministic path, not a fitted SDE or five repeated particle forecasts. All 58 registered reference paths have saved scores; Tables 31 and 32 use the entire prespecified 46-block primary population, not the 12 secondary origins. Each reference is paired with HSM at the same origin, target identity and actual target times. Average the five HSM forecast seeds within each block first, then weight all 46 recording-hash blocks equally; the reference is never counted five times.

Table 31: Same primary origins and saved target slots. All errors are in metres and lower is better. ES evaluates the predictive distribution; ADE/FDE evaluate its mean path. CV is a point mass, so its ES equals displacement error. Counts are saved forecasts, not independent people or fitted models. Descriptive means only; no new confidence interval, significance test or qualification.

Model	Forecasts	Blocks	Weighted ES	ADE	FDE
CV	46	46	824.73	824.73	2095.78
HSM	230	46	477.85	636.11	1326.09

HSM-minus-CV mean differences are -346.87 m for weighted ES, -188.62 m for ADE and -769.69 m for FDE. HSM has strictly lower weighted ES in 34 of 46 blocks. However, the CV reference has lower mean ES at 1 and 5 minutes, whereas HSM has lower mean ES at 15 and 30 minutes. Thus the aggregate benefit is not uniform across horizons.

Table 32: Mean marginal ES in metres at the four nominal target slots; lower is better. Actual retained-clock targets are 49–67, 294–305, 889–910 and 1785–1814 seconds under the unchanged original 30-second tolerance. Neither target interpolation nor a new simulation is used.

Model	1 min	5 min	15 min	30 min
CV	31.17	237.93	934.03	2095.78
HSM	50.57	243.23	609.27	1008.35

These are post-outcome descriptive comparisons of existing scores, not a registered component contrast or an independently audited superiority claim. No inference is made from the 34/46 count; recording hashes do not certify independent participants. The original validation-based $\delta = 41.100259$ m is not recalibrated from this final-evaluation reference. The original saved-output audit, final card inventory and fixed-budget public scalar replay were completed on 8–9 October; the current review rechecks their receipts, not the raw particle arrays. This arithmetic closure does not supply a new superiority test, physical-clock or participant-independence certification, deployment calibration, authorship confirmation or human acceptance.

40.2 Point error at each saved target time

ES evaluates an entire predictive distribution and is not the distance between its mean and the observed position. Table 33 makes that distance explicit for the already-described HSM/RDM/Obs 300 s comparison and the CV reference; it is not a new ranking of all 28 configurations. Use the original saved scoring-disk center $c_{m,b,s,k}$ (the 512-particle mean, or the deterministic reference position) and the same saved target $y_{b,k}$ in the same origin-local scoring frame. No forecast arrays, new particle scores, target interpolation or simulation are needed.

For model m, average errors over its S_m original seeds within each recording-hash block, then equal-weight all 46 blocks. Here $S_m = 5$ for the three probabilistic configurations and $S_m = 1$ for the reference; a reference is not counted five times:

$$\bar{e}_{m,k} = \frac{1}{46} \sum_{b=1}^{46} \frac{1}{S_m} \sum_{s=1}^{S_m} \| c_{m,b,s,k} - y_{b,k} \|_2. \tag{45}$$

Table 33: Mean saved-center displacement error in metres; lower is better. The four columns use the original nominal 1/5/15/30-minute target slots, with actual retained-clock ranges 49–67, 294–305, 889–910 and 1785–1814 s and unchanged 30-second tolerance. Rows are saved forecasts, not independent targets or people. Descriptive values only; no per-horizon inference.

Model	Rows	1 min	5 min	15 min	30 min
HSM	230	73.77	332.31	812.26	1326.09
RDM	230	71.92	323.79	792.99	1299.11
Obs 300 s	230	75.88	340.10	825.76	1335.73
CV	46	31.17	237.93	934.03	2095.78

All 736 selected saved rows match the same 184 target positions. For every row, the arithmetic mean of these four errors and its last error reproduce the original saved ADE and FDE within floating-point roundoff; the maximum discrepancy is below 10^{-12} m. Averaging at each target before taking the Euclidean norm would be a different estimator and is not used.

The CV reference has lower mean point error at 1 and 5 minutes; HSM has lower mean point error at 15 and 30 minutes. RDM has slightly lower descriptive mean errors than HSM at all four slots. Conversely, Obs 300 s has larger mean-position errors than HSM at every slot despite its lower saved aggregate ES. Thus a distribution-score improvement is not automatically an improvement in mean-path location. Here Obs 300 s denotes the training observation interval, not the integration step. The saved-output audit and original paired scalar replay are complete, as recorded in the audit document; the 736-row descriptive identity check above remains distinct from them. No new per-horizon superiority test, participant-independence or physical-clock certification, deployment accuracy or human acceptance follows.

41 Marginal CRPS and within-forecast endpoint error quantiles

Table 34 uses the same four models already compared above, not a new selection by the best diagnostic value. Coordinate CRPS measures the local east/north marginal distributions in metres; lower is better on that marginal, but two good marginals do not certify the joint distribution, dependence structure or predictive-disk calibration. The two coordinate scores are not summed and renamed joint ES.

The lower panel distinguishes mean-position FDE from particle endpoint-error quantiles. For each original forecast, the stored $q_{.50}$, $q_{.90}$ and $q_{.95}$ are quantiles of particle distances to that forecast's observed endpoint. We average those already-saved quantiles over seeds and recording blocks. They are neither population FDE quantiles nor quantiles after pooling particles across cases, and are not radii around the ensemble mean. No best particle is selected. CV is a single point, so all three per-forecast quantiles equal its FDE before aggregation.

The diagnostics show why one score cannot stand for all prediction quality. At nominal 30 minutes, Obs 300 s has lower east/north CRPS than HSM-01 (687.11/524.07 versus 737.72/535.72 m), yet a slightly higher mean-position FDE (1335.73 versus 1326.09 m) and a larger mean within-forecast $q_{.95}$ (2827.57 versus 2346.62 m). RDM's corresponding $q_{.95}$ is 2138.12 m, not a promise that a 2138.12 m predictive disk has 95% target coverage. These means are descriptive; no per-horizon significance, general superiority or causal explanation is added, and the reported long-horizon undercoverage remains a limitation.

The complete `paper/pirc17/secondary-scores-v1/` supplement retains all 120 configuration/mode views and 480 repeated time views from the original 11368 score rows, including five configuration/mode views with failed required rows and null means. Seeds are averaged within origins and then equally across the same recording-hash blocks; CV is not repeated five times. No prediction, particle scoring or resampling is repeated. The anonymous projection binds source export `29cb892f`, audit `8ff917ad` and analysis `d1d67a3b`; actual recorded target-time ranges are retained, without certifying physical UTC or independent participants.

42 Local route illustrations withheld from public review

The owner's local manuscript retains three real-map cases at nominal 1, 5, 15 and 30 minutes. This public review projection intentionally withholds their route maps and case-level diagnostics pending privacy and map-publication permission. No missing illustration is replaced with synthetic data, and no new forecast is generated. The original complete local version is a separate review artifact; this projection does not validate the withheld evidence or establish population terrain effects.

Table 34: Saved marginal CRPS and within-forecast endpoint error quantiles (metres). Lower CRPS is better for that coordinate; two marginals do not determine joint calibration. Each stochastic configuration uses 46 recording-hash blocks and five seeds (230 forecasts); CV uses the same 46 blocks with one deterministic path each. Quantile entries are block/seed means of per-forecast particle error quantiles, not population FDE quantiles, pooled particles or a best-particle path. All targets retain their original recorded elapsed time; no per-horizon hypothesis test.

Model	Nominal min	CRPS east	CRPS north
HSM-01	1	32.71	31.55
HSM-01	5	157.30	153.31
HSM-01	15	390.26	373.30
HSM-01	30	737.72	535.72
RDM	1	32.58	31.51
RDM	5	149.27	142.28
RDM	15	384.50	367.09
RDM	30	736.79	549.33
Obs 300 s	1	37.20	34.78
Obs 300 s	5	147.70	144.49
Obs 300 s	15	368.85	354.71
Obs 300 s	30	687.11	524.07
CV	1	21.37	18.50
CV	5	137.90	160.72
CV	15	561.23	618.27
CV	30	1297.37	1356.32

Model	Mean FDE	Mean $q_{.50}$	Mean $q_{.90}$	Mean $q_{.95}$
HSM-01	1326.09	1564.98	2205.44	2346.62
RDM	1299.11	1419.34	1979.14	2138.12
Obs 300 s	1335.73	1663.28	2585.48	2827.57
CV	2095.78	2095.78	2095.78	2095.78

43 Construction of the saved population descriptions

The duration, speed and source-label tables describe the original candidate population, not a new eligibility screen. Their construction records are collected here so that the main text can focus on the restricted evaluation population and its scientific limitations.

Follow-up support before and after selection. Table 35 describes the same complete candidate population, not only successful forecasts. Follow-up is $D = t_{\text{last}} - t_{\text{origin}}$ for each original window. All 12,370 windows retain their original qualification and selection; invalid clock windows are neither dropped nor clipped. Equal-window descriptions do not make multiple windows of a recording independent participants.

Table 35: Recorded origin-to-last-observation follow-up spans (seconds). The first row is the denominator; the remaining four rows partition all windows. Brackets show interquartile ranges, not confidence intervals. Each original window contributes once.

Window group	Count	Median [Q25,Q75]	Min–max	$\geq$ 1800s
Released	12,370	69.8 [30,203]	2–4183	130
Temporal exclusions	12,243	68 [29,194]	2–2790	3
Coverage exclusions	21	2311 [2108,2559]	1800–3984	21
Eligible, unselected	60	2200 [1947,2570.2]	1816–4183	60
Selected origins	46	2323 [2034,2519]	1808–4143	46

Released windows have median follow-up of only 69.8 s, compared with 2,323 s (about 38.7 minutes) for selected origins. Only 130/12,370 candidates (about 1.05%) have at least 1,800 s of recorded follow-up. The original temporal criteria admit 127, so duration alone is insufficient: three long windows remain in the saved temporal-exclusion group. This statistic does not repair or reattribute those dispositions. Joint map validity excludes another 21 and the frozen selection retains 46 origins. The nominal 30-minute evaluation therefore conditions on a small subset of long, map-supported recordings, not general short-track deployment performance. It neither selects on observed performance nor changes any forecast horizon.

These are relative differences of retained clocks, not certification of physical UTC or actual acquisition durations. Longer follow-up does not extend simulation: forecasts still supply distributions at the original four targets; actual last targets beyond 1,800 s are documented separately in Table 17. Released history-window support in the descriptive file is not the retained last-two/three-point secant buffer. Saved speed is described separately below. Construction details and the complete stage summaries are in Appendix 43.

Saved speed before and after selection. We describe the source table's saved interval-speed scalar, not a velocity recomputed from release positions or clocks. For original window w,

$$s_w = \text{median}_{j=0,\ldots,n_w-1}\, s_{w,j}, \tag{46}$$

where n_w includes every point of the complete released observation window, including its future observations. Table 36 then gives equal-window descriptive quantiles of s_w, with linear quantile interpolation. It is neither a pooled-point nor a duration-weighted speed distribution, and multiple windows from one recording are not independent participants.

Source padding and the distinction from velocity-component norms are specified in Appendix audit record; all original windows remain in this descriptive statistic: no window is removed and no failed-stage distribution is replaced by a successful-subset estimate.

Table 36: Window-median saved interval speed, in m/s according to the retained construction script. Brackets are interquartile ranges, not confidence intervals. The last four rows partition the original denominator. Failures are failures of this descriptive statistic, not forecast failures.

Window group	Count	Median [Q25,Q75]	Min–max	Failed
Released	12,370	1.621 [1.191,1.949]	0.097–4.067	0
Temporal exclusions	12,243	1.626 [1.191,1.954]	0.097–4.067	0
Coverage exclusions	21	1.192 [0.999,1.269]	0.263–1.596	0
Eligible, unselected	60	1.460 [1.268,1.651]	0.569–2.436	0
Selected origins	46	1.486 [1.245,1.636]	0.831–2.554	0

The released and selected medians are 1.621 and 1.486 m/s, respectively. Together with the duration and geography descriptions, this characterizes the restricted evaluation population; it is not a significance test, a physical-speed calibration, or evidence that speed caused any forecast difference. Future observations enter this post-hoc description only, not predictor inputs or revised eligibility. The retained construction and builder scripts are identified separately but were not certified in the original release binding: their documented m/s interpretation does not independently establish physical measurement or historical preprocessing provenance. This is not a distribution of online secant inputs, simulated velocities or velocity outliers. The source correspondence and complete summaries are in Appendix 43.

Geographic scope of selection. Table 37 describes source-country labels for the same 12,370 windows under their original eligibility dispositions. Source-country coverage decreases from 63 (released) to 21 (temporal support), 14 (joint feature validity) and 12 (selected). The display includes all selected country codes and the two unselected countries with the largest released-window counts (HT and NI); remaining countries are pooled. This display rule does not inspect outcomes. Windows are counted first, with recording blocks deduplicated within each stage. A block can have both rejected and admitted windows, so rejection-block counts must not be added to admitted-block counts.

GB, BE and LU supply 33/46 selected origins (71.7%), compared with 493/1094 released recording blocks (45.1%). This describes a compositional change, not a significance test, a causal effect of map coverage or a performance difference. For example, HT contributes 1,731 windows but only two recording blocks: one window passes temporal support and none passes joint validity. None of NI's 265 windows (eight blocks) passes temporal support. Many windows therefore do not imply many independent samples. IT and SI also have eligible blocks but no selected origin; their absence is not a prediction failure.

Country/region names remain harvesting labels, not verified point locations or participant identities. The location metadata was not certified by the original release binding. Among released windows, 465 lack a harvest-area name but retain a country code; all remain in the country counts. Joint-valid and selected windows have no such missing label. All 63 country codes, nested stages and mutually exclusive window exclusions are available with the construction details in Appendix 43. These counts establish neither participant/near-route nor physical-time independence; they do not reweight results or resample intervals. The estimand still concerns the duration-, map-validity- and fixed-block-selected task population, not a representative survey of all 63 source countries.

Recorded-clock correspondence. Only identity and recorded-clock columns are read from bound feature files. Segment-offset bounds locate the origin and last point; original-file point

Table 37: Source-country label distributions through the original final-evaluation funnel. The first three count columns show windows (recording-hash blocks); the last shows selected origins. Nested stages are not additive and do not represent independent participants or country-specific performance.

Source code	Released	Temporal	Joint valid	Selected
AT	46 (9)	1 (1)	1 (1)	1
BE	523 (19)	30 (14)	30 (14)	10
DE	611 (71)	10 (8)	10 (8)	3
ES	570 (23)	6 (3)	3 (2)	2
FR	176 (27)	3 (3)	2 (2)	1
GB	4880 (451)	37 (29)	37 (29)	17
HT	1731 (2)	1 (1)	0 (0)	0
IE	104 (9)	2 (2)	1 (1)	1
LU	438 (23)	9 (7)	9 (7)	6
NI	265 (8)	0 (0)	0 (0)	0
NL	228 (22)	2 (1)	2 (1)	1
PL	24 (10)	3 (2)	3 (2)	2
RU	639 (60)	6 (6)	2 (2)	1
US	552 (68)	2 (2)	2 (2)	1
Other source countries	1583 (292)	15 (10)	4 (2)	0
Total	12370 (1094)	127 (89)	106 (73)	46

indexes are not segment offsets. All 1,125 used files match their bindings and row counts, and all 12,370 windows are retained, including invalid clock windows without clipping. HSM stage-specific history, follow-up and whole-window clock summaries and source bindings are retained in `paper/pirc17/final-cohort-duration-description-v1.json`. This clock-only description decodes no positions, speeds, map-feature values or prediction arrays. Released history-window support is not the retained last-two/three-point predictor buffer.

Saved-scalar correspondence. Source relative time is used only for the original parent-ordinal correspondence, not to estimate speed using the release clock. Both source and alignment files match their original bindings. Ranking precedes refined-window selection; complete parent counts, unique order and per-window identity/point coverage are checked. HSM stage summaries and source identities are retained in `paper/pirc17/final-cohort-speed-description-v1.json`. No coordinates, centered velocity components, forecast arrays or scores are read. The retained construction and builder scripts were not certified in the original release binding; file correspondence does not certify physical measurement or historical preprocessing provenance.

Source-label correspondence and scope. Every original saved disposition is joined to released sample identities, the original recording–harvest-group mapping and the same source-location metadata, without reading coordinates, speeds, prediction errors or map pixels. All 63 country codes, nested stages, mutually exclusive window exclusions and source bindings are retained in `paper/pirc17/final-cohort-geography-description-v1.json`. These three descriptions change no qualification or selection and involve no fitting, clipping, simulation, scientific rescoring or resampling. They neither replace the independent saved-output audit nor establish participant, near-route or physical-time independence.

44 Scientific names and implementation inventory

The frozen trajectory cohort, feature specification and representation/capacity selection are internally named PIRC-20, PIRC-21 and PIRC-22. The method inventory is named NEX326; these identifiers name retained research records, not additional model families. A slot means one configuration binding.

The NEX326 design has 22 conceptual arms and 36 execution slots. The registered exemption for arms 13, 17 and 22 covers eight slots, leaving 28 required slots. All 36 entries retain a disposition and source. Exemption is an administrative scope decision, not evidence of a harmful or ineffective component.

Table 38 lists the complete design inventory. A slot identifier combines its arm number with the subconfiguration label. The counts are configuration counts, not successful forecasts or independent replications. In arm 6, `dt` labels denote the observation/training and history cadence in seconds, not the numerical step or forecast duration. The independent ten-configuration terrain study remains required even though the historical arm-17 condition variants are exempt.

Table 38: All 22 arms and 36 registered slots. Roles describe the frozen design, not empirical outcomes. The six HSM anchors, 21 candidates and one identical-configuration replay constitute the 28 required slots; the other eight retain approved EXCLUDED dispositions.

Arm	Subconfiguration labels	Registered role	Slots
1	`full`	HSM anchor: persistent modes	1
2	`Heading/redraw`	Candidate against arm 1	1
3	`Single Gaussian`	Candidate against arm 1	1
4	`RDM`	Candidate against arm 1	1
5	`Speed/radial`	Candidate against arm 1; explicit features	1
6	`Obs 30 s`, `Obs 60 s`, `Obs 120 s`, `Obs 300 s`, `Obs 600 s`	Four candidates against arm 1; `Obs 60 s` repeats HSM	5
7	`full`	HSM anchor: objective and score	1
8	`QMLE label`	Candidate against arm 7	1
9	`mixed`, `ES calibration`	Two candidates against arm 7	2
10	`2D MC score`, `2D closed score`	Two score-route candidates against arm 7	2
11	`full`	HSM anchor: training and adaptation	1
12	`scratch`	Candidate against arm 11	1
13	`animal_pretrain`	Approved EXCLUDED; no transfer-effect claim	1
14	`reptile`	Candidate against arm 11; first-order initialization	1
15	`Drift-only`, `Two-step`	Two candidates against arm 11; implementation limits apply	2
16	`full`	HSM anchor: solar conditioning held fixed	1
17	`uncond`, `is_day`, `weather`, `terrain`	Approved EXCLUDED; not the separate terrain study	4
18	`full`	HSM anchor: split integration	1
19	`em`, `euler`	Two aliases against arm 18; same update	2
20	`full`	HSM anchor: conditional Gaussian moments	1
21	`mc`, `crn`	Two propagation candidates against arm 20	2
22	`doob`, `sb`, `soft_endpoint`	Approved EXCLUDED; no bridge-effect claim	3

45 Saved fitting population and development diagnostics

We read the 16 saved method fits and ten saved terrain fits, checking the original file/content hashes and inventory bindings without fitting or forecasting again. Table 39 gives actual retained method transitions after resampling and removal of incomplete tails. All intervals use the same 328 training, 76 adaptation and 81 validation windows; transition totals are not independent participant counts. These intervals are fitting intervals τ, not simulation steps h or a change to the registered forecast duration. The original scoring observations are unchanged by fitting resampling.

Table 39: Retained method-fitting transitions from the saved records. Each row is per fit at the indicated interval, not a sum over method slots.

τ (s)	Training	Adaptation	Validation
30	52,170	11,567	12,340
60	26,009	5,762	6,149
120	12,918	2,861	3,055
300	5,079	1,124	1,196
600	2,467	545	578

The HSM fit's recorded training-sample label is 31,771, the sum of its 26,009 training and 5,762 adaptation transitions. Scratch records only the 5,762 adaptation transitions. Reptile records 111,887 algorithm exposures, which include repeated uses of data; this is not 111,887 unique observations or additional trajectories. Likewise, 28 prediction configurations reuse 16 method parameter fits rather than constituting 28 independently trained algorithms. The saved selected covariance multiplier is 1 for HSM and 1.7 for the RDM-kernel fit; mixed and pure-ES record a selected drift fraction of 0.5. These are recovered fit/calibration settings, not new validation choices or final-evaluation tuning.

All ten terrain fits use 404 original first-future training transitions and 81 validation transitions, without method-style resampling. Their shared baseline rate MSE is 0.145796 on training and 0.077077 on validation, in m^2/s^2. Table 40 reports the correction fits, not final position errors or energy scores.

Table 40: Saved terrain development diagnostics. MSE is rate error (m^2/s^2); κ_{aug} is the condition number of the regularized augmented solver, and the last column gives the smallest/largest eigenvalues of Q (m^2/s).

Configuration	Columns	Train MSE	Validation MSE	κ_{aug}	Q eigenvalues
B (history)	4	0.145796	0.077077	173.2	1.0898 / 1.6489
A (all groups)	56	0.136666	0.079117	1047.9	1.0404 / 1.4673
B + road	14	0.142237	0.078151	513.7	1.0676 / 1.5927
B + river	14	0.144073	0.078681	756.7	1.0801 / 1.5734
B + cover	12	0.144966	0.078132	277.5	1.0843 / 1.6408
B + surface	20	0.143452	0.076193	346.5	1.0737 / 1.5987
A - road	38	0.140826	0.080224	840.8	1.0646 / 1.5094
A - river	46	0.138394	0.076226	768.0	1.0477 / 1.5316
A - cover	40	0.138083	0.078583	932.8	1.0403 / 1.4771
A - surface	40	0.139445	0.078089	1005.6	1.0624 / 1.5210

The full-terrain correction lowers training MSE by about 6.3% but raises validation MSE by about 2.6% relative to the shared baseline. This cautions against equating improved training fit

with improved out-of-sample forecasts; it neither establishes overfitting as the cause nor identifies a terrain group's final contribution. Diffusion eigenvalues are positive in all ten fits, and recorded relative gradient norms range from 3.88×10^{-17} to 5.13×10^{-14}. The augmented ridge ranks equal the number of correction columns plus the intercept. Positive ridge penalties make this possible even for a rank-deficient raw design, so neither full augmented rank nor these stationarity checks exclude redundant constant mask columns, scale imbalance, or extrapolation at predicted map locations. The saved fit records themselves do not provide raw-column ranks, mask-validity rates or scale summaries. We therefore recover those separately from their original input rows below, rather than inferring them from solver success. The route-free saved-fit projection is `paper/pirc17/fit-diagnostics-v1.json`. It preserves source hashes but exports no fitted coefficient matrices, trajectory identifiers or private paths. These development-fit diagnostics are separate from the completed saved-forecast audit and do not authorize a final terrain-effect claim.

Original input design, masks and regularization. We read the original 404 training and 81 validation origin rows from 397 existing feature files, verifying selected-file hashes against the original training identity. The ten saved fits bind that exact original input identity; its development sources and column layouts also match the restored continuation. The unchanged frozen Heading/redraw transforms reconstruct the numeric-plus-mask design without map queries, refitting, forecasting, or final-evaluation numeric reads. This is a narrow source/design description, not another validation of the entire 7,618-file snapshot.

All 28 full-model numeric columns are valid at every training and validation origin, so every selected mask is a constant-one column in these two designs. Table 41 reports the actual unregularized design, including masks and the solver intercept. Rank uses float64 SVD with tolerance $\max(n, p)\epsilon_{64}s_{\max}$. Every design is deficient; its full condition number is not reported as finite. The displayed $\kappa_+ = s_{\max}/s_{\text{smallest retained}}$ concerns only the nonzero numerical spectrum, not an invertible raw matrix.

Table 41: Reconstructed original terrain-origin designs. K is also the number of constant-one masks; raw width is $2K + 1$. Training/validation have 404/81 rows. These ranks differ from augmented ridge ranks.

Configuration	K	Raw width	Train rank	Validation rank	Train κ_+
B (history)	2	5	3	3	2.5
A (all groups)	28	57	27	25	808.1
B + road	7	15	8	8	17.1
B + river	7	15	8	8	75.7
B + cover	6	13	6	6	28.8
B + surface	10	21	11	11	265.9
A - road	19	39	19	19	644.8
A - river	23	47	22	20	586.9
A - cover	20	41	21	21	720.3
A - surface	20	41	19	17	467.6

Constant masks and the penalized intercept are redundant. Restricting attention to their constant subspace, a constant correction c can be distributed over $K+1$ equally penalized coefficients:

$$\min_{\sum_{j=1}^{K+1} a_j = c} \sum_{j=1}^{K+1} a_j^2 = \frac{c^2}{K+1}. \tag{47}$$

Thus its ridge term is $\lambda c^2/\{2(K+1)\}$: the base and full mask-and-bias subspaces have denominators 3 and 29 despite the same λ. Other feature dependencies can also affect the complete coefficient representation; this equation is not an exact reduction of the entire fitted model to one scalar penalty. The frozen terrain comparisons therefore change feature information, dimension and regularized representation together. They do not isolate the information in a terrain group under an otherwise identical effective constant penalty.

Input scales also differ: training population standard deviations are 0.0387 for the upward normal n_u, 0.1274 for slope β, 2.0781 for road log-distance, and 5.2308 for the east component of river log-distance times direction. They use the original representations without new centering or scale fitting. HSM per-column minima, maxima, means, standard deviations and mask frequencies for both roles are saved in `paper/pirc17/feature-design-description-v1.json`. This confirms the constant-mask concern, but neither quantifies its contribution to forecast differences nor proves that scaling caused a failure. No masks, penalties, parameters or experimental outcomes are changed after inspecting these diagnostics. Validity at observed development origins does not guarantee validity at future simulated map queries.

46 Complete saved five-seed differences

The table retains every original primary-method comparison and its named control; no terrain, secondary-origin or failed-family subset is added. Sign counts use the full saved values, without rounding, a zero tolerance or a new practical-effect threshold. The range of five means is not a confidence interval. Original block-level intervals remain in Table 25; the full-precision, route-free projection is `paper/pirc17/method-seed-stability-v1.json`.

Table 42: Original five simulation-seed mean differences in weighted ES (m). Each entry averages the same 46 recording blocks. S1–S5 are 20260814–20260818 in order; the final column counts negative/positive/exact-zero entries before display rounding. Negative favours the candidate against its named HSM reference. Scientific notation e-5 means times ten to the power minus five. These are not confidence intervals or five independent studies.

Candidate	Control	S1	S2	S3	S4	S5	−/+/0
Model structure							
Heading/redraw	HSM-01	32.60	29.70	33.89	33.83	34.75	0/5/0
Single Gaussian	HSM-01	21.65	19.02	21.72	23.03	24.64	0/5/0
RDM	HSM-01	-5.00	-4.27	-1.08	-2.40	-1.39	5/0/0
Speed/radial	HSM-01	57.28	56.49	58.30	58.59	59.13	0/5/0
Observation interval							
30 s observations	HSM-01	8.39	6.93	7.24	8.72	11.54	0/5/0
120 s observations	HSM-01	-10.76	-14.89	-10.32	-11.07	-10.74	5/0/0
300 s observations	HSM-01	-22.48	-25.66	-25.48	-20.08	-23.32	5/0/0
600 s observations	HSM-01	-22.60	-24.92	-21.92	-28.96	-25.05	5/0/0
Objective / score							
QMLE label	HSM-07	1.90	-0.41	1.66	0.58	-2.35	2/3/0
Mixed calibration	HSM-07	-4.13	-9.30	-4.58	-3.96	-10.78	5/0/0
ES calibration	HSM-07	-4.40	-6.13	-5.00	-1.56	-9.49	5/0/0
2D MC score	HSM-07	0	0	0	0	0	0/0/5
2D closed score	HSM-07	0	0	0	0	0	0/0/5
Training / adaptation							
Scratch	HSM-11	2.33	-0.43	3.20	5.58	2.33	1/4/0
Reptile	HSM-11	2.52	-1.39	2.61	3.88	3.36	1/4/0
Drift-only	HSM-11	0.50	-1.12	-0.81	3.84	0.11	2/3/0

Table 42 (continued)

Candidate	Control	S1	S2	S3	S4	S5	$-/+/0$
Two-step	HSM-11	-1.06	-1.90	-1.82	1.40	3.19	3/2/0
Numerical propagation							
Euler-M.	HSM-18	0.02542	0.02503	0.02542	0.02621	0.02519	0/5/0
Euler alias	HSM-18	0.02542	0.02503	0.02542	0.02621	0.02519	0/5/0
MC	HSM-20	-0.19	-1.77	2.03	-0.93	-3.04	4/1/0
CRN	HSM-20	-5.06e-5	-6.80e-5	-6.10e-5	-6.47e-5	-3.74e-5	5/0/0

47 Saved resampling tail diagnostics

The following unchanged table and its caption describe the earlier snapshot. The current completed mechanism-analysis status is disclosed in Section 38; independent saved-output and fixed-budget aggregate replay are complete. The historical caption describes its captured stage, not today's mechanism or audit status. Second-stream endpoints and sampling indices absent from the historical projection are not fabricated.

Table 43: Historical tail diagnostics for all 21 method contrasts. The saved-output audit and public paired replay are now complete; mechanisms remain unevaluated. The original two-stream maximum endpoint shift (m) is retained: full second-stream endpoints were not saved in this projection and are not independently reconstructed here. T denotes finite intervals and shift within one quarter of the practical margin, not scientific acceptance.

Candidate	p_0	p_{Holm}	Shift (m)	Stable
`Heading/redraw`	0.00050	0.00200	0.565	T
`Single Gaussian`	0.16392	0.32784	3.4	T
`RDM`	0.71114	0.71114	1.81	T
`Speed/radial`	0.01799	0.05397	0.647	T
`Obs 30 s`	0.00050	0.00200	0.22	T
`Obs 120 s`	0.00100	0.00300	0.662	T
`Obs 300 s`	0.01149	0.02299	1.01	T
`Obs 600 s`	0.11594	0.11594	3.04	T
`QMLE label`	0.76412	1.00000	0.223	T
`Mixed calibration`	0.10995	0.54973	0.278	T
`ES calibration`	0.20990	0.83958	0.455	T
`2D MC score`	1.00000	1.00000	0	T
`2D closed score`	1.00000	1.00000	0	T
`Scratch`	0.40630	1.00000	0.0682	T
`Reptile`	0.63518	1.00000	1.47	T
`Drift-only`	0.49975	1.00000	0.02	T
`Two-step`	0.96852	1.00000	0.11	T
`Euler-M.`	0.00700	0.02799	0.00172	T
`Euler alias`	0.00700	0.02799	0.00172	T
`MC`	0.36532	0.36532	0.146	T
`CRN`	0.08346	0.16692	9.58×10^{-6}	T

48 Saved method probability and diffusion parameters

Table 44 lists the original 16 parameter fits reused by 28 prediction configurations. Mode probabilities follow the saved index order, not matched behavioural categories across fits or regions. Single-mode rows have probability one. Display rounding is not a renormalization. The last column is the minimum and maximum eigenvalue over all modes of that fit, not the eigenvalues of a pooled predictive covariance. The full projection keeps both R_m and Q_m spectra for every mode,

including the PSD tolerance and zero recorded negative-roundoff repairs. The method fitting and noise-embedding sources match their saved bindings; no intermediate fit is reconstructed and no forecast is rescored. The missing per-role frequencies limit inspection of sparse modes and cannot be replaced by balanced segment counts or blended probabilities.

Table 44: Original saved method parameters. τ is the fitting interval in seconds; π lists final mode probabilities; the Q envelope is in m^2/s. These are not forecast scores or sample counts.

Representative fit	τ (s)	π (index order)	Q eigenvalue envelope
`HSM-01`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470
`Heading/redraw`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470
`Single Gaussian`	60	1.0000	45.529 / 51.942
`RDM`	60	0.3112, 0.2792, 0.4096	30.496 / 104.088
`Speed/radial`	60	1.0000	43.828 / 48.535
`Obs 120 s`	120	0.3348, 0.3314, 0.3339	62.597 / 99.070
`Obs 30 s`	30	0.3349, 0.3301, 0.3350	19.034 / 28.435
`Obs 300 s`	300	0.3350, 0.3300, 0.3350	123.274 / 216.263
`Obs 600 s`	600	0.3437, 0.3336, 0.3227	214.844 / 353.666
`QMLE label`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470
`Mixed calibration`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470
`ES calibration`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470
`Scratch`	60	0.3402, 0.3299, 0.3299	34.148 / 54.994
`Reptile`	60	0.3547, 0.3329, 0.3124	35.552 / 58.359
`Drift-only`	60	0.3350, 0.3302, 0.3348	36.948 / 52.900
`Two-step`	60	0.3350, 0.3302, 0.3348	35.195 / 53.470

49 Original Reptile task accounting

Source-field grouping rule. The adapter first strips the source `city` string and uses it unless empty or case-insensitive `nan`; otherwise it uses the unstripped `region` string or `unknown`. Groups use exact strings, without a country key or case normalization. All 328 training segments use the city branch, but these are harvesting labels, not certified cities, administrative regions or independent participants. The saved algorithm and parameter-flow description is `paper/pirc17/method-algorithm-description-v1.json`; the task-membership and exposure correspondence is retained below.

Table 45 retains every eligible task, in the same lexicographic label order as the executed algorithm. T01–T30 are anonymous display keys, not geographic or participant identities. Source labels are constant within each of the 266 used training recordings; all 328 training segments use the city-first branch. Exact label strings define 84 groups: 30 eligible groups with 258 segments and 20,029 transitions, and 54 smaller groups with 70 segments and 5,980 transitions. Smaller groups are omitted from task visits only, not from the global initialization's 328 segments and 26,009 transitions.

Table 45: All original eligible Reptile tasks. Each task is visited once in each of four epochs. Mode columns are deterministic within-task heading-rank segment cardinalities, not transition frequencies or final mixture probabilities.

Task	Segments	Transitions/pass	Mode 0	Mode 1	Mode 2
T01	3	177	1	1	1
T02	7	610	3	2	2
T03	3	333	1	1	1
T04	75	5,324	25	25	25
T05	33	2,476	11	11	11
T06	4	311	2	1	1
T07	4	279	2	1	1
T08	3	203	1	1	1
T09	4	312	2	1	1
T10	5	342	2	2	1
T11	10	1,012	4	3	3
T12	4	321	2	1	1
T13	5	575	2	2	1
T14	21	1,628	7	7	7
T15	3	271	1	1	1
T16	3	239	1	1	1
T17	3	212	1	1	1
T18	3	253	1	1	1
T19	5	467	2	2	1
T20	5	400	2	2	1
T21	3	285	1	1	1
T22	9	687	3	3	3
T23	3	279	1	1	1
T24	4	350	2	1	1
T25	8	670	3	3	2
T26	3	188	1	1	1
T27	5	349	2	2	1
T28	11	834	4	4	3
T29	5	374	2	2	1
T30	4	268	2	1	1
Total	258	20,029	94	86	78

Eligible task sizes range from 3 to 75 segments and from 177 to 5,324 transitions per visit. Ten tasks have only three segments. Equal visits and the same outer step do not weight updates by task transition counts, nor imply equal net influence: updates are sequential and task-dependent. The 120 task visits and their repeated residual diagnostics are not 120 independent held-out studies. Per-task rank counts do not recover the within-mode transition frequencies, direction labels or RDM residual groups.

The retained training counter reconciles as $26{,}009 + 4 \times 20{,}029 + 5{,}762 = 111{,}887$. It counts one task transition set per visit, not every reuse by five inner steps or validation calibration; it is neither total processing cost nor the number of distinct observations. No raw place labels, member identities or coefficients are exported. Source-only label columns and original saved accounting suffice for this description, without resampling, residual reconstruction or a new fit. The source-bound record is `paper/pirc17/reptile-task-description-v1.json`. Pre-meta/pre-target parameters and actual multimode/RDM transition labels remain unavailable.

50 Original terrain fitting intervals

Table 46 describes the recorded origin-to-first-future gaps used by all ten terrain fits. Each window contributes one transition; 404 training rows estimate drift and diffusion, while 81 validation rows evaluate the saved development fit and do not enter Equation (6). These short fitting transitions are not the 30-minute forecast horizon. Quantiles use linear interpolation of ordered gaps and are descriptive, not confidence intervals or evidence of independent participants.

Table 46: Original terrain development intervals in seconds. IQR gives the 25th–75th percentiles; no clock, trajectory or fitted parameter is changed.

Role	Rows	Minimum	Median [IQR]	Mean	Maximum
Training	404	1.00	3.00 [1.00, 6.00]	5.51	38.00
Validation	81	1.00	2.00 [1.00, 8.00]	6.63	31.00

The description reads clock and identity columns of selected development segments in the same 397 feature files as the original design diagnostics. It uses adjacent segment offsets, not original-file row numbers as offsets, and subtracts recorded integer nanoseconds before conversion to seconds. Only two ticks per transition enter the summary. Coordinates, rate residuals, maps and final forecast arrays are not decoded. The route-free projection `paper/pirc17/terrain-fit-scope-description-v1.json` binds the original development population, fitting sources and ten unchanged saved Q spectra. It neither re-estimates residual covariance nor certifies physical clock provenance.

51 Map coordinate, sampling and validity details

For longitude and latitude in degrees, the forecast frame is anchored at the visible origin (ℓ_0, φ_0). With $k = \pi r_\oplus/180$, $r_\oplus = 6371008.8$ m, it uses

$$x_e = k\cos(\pi\varphi_0/180)(\ell - \ell_0), \qquad x_n = k(\varphi - \varphi_0). \tag{48}$$

The cosine converts degrees to radians internally. Online vector geometry uses the same scale formula but anchors it at the centre of the containing 3-degree cell, not at the forecast origin or a future-route mean. Shapely proposes candidate lines and DuckDB resolves projected distance and WKT ties, with a 500000 m projected cutoff. Overture road classes bridleway, cycleway, footway, path, pedestrian, steps and track are excluded from this road channel. Only registered Overture cell files and intersecting HydroRIVERS archives are loaded; there is no neighbouring-cell expansion or online download. The cutoff is not a guarantee of a globally closest feature within 500 km.

GeoTIFF queries transform EPSG:4326 coordinates into the raster CRS, apply the inverse affine, and subtract half a pixel to obtain height pixel-centre coordinates. SRTM HGT is decoded as a square big-endian int16 grid; its fractional coordinates use side -1 samples per degree, with no GeoTIFF half-pixel shift. SRTM values at or below -30000 m become missing; GeoTIFF masks and non-finite heights become missing. A bilinear evaluation requires all four neighbouring samples to be finite and in the same tile, even when one interpolation weight is zero. No edge halo or adjacent-tile stitching repairs a missing stencil.

For geographic rasters, the metre-per-degree scales at query latitude φ, writing $\psi = \pi\varphi/180$ in radians, are

$$\begin{aligned} m_N &= 111132.92 - 559.82\cos(2\psi) + 1.175\cos(4\psi) - 0.0023\cos(6\psi), \\ m_E &= \left|111412.84\cos\psi - 93.5\cos(3\psi) + 0.118\cos(5\psi)\right|. \end{aligned} \tag{49}$$

GeoTIFF row/column spacing multiplies these scales by absolute affine increments; SRTM divides by side -1. At fractional raster row r, column c, and metre spacings h_N, h_E, the gradient is

$$\gamma_e = \frac{H(r, c+1) - H(r, c-1)}{2h_E}, \qquad \gamma_n = \frac{H(r-1, c) - H(r+1, c)}{2h_N}. \tag{50}$$

Each H is bilinearly evaluated. Surface validity also requires all nine evaluations at row/column offsets $-1, 0, 1$, because its common status checks slope, aspect, normal, curvature and roughness, including quantities not selected as conditioner columns. The separately computed 250 m local elevation neighbourhood is not a selected direct terrain input. A gradient magnitude at most 10^{-12} gives zero aspect components, not a physically identified flat-terrain direction.

WorldCover transforms to raster CRS and floors inverse-affine row/column coordinates without height's half-pixel shift. Its masked or zero classes are invalid, not valid unknown-cover observations. Missing source coverage or no admitted line leaves the corresponding status missing; raster-edge or NoData stencils do not become flat zero-height terrain. The encoder then casts canonical numeric fields to float32, applies the frozen transforms, zero-imputes invalid numeric columns and supplies their explicit masks. These are implementation rules, not measured deployment missingness rates. The route-free `paper/pirc17/map-provider-description-v1.json` records receipt declarations, whole-catalog counts and exact source bindings; it exports no per-tile extent, trajectory identity or private map-use trace.

52 Complete saved method tables

52.1 Complete absolute method values

Table 47 retains all 28 execution identities from the same pinned stage snapshot. Figure 12 omits only the six extra HSM/replay roles from its display, not their predictions from the source. They are neither averaged nor assumed to have equal finite scores. Every row uses all 46 recording blocks and five forecast seeds (230 predictions); these are not 230 independent targets. Mean values are rounded for display, with full precision retained in `paper/pirc17/preliminary-method-statistics-v1.json`. Lower is better for each column, but ES, ADE and FDE are different quantities. This is not an absolute-score confidence interval or a global best-model test. Saved parameter equality and different random streams remain relevant (Table 3).

Table 47: All 28 primary-method absolute stage means in metres; no new forecasts or statistical inference.

Execution identity	Weighted ES	ADE	FDE
`HSM-01`	477.9	636.1	1326.1
`Heading/redraw`	510.8	635.2	1324.2
`Single Gaussian`	499.9	622.9	1316.4
`RDM`	475.0	622.0	1299.1
`Speed/radial`	535.8	647.6	1403.1
`Obs 30 s`	486.4	631.8	1317.6
`Obs 120 s`	466.3	638.7	1329.8
`Obs 300 s`	454.4	644.4	1335.7
`Obs 600 s`	453.2	651.5	1340.9
`QMLE label`	477.0	635.6	1325.1
`Mixed calibration`	470.1	630.3	1315.7
`ES calibration`	471.4	631.4	1318.5

Continued on the next page

Table 47 (continued)

Execution identity	Weighted ES	ADE	FDE
`2D MC score`	476.7	634.8	1322.1
`2D closed score`	476.7	634.8	1322.1
`Scratch`	478.6	640.1	1339.0
`Reptile`	478.2	630.0	1320.8
`Drift-only`	476.5	634.4	1320.6
`Two-step`	475.9	634.3	1322.4
`Euler-M.`	477.2	635.2	1325.0
`Euler alias`	477.2	635.2	1325.0
`MC`	475.5	633.7	1320.0
`CRN`	476.3	634.2	1320.9
Additional HSM/replay identities, omitted only from the dot figure			
`Obs 60 s`	477.9	636.1	1326.1
`HSM-07`	476.7	634.8	1322.1
`HSM-11`	476.0	634.0	1320.7
`HSM-16`	475.9	633.9	1321.2
`HSM-18`	477.2	635.2	1325.0
`HSM-20`	476.3	634.2	1320.9

52.2 Complete method horizon profiles

Every original primary configuration is shown separately, including the six extra reference/replay roles and both score-only routes. Each cell is marginal ES in metres / nominal 90% disk coverage in percent, over the same 46 blocks and five simulation seeds. These are descriptive means, not paired tests, confidence intervals or independent algorithms. Target slots retain the original observation-time tolerance; equal display rounding does not certify equal full-precision scores.

Table 48: All 28 saved primary configuration profiles. Cells: ES (m) / 90% disk coverage (%). ES lower is better; coverage should be interpreted against 90% and region size, not maximized alone. Each row uses 230 saved scores, not 230 independent targets.

Configuration	1 min	5 min	15 min	30 min
Model structure / HSM-01				
HSM-01	50.57 / 95.65	243.23 / 37.83	609.27 / 35.22	1008.35 / 46.52
Heading/redraw	50.37 / 95.65	249.10 / 24.78	652.09 / 10.87	1091.67 / 21.74
Single Gaussian	48.61 / 95.65	237.81 / 31.74	627.23 / 17.83	1085.82 / 32.61
RDM	50.35 / 96.96	228.87 / 66.52	598.70 / 34.78	1022.19 / 45.22
Speed/radial	48.21 / 91.30	239.20 / 32.17	658.21 / 13.48	1197.62 / 21.74
Observation interval / HSM-01				
30 s observations	51.36 / 72.61	251.36 / 23.04	621.77 / 24.35	1021.17 / 38.26
60 s (HSM-01 replay)	50.57 / 95.65	243.23 / 37.83	609.27 / 35.22	1008.35 / 46.52
120 s observations	51.53 / 98.26	233.98 / 70.87	592.55 / 54.78	987.12 / 56.96
300 s observations	56.58 / 100.00	229.28 / 96.52	574.53 / 71.74	957.40 / 64.35
600 s observations	63.78 / 100.00	234.32 / 99.13	572.65 / 92.61	941.89 / 79.13
Objective / score / HSM-07				
HSM-07	50.52 / 95.65	242.28 / 37.83	608.76 / 33.48	1005.15 / 45.65
QMLE label	50.33 / 95.65	242.14 / 38.70	608.85 / 34.35	1006.48 / 46.52
Mixed calibration	50.16 / 95.65	238.69 / 40.43	600.36 / 34.78	991.29 / 46.52
ES calibration	50.09 / 95.65	238.94 / 39.13	601.74 / 35.22	994.67 / 45.65
2D MC score	50.52 / 95.65	242.28 / 37.83	608.76 / 33.48	1005.15 / 45.65
2D closed score	50.52 / 95.65	242.28 / 37.83	608.76 / 33.48	1005.15 / 45.65
Training / adaptation / HSM-11				
HSM-11	50.37 / 95.65	242.17 / 38.70	607.73 / 34.78	1003.68 / 46.52
Scratch	50.65 / 95.65	242.97 / 40.43	607.79 / 37.83	1012.95 / 46.52

Table 48 (continued)

Configuration	1 min	5 min	15 min	30 min
Reptile	50.15 / 95.65	239.46 / 37.39	602.72 / 33.04	1020.39 / 41.74
Drift-only	50.57 / 95.65	242.58 / 39.57	608.70 / 34.35	1004.12 / 46.96
Two-step	50.39 / 95.65	241.93 / 37.83	606.76 / 35.65	1004.71 / 46.96
Numerical / HSM-18, HSM-20				
HSM-18	50.48 / 95.65	242.54 / 38.26	608.30 / 32.61	1007.57 / 45.65
Euler-M. (HSM-18 control)	50.48 / 95.65	242.56 / 38.26	608.34 / 32.61	1007.61 / 45.65
Euler alias	50.48 / 95.65	242.56 / 38.26	608.34 / 32.61	1007.61 / 45.65
HSM-20	50.46 / 95.65	242.39 / 37.83	607.85 / 33.48	1004.44 / 46.96
MC (HSM-20 control)	50.48 / 95.65	242.05 / 36.52	607.04 / 34.35	1002.46 / 46.09
CRN (HSM-20 control)	50.46 / 95.65	242.39 / 37.83	607.85 / 33.48	1004.44 / 46.96
Additional reference (not a family)				
HSM-16 (extra reference)	50.29 / 95.65	242.06 / 37.83	607.10 / 33.48	1004.17 / 45.65

52.3 Complete primary predictive-region sizes

Table 49: All 28 original primary configurations. Each cell: mean radius (m) / mean area (km^2) / empirical coverage (%) of the nominal 90% marginal disk. Each row uses the same 46 blocks and five seeds (230 saved scores), not 230 independent targets. Area averages individual disk areas; it is not the area of the mean radius. Descriptive only; all four probability levels remain in the saved aggregate.

Configuration	1 min	5 min	15 min	30 min
Model structure / HSM-01				
HSM-01	118 / 0.044 / 95.7	305 / 0.294 / 37.8	665 / 1.415 / 35.2	1184 / 4.534 / 46.5
Heading/redraw	117 / 0.043 / 95.7	268 / 0.226 / 24.8	482 / 0.731 / 10.9	720 / 1.629 / 21.7
Single Gaussian	116 / 0.042 / 95.7	267 / 0.224 / 31.7	492 / 0.760 / 17.8	765 / 1.840 / 32.6
RDM	151 / 0.072 / 97.0	356 / 0.400 / 66.5	648 / 1.321 / 34.8	990 / 3.079 / 45.2
Speed/radial	113 / 0.040 / 91.3	253 / 0.201 / 32.2	441 / 0.611 / 13.5	630 / 1.247 / 21.7
Observation interval / HSM-01				
30 s observations	89 / 0.025 / 72.6	244 / 0.190 / 23.0	567 / 1.039 / 24.3	1052 / 3.619 / 38.3
60 s (HSM-01 replay)	118 / 0.044 / 95.7	305 / 0.294 / 37.8	665 / 1.415 / 35.2	1184 / 4.534 / 46.5
120 s observations	157 / 0.077 / 98.3	386 / 0.469 / 70.9	797 / 2.019 / 54.8	1363 / 5.969 / 57.0
300 s observations	222 / 0.155 / 100.0	528 / 0.879 / 96.5	1036 / 3.395 / 71.7	1693 / 9.109 / 64.3
600 s observations	283 / 0.252 / 100.0	666 / 1.395 / 99.1	1267 / 5.058 / 92.6	1999 / 12.609 / 79.1
Objective / score / HSM-07				
HSM-07	118 / 0.044 / 95.7	303 / 0.291 / 37.8	664 / 1.410 / 33.5	1184 / 4.537 / 45.7
QMLE label	118 / 0.044 / 95.7	304 / 0.293 / 38.7	666 / 1.418 / 34.3	1187 / 4.558 / 46.5
Mixed calibration	118 / 0.044 / 95.7	302 / 0.287 / 40.4	661 / 1.378 / 34.8	1187 / 4.455 / 46.5
ES calibration	117 / 0.043 / 95.7	302 / 0.287 / 39.1	662 / 1.380 / 35.2	1184 / 4.429 / 45.7
2D MC score	118 / 0.044 / 95.7	303 / 0.291 / 37.8	664 / 1.410 / 33.5	1184 / 4.537 / 45.7
2D closed score	118 / 0.044 / 95.7	303 / 0.291 / 37.8	664 / 1.410 / 33.5	1184 / 4.537 / 45.7
Training / adaptation / HSM-11				
HSM-11	118 / 0.044 / 95.7	304 / 0.293 / 38.7	665 / 1.414 / 34.8	1183 / 4.535 / 46.5
Scratch	119 / 0.045 / 95.7	311 / 0.306 / 40.4	687 / 1.525 / 37.8	1237 / 5.021 / 46.5
Reptile	119 / 0.044 / 95.7	301 / 0.286 / 37.4	645 / 1.334 / 33.0	1141 / 4.227 / 41.7
Drift-only	118 / 0.044 / 95.7	304 / 0.293 / 39.6	665 / 1.415 / 34.3	1184 / 4.535 / 47.0
Two-step	118 / 0.044 / 95.7	305 / 0.294 / 37.8	666 / 1.418 / 35.7	1187 / 4.561 / 47.0
Numerical / HSM-18, HSM-20				
HSM-18	118 / 0.044 / 95.7	304 / 0.292 / 38.3	663 / 1.407 / 32.6	1186 / 4.551 / 45.7
Euler-M. (HSM-18 control)	118 / 0.044 / 95.7	304 / 0.292 / 38.3	663 / 1.406 / 32.6	1185 / 4.549 / 45.7
Euler alias	118 / 0.044 / 95.7	304 / 0.292 / 38.3	663 / 1.406 / 32.6	1185 / 4.549 / 45.7
HSM-20	118 / 0.044 / 95.7	305 / 0.295 / 37.8	665 / 1.415 / 33.5	1186 / 4.553 / 47.0
MC (HSM-20 control)	118 / 0.044 / 95.7	303 / 0.291 / 36.5	663 / 1.407 / 34.3	1182 / 4.526 / 46.1
CRN (HSM-20 control)	118 / 0.044 / 95.7	305 / 0.295 / 37.8	665 / 1.415 / 33.5	1186 / 4.553 / 47.0

Table 49 (continued)

Configuration	1 min	5 min	15 min	30 min
Additional reference (not a family)				
HSM-16 (extra reference)	118 / 0.044 / 95.7	304 / 0.293 / 37.8	663 / 1.407 / 33.5	1181 / 4.521 / 45.7

52.4 Coverage across all saved probability levels

Table 50: All 28 original configuration identities at every saved probability level and target slot. Ranges are minima–maxima of configuration means, not confidence intervals, individual-target ranges or population uncertainty. Each configuration uses the same 46 blocks and five seeds; aliases and HSM controls remain included, so counts are not independent algorithms or evidence replications. The final column counts configuration means below/equal to/above the nominal probability, using unrounded values (floating equality tolerance 10^{-12} only, not a statistical calibration margin). Exact agreement at one level does not certify calibration. Area is the mean of individual disk areas; no averaging over configurations, new tests or recalibration.

Minutes	Nominal (%)	Coverage range (%)	Area range (km^2)	Below/equal/above
1	50	22.17–97.83	0.008–0.075	24 / 0 / 4
1	80	45.65–100.00	0.017–0.176	2 / 0 / 26
1	90	72.61–100.00	0.025–0.252	1 / 0 / 27
1	95	93.04–100.00	0.032–0.331	1 / 0 / 27
5	50	2.17–52.17	0.060–0.422	27 / 0 / 1
5	80	15.65–97.83	0.138–0.975	26 / 0 / 2
5	90	23.04–99.13	0.190–1.395	26 / 0 / 2
5	95	29.13–100.00	0.240–1.822	26 / 0 / 2
15	50	0.43–38.26	0.184–1.602	28 / 0 / 0
15	80	6.52–73.04	0.427–3.602	28 / 0 / 0
15	90	10.87–92.61	0.611–5.058	27 / 0 / 1
15	95	16.09–97.83	0.794–6.496	27 / 0 / 1
30	50	6.52–50.00	0.375–4.360	27 / 1 / 0
30	80	15.65–68.70	0.872–9.219	28 / 0 / 0
30	90	21.74–79.13	1.247–12.609	28 / 0 / 0
30	95	26.52–87.39	1.632–15.959	28 / 0 / 0

References

[1] D. S. Johnson, J. M. London, M.-A. Lea and J. W. Durban (2008). Continuous-time correlated random walk model for animal telemetry data. *Ecology*, 89(5), 1208–1215. `https://doi.org/10.1890/07-1032.1`.

[2] Z. Ghahramani and G. E. Hinton (2000). Variational learning for switching state-space models. *Neural Computation*, 12(4), 831–864. `https://doi.org/10.1162/089976600300015619`.

[3] T. Avgar, J. R. Potts, M. A. Lewis and M. S. Boyce (2016). Integrated step selection analysis: bridging the gap between resource selection and animal movement. *Methods in Ecology and Evolution*, 7(5), 619–630. https://doi.org/10.1111/2041-210X.12528.

[4] (2017). Corrigendum. [Correction to Avgar et al. (2016)]. *Methods in Ecology and Evolution*, 8(9), 1168. https://doi.org/10.1111/2041-210X.12725.

[5] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese and A. Alahi (2018). Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2255–2264. https://openaccess.thecvf.com/content_cvpr_2018/html/Gupta_Social_GAN_Socially_CVPR_2018_paper.html.

[6] T. Salzmann, B. Ivanovic, P. Chakravarty and M. Pavone (2020). Trajectron++: Dynamically-Feasible Trajectory Forecasting with Heterogeneous Data. In *Computer Vision – ECCV 2020*, Lecture Notes in Computer Science, 12363, 683–700. Springer. https://doi.org/10.1007/978-3-030-58523-5_40.

[7] P. Kidger, J. Foster, X. Li and T. J. Lyons (2021). Neural SDEs as Infinite-Dimensional GANs. *Proceedings of the 38th International Conference on Machine Learning*, Proceedings of Machine Learning Research, 139, 5453–5463. https://proceedings.mlr.press/v139/kidger21b.html.

[8] T.-T. Gao, B. Barzel and G. Yan (2024). Learning interpretable dynamics of stochastic complex systems from experimental data. *Nature Communications*, 15, 6029. https://doi.org/10.1038/s41467-024-50378-x.

[9] T. Gneiting and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. https://doi.org/10.1198/016214506000001437.

[10] T. Gneiting, L. I. Stanberry, E. P. Grimit, L. Held and N. A. Johnson (2008). Assessing probabilistic forecasts of multivariate quantities, with an application to ensemble predictions of surface winds. *TEST*, 17, 211–235. https://doi.org/10.1007/s11749-008-0114-x.

[11] P. E. Kloeden and E. Platen (1992). *Numerical Solution of Stochastic Differential Equations*. Springer. https://doi.org/10.1007/978-3-662-12616-5.

[12] A. Nichol, J. Achiam and J. Schulman (2018). On First-Order Meta-Learning Algorithms. [Preprint], *arXiv:1803.02999v3*. https://arxiv.org/abs/1803.02999v3.

[13] S. Holm (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. https://www.jstor.org/stable/4615733.

[14] A. C. Davison and D. V. Hinkley (1997). *Bootstrap Methods and their Application*. Cambridge University Press. https://doi.org/10.1017/CBO9780511802843.

[15] NOAA Global Monitoring Division (n.d.). *General Solar Position Calculations*. [Online documentation]. Accessed 7 October 2026. https://gml.noaa.gov/grad/solcalc/solareqns.PDF.

[16] D. Zanaga et al. (2022). *ESA WorldCover 10 m 2021 v200*. [Dataset], version v200. https://doi.org/10.5281/zenodo.7254221.

[17] Overture Maps Foundation (n.d.). *Attribution and Licensing*. [Online documentation]. Accessed 6 October 2026. https://docs.overturemaps.org/attribution/.

[18] B. Lehner and G. Grill (2013). Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15), 2171–2186. https://doi.org/10.1002/hyp.9740.

[19] HydroSHEDS (n.d.). *HydroRIVERS product documentation and licence*. [Online documentation], version v1.0. Accessed 6 October 2026. https://www.hydrosheds.org/products/hydrorivers.

[20] Mapzen / Tilezen (2017). *Terrain Tiles: Types of Terrain Tiles and Skadi format*. [Dataset distribution documentation]. Document revision e3d4351; accessed 7 October 2026. https://registry.opendata.aws/terrain-tiles/. https://github.com/tilezen/joerd/blob/e3d4351ee3e5be333e23f47cb500e9a7c310656a/docs/formats.md.

[21] Mapzen / Tilezen (2017). *Terrain Tiles: Attribution*. [Online documentation]. Document revision d8f587b; accessed 7 October 2026. https://github.com/tilezen/joerd/blob/d8f587b73d26e0a0c42cdccd9ae8c55de4197763/docs/attribution.md.

[22] Copernicus Programme / Sinergise (2021). *Copernicus Digital Elevation Model: GLO-30 Public*. [Dataset distribution documentation], 2021 release; accessed 7 October 2026. https://registry.opendata.aws/copernicus-dem/.

[23] Overture Maps Foundation (2026). *2026-08-19 release notes*. [Dataset release documentation]. Data version 2026-08-19.0; accessed 7 October 2026. https://docs.overturemaps.org/blog/2026/08/19/release-notes/.